

INTRODUCTION

• Artificial Intelligence and Its Growth

Artificial Intelligence (AI) refers to computational systems capable of performing tasks associated with intelligence, including **learning, reasoning, prediction, language processing, decision-making, and problem-solving**. Modern AI has expanded from primarily task-specific systems toward increasingly capable generative, conversational, multimodal, and agentic systems.

This development has changed not only **what AI can do**, but also **how humans use AI**. Natural-language interfaces allow users to communicate requirements, provide context, ask follow-up questions, correct misunderstandings, and continue an interaction without specialized programming knowledge.

AI is consequently becoming increasingly human-facing across areas such as **education, personal assistance, research, customer support, healthcare-related applications, and robotics**.

This development creates two related but distinct questions:

Technical Capability — *What can an AI system do?*

Interaction Capability — *How effectively can it use those capabilities while interacting with a human?*

CORE DISTINCTION

Greater task capability does not necessarily imply equally effective human interaction.

This distinction may become increasingly relevant as AI research progresses toward broader and more general-purpose systems. Effective interaction does not itself constitute Artificial General Intelligence (AGI), but the ability to translate increasingly broad capabilities into appropriate human interaction may represent an important complementary capability of future AI systems.

• Importance of Human–AI Interaction

For an interactive AI system, possessing the correct information or capability is only part of successful operation. The system must also interpret **what the user actually requires** and translate its capabilities into an interaction appropriate to that requirement.

Consider an AI tutor responding to:

“I don't understand electric dipole moment.”

Providing the correct definition and equation demonstrates knowledge, but it does not necessarily demonstrate that the AI understood **why the learner is confused**, selected an appropriate explanation level, communicated clearly, or adapted when the learner required further assistance.

Similar interaction requirements appear in different forms across AI applications:

Application	Interaction Requirement
AI Tutor	Understand difficulty and explain appropriately
Personal Assistant	Interpret requirements, context, and relevant preferences
Customer-Support AI	Identify the actual problem and guide resolution
Research Assistant	Understand intent and communicate relevant information
Interactive Robot	Interpret human requirements and respond appropriately

The task changes, but the underlying interaction challenge remains.

An AI may possess the capability required to solve a problem while still failing to translate that capability into an effective interaction with the human using it.

Human–AI interaction therefore introduces an evaluation perspective beyond isolated task completion.

• The Existing Evaluation Problem

AI evaluation already examines numerous important properties, including **accuracy, reasoning, robustness, safety, instruction-following, helpfulness, and conversational performance**.

Modern evaluation frameworks also increasingly examine models across multiple dimensions rather than relying on a single performance measure.

However, properties relevant to effective human interaction may appear across several different evaluations rather than as one explicitly defined construct.

For example, an AI response may be technically correct while the system:

- misunderstands the user's actual requirement;
- communicates at an unsuitable level;
- fails to use relevant context;
- adapts poorly to available user information;
- behaves inconsistently across interactions; or
- responds inadequately as the conversation develops.

Correctness and effective interaction are therefore related, but they are not equivalent.

The problem addressed in this work is consequently narrower than claiming that existing AI evaluation ignores interaction.

The central research problem is whether the AI-side capability to build and maintain effective connection during human–AI interaction can be defined as a distinct multidimensional construct and evaluated systematically.

This produces the principal research question:

Can an AI system's connection-building capability be defined, decomposed into measurable dimensions, evaluated across interactions, and represented through a structured metric?

• Introducing BOND

This work proposes **BOND**, an evaluation framework designed around **AI connection-building capability during human–AI interaction**.

BOND evaluates how effectively an AI system builds and maintains connection with a human through its behavior during interaction.

BOND deliberately evaluates the interaction from the **AI side**.

It does **not** attempt to measure the complete mutual relationship between a particular human and an AI. Such a measurement would require variables from both participants and could vary substantially according to individual human characteristics, expectations, communication styles, and circumstances.

BOND instead establishes a clear evaluation direction:

Human → provides the interaction context

AI → remains the evaluation target

The framework decomposes AI connection-building capability into five core factors:

Factor	Primary Focus
Understanding (U)	Understanding the user's actual requirement, meaning, and relevant context
Clarity (C)	Communicating information in an understandable and usable manner
Personalization (P)	Appropriately adapting interaction using relevant user- or situation-specific information
Reliability (R)	Maintaining dependable and consistent behavior
Engagement & Responsiveness (ER)	Responding appropriately as interaction develops and maintaining useful continuity

These factors are not proposed as separate forms of intelligence. Together, they provide different perspectives from which **AI-side connection-building behavior** can be evaluated.

BOND is also not an AGI metric. A high BOND value does not establish general intelligence. Rather, BOND investigates one potentially relevant dimension of increasingly general and human-

centered AI systems: **how effectively their capabilities are translated into interaction with humans.**

• Scope of This Work

This research develops BOND from a conceptual construct toward a **defined, measurable, testable, and comparable evaluation framework.**

The work proceeds through five major stages:

- 1. Define** → Establish the meaning, boundaries, and principles of AI connection-building capability.
- 2. Decompose** → Develop the five core factors and their underlying components.
- 3. Measure** → Construct the mathematical model, scoring intervals, weighting structure, and benchmark methodology.
- 4. Test** → Evaluate BOND across controlled interactions and compare conventional AI behavior with an interaction-oriented implementation.
- 5. Validate** → Examine the framework through experimental evidence and multidisciplinary expert review.

The framework is intended for investigation across interactive systems such as **conversational AI, AI tutors, personal assistants, customer-support systems, voice assistants, interactive robots, and future general-purpose AI systems.**

RESEARCH OBJECTIVE

The objective of BOND is not merely to create another AI score, but to determine whether AI connection-building capability can be transformed from an abstract interaction concept into a systematically defined and experimentally evaluable construct.

The following chapters establish the theoretical foundation, evaluation structure, mathematical framework, testing methodology, validation process, applications, limitations, and future development of BOND.

• 1.1 AI Evolution

History of Artificial Intelligence

The development of Artificial Intelligence began with the idea that aspects of **human reasoning, learning, and problem-solving could be represented computationally.** In 1950, Alan Turing proposed the question “*Can machines think?*” and introduced what later became known as the **Turing Test**, an early framework for considering machine intelligence.

The term **Artificial Intelligence** was formally introduced during the **Dartmouth workshop of 1956**, commonly regarded as a foundational event in AI as an academic field.

Early AI relied heavily on **symbolic reasoning and rule-based systems**, where human-defined rules represented knowledge and guided decisions. This approach later contributed to **expert systems**, which attempted to reproduce specialist decision-making within restricted domains.

AI subsequently shifted increasingly toward **Machine Learning (ML)**, where systems learn patterns from data rather than depending entirely on explicitly programmed rules. Improvements in computational power, data availability, and learning algorithms accelerated this transition.

The development of **Deep Learning**, particularly neural-network-based approaches, produced major advances in areas such as computer vision, speech recognition, natural-language processing, and pattern recognition.

More recently, **Transformers, Large Language Models (LLMs), generative AI, multimodal models, and AI agents** have expanded AI from specialized prediction and classification toward systems capable of generating content, reasoning across tasks, interacting through natural language, processing multiple information types, and performing multi-step actions.

AI Evolution — Compact Timeline

Period	Major Development	Main Characteristic
1950s	Foundations of AI	Machine intelligence becomes a formal research question
1956 onward	Symbolic AI	Rules, logic and symbolic reasoning
1970s–1980s	Expert Systems	Domain-specific knowledge and decision rules
1990s–2000s	Machine Learning	Learning patterns from data
2010s	Deep Learning	Large neural networks and representation learning
Late 2010s–2020s	Transformers & Generative AI	Language generation and broader generative capabilities
Current direction	Multimodal & Agentic AI	Multiple modalities, interaction and multi-step task execution

CORE SHIFT

AI has progressively moved from **explicitly programmed rules** toward systems that **learn from data, generate information, interact with humans, and increasingly perform complex multi-step tasks**.

• Current Artificial Intelligence

Current AI is dominated by **data-driven models**, particularly deep neural networks, foundation models, and generative AI systems. Unlike traditional rule-based systems, these models can learn complex patterns from large datasets and apply those learned representations across a wide range of tasks.

Modern AI can be understood through several major capabilities:

Capability	Current AI Can
Language	Understand and generate natural language
Vision	Analyze images, video, objects, and visual patterns
Audio	Process speech, sound, and voice
Generation	Generate text, images, audio, video, and code
Reasoning	Perform increasingly complex problem-solving and structured reasoning
Multimodality	Work across multiple information types within one system
Tool Use	Use software tools and external systems to complete tasks
Interaction	Maintain conversational context and respond across multiple turns
Agency	Plan and execute parts of multi-step tasks with varying levels of autonomy

Foundation Models and LLMs

A major development in current AI is the rise of **foundation models**—large models trained on broad datasets that can be adapted or applied to many downstream tasks.

Large Language Models (LLMs) are an important example. Rather than being designed for only one fixed task, an LLM can support activities such as **question answering, summarization, translation, coding, reasoning, tutoring, and conversation** through a common language interface.

Current Limitation

Despite rapid progress, current AI remains imperfect. Systems can produce incorrect information, misunderstand context, behave inconsistently, struggle with long or complex tasks, and vary significantly in performance across domains.

CORE IDEA

Modern AI combines increasingly broad capabilities within interactive systems, but broad capability does not guarantee reliable understanding, adaptation, or effective human interaction.

This creates an important transition: as AI becomes more capable, multimodal, agentic, and human-facing, research increasingly asks not only how current systems perform, but **what the next generation of AI may require**.

• Future Artificial Intelligence

The future development of AI is expected to focus not only on increasing computational capability, but also on creating systems that are **more general, adaptive, autonomous, multimodal, reliable, and effective in human environments**.

Several directions are particularly important:

Direction	Goal
General-Purpose AI	Perform effectively across a broader range of tasks and domains
AI Agents	Plan, use tools, and execute multi-step tasks with greater autonomy
Multimodal AI	Integrate language, vision, audio, video, and other information
Adaptive AI	Adjust behavior according to changing context and requirements
Embodied AI & Robotics	Connect AI reasoning with perception and action in physical environments
Reliable AI	Improve consistency, robustness, safety, and dependability
Human-Centered AI	Improve how AI systems support and interact with humans

Artificial General Intelligence

Artificial General Intelligence (AGI) generally refers to the goal of developing AI with broad and flexible capabilities rather than intelligence restricted to a narrow set of predefined tasks.

However, **general capability and effective human interaction are not identical**.

A future system may become capable across many domains, but humans using that system could still differ in **knowledge, goals, communication style, context, and requirements**. The AI would therefore need to determine not only *what* it can do, but also *how* its capabilities should be applied within a particular interaction.

FUTURE CHALLENGE

As AI becomes more general, the challenge may increasingly shift from simply expanding capability toward effectively adapting and applying that capability within changing human contexts.

This does not mean that human-interaction capability alone leads to AGI. Rather, it may represent **one complementary component** required for increasingly general AI systems to operate effectively alongside humans.

Future AI may therefore be defined not only by greater intelligence, but also by how effectively that intelligence can be translated into useful, adaptive, and reliable interaction.

1.2 AI in Different Fields

Artificial Intelligence is increasingly applied across fields where systems must not only process information but also **support decisions, perform tasks, generate outputs, or interact directly with humans**. The role and requirements of AI therefore change according to the environment in which it operates.

• Education

AI in education can support **personalized learning, tutoring, assessment, content generation, feedback, translation, and administrative tasks**.

An AI tutor, for example, can explain concepts, generate practice questions, evaluate responses, and assist learners at different stages of learning. However, educational interaction cannot depend only on whether an answer is technically correct.

Two students asking the same question may differ in:

Knowledge level • Learning difficulty • Previous understanding • Required depth • Preferred explanation

Therefore, an educational AI may need to **identify the learner's requirement and adapt its explanation accordingly**.

KEY IDEA

In education, effective AI requires not only knowledge of the subject, but also the ability to communicate that knowledge according to the learner's needs.

• Healthcare

AI is used in healthcare-related systems for tasks such as **medical-image analysis, clinical decision support, documentation, information retrieval, risk prediction, and patient-facing assistance**.

In human-facing applications, communication becomes particularly important because users may interpret AI-generated information differently depending on their knowledge and situation.

AI systems in this domain therefore require strong attention to **accuracy, reliability, appropriate communication, uncertainty, privacy, and human oversight**.

AI can support healthcare professionals and patients, but high-stakes medical decisions require appropriate professional oversight rather than unrestricted dependence on AI output.

• Robotics

AI enables robots to move beyond fixed programmed actions toward systems capable of **perception, decision-making, navigation, object recognition, planning, and interaction with humans and their environment**.

In human-facing robotics, physical capability alone is insufficient. A robot may need to interpret instructions, understand relevant context, respond to changing situations, and communicate its actions appropriately.

This becomes important in areas such as **service robots, industrial collaboration, assistive robotics, autonomous systems, and future personal robots.**

KEY IDEA

For interactive robots, intelligence must connect perception and action with an appropriate understanding of human requirements.

• Customer Support

AI-based customer-support systems are used to **answer questions, retrieve information, classify problems, automate routine requests, and assist users in resolving issues.**

However, customer queries are rarely identical. Different users may describe the same problem using different words, provide incomplete information, or require different levels of explanation.

An effective support AI therefore needs to:

Identify the actual problem → use relevant context → provide clear guidance → respond to follow-up information → maintain consistency until resolution.

A technically correct response may still be ineffective if it fails to address the user's actual problem.

KEY IDEA

In customer support, successful AI interaction is determined not only by providing information, but by helping the user move from a problem toward an appropriate resolution.

• Research

AI is increasingly used in research to support **literature analysis, data processing, pattern recognition, simulation, coding, hypothesis exploration, scientific writing assistance, and information retrieval.**

AI can help researchers process large amounts of information more efficiently, but research assistance requires more than simply generating information. The system must understand the **research question, domain, required depth, constraints, and context** before producing a useful response.

For example, the same scientific concept may need to be explained differently when the user requires a **basic overview, mathematical derivation, literature comparison, experimental design, or technical analysis.**

KEY IDEA

In research, AI becomes more useful when it can connect broad computational capabilities with the specific objective and context of the researcher.

Human verification remains important, particularly where AI-generated information influences **scientific conclusions, citations, methodology, or experimental decisions.**

• Daily Life

AI has also become part of everyday digital environments through **virtual assistants, search and recommendation systems, navigation, translation, communication tools, smart devices, content platforms, and generative AI assistants.**

Daily-life interaction presents a unique challenge because users differ substantially in their **goals, knowledge, preferences, communication styles, and circumstances.**

A useful personal AI may therefore need to move beyond providing generic responses and appropriately use available context to determine **what the user needs and how the response should be delivered.**

KEY IDEA

As AI becomes part of everyday life, interaction quality becomes relevant not only to specialized applications but also to ordinary human–AI communication.

AI Across Fields — Common Pattern

Although these fields differ significantly, a common pattern appears:

Education → learner-specific communication

Healthcare → careful and reliable assistance

Robotics → context-aware human interaction

Customer Support → problem-oriented communication

Research → objective- and context-aware assistance

Daily Life → adaptive everyday interaction

CORE OBSERVATION

The task performed by AI changes across fields, but the need to understand human requirements and translate AI capability into appropriate interaction repeatedly appears across human-facing applications.

1.3 Human–AI Interaction

• What is Human–AI Interaction?

Human–AI Interaction (HAI) refers to the process through which humans and AI systems exchange information, instructions, responses, feedback, and actions while working toward a goal or continuing an interaction.

Traditional human–computer interaction often involves predefined commands, menus, buttons, or interfaces. Modern AI introduces a more dynamic form of interaction in which users can communicate through **natural language, voice, images, gestures, and contextual information**.

Human–AI interaction can therefore take multiple forms:

Interaction Type	Example
Text	Conversational AI and chatbots
Voice	Voice assistants
Visual	Image- and video-based AI systems
Physical	Human–robot interaction
Multimodal	Combined text, voice, image, and other inputs
Continuous	Multi-turn assistants and AI agents

Interaction as a Process

Human–AI interaction is not always a single:

Input → Output

In many systems it develops as:

Human Requirement → AI Interpretation → Response → Human Feedback/New Context → AI Adaptation → Continued Interaction

This makes interaction a **dynamic process**. The meaning of later responses may depend on information established earlier in the conversation.

CORE IDEA

Human–AI interaction is not defined only by the information exchanged, but also by how the AI interprets and responds to changing human requirements throughout the interaction.

This distinction becomes especially important in systems designed for **repeated, personalized, or long-term interaction**, where the AI may need to maintain relevant context rather than treating every message as an entirely independent request.

• Role of AI in Human–AI Interaction

In Human–AI Interaction, AI acts as the **computational participant** that receives human input, interprets available information, processes the requirement, and generates an appropriate response or action.

Its role can be represented as:

Receive → Interpret → Process → Respond → Adapt

1. Receive

The AI receives information through **text, voice, images, sensor data, gestures, or other supported inputs.**

2. Interpret

The system determines the meaning of the input and identifies the **user's requirement, relevant context, and intended task.**

3. Process

AI applies its available **knowledge, reasoning, learned patterns, tools, and computational capabilities** to determine an appropriate response or action.

4. Respond

The result must then be translated into a form that the human can **understand, use, or act upon.**

5. Adapt

When new information, feedback, corrections, or requirements appear, an interactive AI may need to **adjust its subsequent behavior** rather than treating the interaction as unchanged.

CORE IDEA

The role of AI in interaction is not limited to generating an output; it includes interpreting human input and translating computational capability into an appropriate response or action.

For interactive AI, failure at any stage can affect the overall interaction. A system may possess sufficient knowledge but **misinterpret the requirement**, or understand the requirement but **communicate the result ineffectively.**

Therefore, effective Human–AI Interaction depends on how successfully these stages operate together.

• Importance of Human–AI Interaction

As AI becomes more capable and increasingly integrated into human-facing systems, the effectiveness of interaction influences **how successfully AI capabilities are delivered to users.**

Strong technical performance alone may not ensure successful interaction. The system must also handle differences in **human intent, context, knowledge, communication, and changing requirements.**

Three aspects make Human–AI Interaction particularly important:

1. Capability Utilization

An AI's capabilities become useful only when the system can correctly interpret when and how they should be applied to the user's requirement.

2. Communication

The output must be presented in a form that the user can appropriately understand and use. The same information may require different explanations depending on the interaction context.

3. Continuity and Adaptation

In multi-turn or repeated interactions, new information can change the meaning of earlier requirements. AI therefore needs to respond appropriately as the interaction develops.

KEY DISTINCTION

AI capability determines what a system can potentially do; Human–AI Interaction influences how effectively that capability reaches and serves the human user.

This importance increases in systems where AI operates as an **assistant, tutor, collaborator, support system, or interactive robot**, because successful operation depends partly on continued communication rather than isolated outputs.

Human–AI Interaction therefore acts as a bridge between AI capability and its effective application in human environments.

1.4 Need for Better Interaction

• Why Better Interaction Is Needed

As AI systems become more capable, their usefulness increasingly depends on how effectively those capabilities are applied within human interaction. A powerful AI may still provide an ineffective experience if it **misunderstands intent, ignores relevant context, communicates poorly, fails to adapt, or behaves inconsistently**.

Better interaction is therefore required for several reasons:

Understanding Human Requirements — Users may express incomplete, ambiguous, or context-dependent requirements. AI must identify what is actually needed.

Appropriate Communication — Correct information must be communicated at a level and in a form suitable for the interaction.

Adaptation — Different users and situations may require different responses rather than one fixed interaction pattern.

Consistency — AI should maintain relevant context and dependable behavior as an interaction develops.

Effective Collaboration — As AI increasingly assists humans rather than merely executing isolated commands, successful outcomes can depend on how well the system and user communicate throughout the task.

CORE IDEA

Improving AI capability increases what a system can do; improving interaction increases how effectively those capabilities can be applied for humans.

• Importance for Future AI

Future AI systems are likely to operate across broader tasks, environments, and forms of interaction. **General-purpose assistants, AI agents, interactive robots, adaptive educational systems, and other human-facing AI** may interact with users repeatedly and under changing conditions.

As the range of AI capabilities expands, interaction itself can become more complex. A system may need to determine:

What does the user need? → What context matters? → Which capability should be used? → How should the result be communicated? → Does the interaction need to adapt?

This creates an important future requirement:

More capable AI does not eliminate the interaction problem; broader capability may make effective interaction even more important.

For increasingly general-purpose AI, interaction capability should therefore be considered alongside improvements in **reasoning, knowledge, autonomy, multimodality, reliability, and safety**.

The future challenge is not only to develop AI that can perform more tasks, but also to develop systems capable of applying those abilities appropriately while working with humans.

This need establishes the foundation for investigating whether **AI-side interaction and connection-building capability can be systematically defined and evaluated**—the central problem addressed by BOND.

CHAPTER 2 — EXISTING AI EVALUATION METHODS

• 2.1 Traditional Metrics

Before modern conversational and generative AI systems, AI and machine-learning evaluation largely relied on **task-specific quantitative metrics**. These metrics remain important because they provide objective and reproducible methods for measuring particular aspects of system performance.

However, each metric answers a **specific evaluation question** rather than representing the complete quality of an AI system.

Accuracy

Accuracy measures the proportion of predictions that a system classifies correctly among all evaluated predictions.

Accuracy = Correct Predictions / Total Predictions

For example, if a classifier produces 100 predictions and 90 are correct, its accuracy is **90%**.

Best suited for: Classification tasks, particularly when classes are reasonably balanced.

Limitation: Accuracy can hide poor performance on less frequent classes when the dataset is highly imbalanced.

Accuracy answers: *“How often was the system correct?”*

Precision

Precision measures how many items predicted as positive are actually positive.

Precision = True Positives / (True Positives + False Positives)

High precision means the system produces relatively few **false-positive predictions**.

Example: If an AI identifies 20 documents as relevant but only 15 are actually relevant:

Precision = 15 / 20 = 0.75

The Stanford Information Retrieval text similarly defines precision as the fraction of retrieved documents that are relevant.

Precision asks: *“When the system predicted positive, how often was it correct?”*

Recall

Recall measures how many of the actually positive or relevant items were successfully identified by the system.

Recall = True Positives / (True Positives + False Negatives)

If 20 relevant documents exist but the system retrieves only 15:

Recall = 15 / 20 = 0.75

In information retrieval, recall is correspondingly defined as the fraction of relevant documents that are successfully retrieved.

Recall asks: *“Of everything the system should have identified, how much did it successfully find?”*

Precision vs. Recall

Metric	Primary Question
Accuracy	How often is the system correct overall?
Precision	How reliable are its positive predictions?
Recall	How much of the relevant information did it successfully identify?

These metrics remain fundamental for evaluating specific prediction and retrieval tasks. However, they do not directly measure properties such as **communication quality, contextual adaptation, personalization, or multi-turn interaction.**

CORE DISTINCTION

Traditional metrics can precisely measure specific forms of task performance without necessarily measuring the broader quality of interaction between an AI system and a human.

BLEU

BLEU (Bilingual Evaluation Understudy) was introduced in 2002 as an automatic metric for evaluating **machine translation**. It compares a machine-generated translation with one or more human reference translations by measuring overlap in short word sequences called **n-grams**.

BLEU primarily uses **modified n-gram precision**, together with a **brevity penalty** that discourages translations from receiving high scores simply by producing very short outputs.

Higher BLEU score → greater n-gram similarity to the reference translation.

Best suited for:

Machine translation and other reference-based text-generation settings where lexical overlap is informative.

Major Limitation:

A generated response can express the correct meaning using different wording and still receive a lower BLEU score. Therefore, lexical similarity does not necessarily represent **semantic quality, usefulness, or interaction quality.**

BLEU asks: *“How closely does the generated text overlap with reference text?”*

ROUGE

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) was introduced for automatic evaluation of **text summarization**. It compares a generated summary with human-written reference summaries using different forms of textual overlap.

Common variants include:

ROUGE-N — overlap of n-grams.

ROUGE-L — similarity based on the longest common subsequence.

ROUGE-S — skip-bigram based comparison.

Unlike BLEU's original emphasis on precision for translation evaluation, ROUGE was designed with a stronger **recall-oriented perspective** for summarization.

Best suited for:

Reference-based evaluation of summaries and related text-generation tasks.

Major Limitation:

Like BLEU, ROUGE depends strongly on comparison with reference text. A response can be useful and semantically appropriate while using substantially different wording from the reference.

ROUGE asks: “How much important reference content appears in the generated text?”

Traditional Metrics — Overall Perspective

Metric	Primarily Measures	Originally/Commonly Used For
Accuracy	Overall correctness	Classification
Precision	Correctness of positive predictions	Classification / Retrieval
Recall	Coverage of relevant positives	Classification / Retrieval
BLEU	Reference n-gram overlap	Machine Translation
ROUGE	Reference-content overlap	Summarization

These metrics are valuable because they convert specific performance characteristics into **quantitative and reproducible measurements**.

However:

CORE LIMITATION

A metric designed to evaluate correctness, classification, retrieval, or reference-text similarity does not by itself determine whether an AI understands a particular user, adapts appropriately, communicates effectively, or maintains quality across an interaction.

This does **not** make traditional metrics obsolete. Instead, it demonstrates that **different evaluation objectives require different metrics**.

2.2 Modern AI & LLM Evaluation

As Large Language Models became capable of producing **open-ended, conversational, and multi-turn responses**, evaluation became more difficult than checking a single predefined correct answer.

Modern evaluation therefore increasingly uses **human judgment, model-based evaluation, multi-turn benchmarks, and pairwise preference testing** alongside traditional metrics. Research introducing MT-Bench and Chatbot Arena specifically identified the difficulty of evaluating open-ended chat assistants using conventional benchmarks alone.

• Human Evaluation

Human evaluation asks people—often users or expert annotators—to directly assess AI outputs according to defined criteria or preferences.

Humans can evaluate qualities that may be difficult to capture through simple lexical metrics, including:

Helpfulness • Relevance • Clarity • Instruction-following • Overall preference

Evaluation may use:

Rating → Give a response a numerical score.

Ranking → Rank several responses.

Pairwise Comparison → Choose the better of two responses.

ADVANTAGE

Human evaluation can capture subjective and context-dependent qualities that automatic reference-based metrics may miss.

However, human evaluation can be **expensive, slow, variable between evaluators, and difficult to scale** across very large numbers of model responses.

• LLM-as-a-Judge

To make open-ended evaluation more scalable, researchers increasingly use another capable LLM as an **evaluator or judge**.

The judge receives an AI response—sometimes together with the original question, reference answer, rubric, or competing response—and produces a **score, comparison, or evaluation**.

A simplified process is:

Prompt → Candidate Response(s) → LLM Judge → Score / Preference

The MT-Bench research found strong LLM judges could show substantial agreement with human preferences, suggesting model-based judging can provide a scalable approximation of some forms of human evaluation.

However, LLM judges are not neutral measurement instruments. Documented issues include **position bias, verbosity bias, self-enhancement bias, and limitations in reasoning**. Later research has continued to find systematic gaps between LLM and human judgments.

CORE IDEA

LLM-as-a-Judge increases the scalability of open-ended evaluation, but the reliability of the evaluator itself must also be examined.

• MT-Bench

MT-Bench was introduced as a benchmark for evaluating chat assistants using **multi-turn questions** and LLM-based judging.

Unlike evaluations focused only on isolated responses, MT-Bench tests models across conversations where later turns can depend on earlier interaction. It was introduced specifically alongside research examining LLM judges and human preferences.

This represents an important development:

Single-turn evaluation → Multi-turn conversational evaluation

MT-Bench is therefore particularly relevant to interaction-oriented research because it recognizes that model behavior can change when conversation develops across multiple turns.

However, its primary objective is still evaluation of **chat-assistant response quality and capability**, rather than explicitly constructing a multidimensional measure of AI-side connection-building.

MT-Bench demonstrates that conversation itself can become an evaluation environment rather than treating every AI response as an independent event.

• Chatbot Arena

Chatbot Arena was introduced as an open evaluation platform for comparing LLMs using **human preferences**. Instead of requiring one predefined correct answer, users compare responses from two anonymous models and select which response they prefer.

The basic process is:

User Prompt → Model A + Model B → Anonymous Responses → Human Preference → Model Ranking

This approach provides an important advantage: models are evaluated using **real user prompts and direct human judgments**, allowing evaluation of open-ended responses that may have several valid answers.

The original Chatbot Arena paper reported that crowdsourced human votes showed agreement with expert raters and used these pairwise preferences to statistically rank models.

Primary strength:

Captures **overall human preference** under realistic and open-ended interactions.

Limitation:

A preference vote primarily identifies **which response or model a user prefers overall**. By itself, it does not explain precisely *why* the interaction succeeded or failed across dimensions such as understanding, personalization, reliability, or responsiveness.

Arena asks: “Which model does the human prefer?”

• Broad Capability Benchmarks

Modern AI is also evaluated through benchmarks designed to test performance across **multiple subjects and capabilities** rather than one specialized task.

A major example is **MMLU (Massive Multitask Language Understanding)**, which evaluates models across **57 subjects**, including mathematics, history, computer science, law, and other academic and professional domains. It was designed to examine broad knowledge and problem-solving ability.

Such benchmarks help answer:

What does the model know?

Across how many domains can it perform?

How effectively can it solve predefined problems?

They are therefore useful for measuring the **breadth of AI capability**.

However, strong performance on knowledge or reasoning benchmarks does not directly establish how effectively the same model interacts with an individual human.

CAPABILITY $\neq$ INTERACTION

Knowing how capable a model is across tasks does not automatically tell us how effectively those capabilities are translated into human interaction.

• Multidimensional Evaluation

Modern evaluation is also moving beyond single benchmark scores.

For example, **HELM (Holistic Evaluation of Language Models)** evaluates language models across multiple scenarios and metrics rather than reducing evaluation to accuracy alone. Its framework explicitly emphasizes broad coverage, multiple measurements, transparency, and recognition that no evaluation can capture everything.

HELM-related evaluation has also examined instruction-following systems using dimensions such as **helpfulness, understandability, completeness, conciseness, and harmlessness**.

This point is particularly important for BOND:

Existing AI research already recognizes that model quality is multidimensional.

Therefore, BOND should **not** claim that multidimensional AI evaluation is new.

Its proposed contribution must instead be narrower: defining and evaluating a particular multidimensional construct centered on **AI-side connection-building capability**.

Modern Evaluation — Overall Perspective

Method	Main Evaluation Target
Human Evaluation	Human judgment of model outputs
LLM-as-a-Judge	Scalable model-based evaluation
MT-Bench	Multi-turn chat-assistant quality
Arena	Human preference between models
MMLU	Broad knowledge and problem-solving
HELM	Multidimensional model evaluation

Modern evaluation therefore captures far more than traditional accuracy or text-overlap metrics.

But this creates the real question for BOND—not whether interaction-related evaluation exists, but whether the specific construct BOND proposes is already adequately measured.

2.3 Limitations of Existing AI Evaluation Methods

Existing AI evaluation methods provide valuable measurements of **correctness, task capability, language quality, human preference, conversational performance, and other model characteristics**. However, no single evaluation method is intended to represent every dimension of an AI system.

For interaction-oriented evaluation, several limitations become particularly relevant.

• Task-Specific Measurement

Traditional metrics generally measure a specific property.

Accuracy → correctness

Precision → positive prediction quality

Recall → relevant-item coverage

BLEU/ROUGE → reference-text overlap

These measurements can be highly useful for their intended tasks, but they provide limited information about **how an AI behaves while interacting with a human**.

A metric can accurately measure one property without representing the complete quality of an AI interaction.

• Single-Turn Evaluation

Many evaluations examine a model through an individual **prompt** → **response** pair.

However, interactive AI frequently operates through multiple turns:

Request → **Response** → **Feedback** → **Clarification** → **Adaptation** → **Continued Interaction**

Important behaviors may therefore become visible only as the conversation develops.

Multi-turn benchmarks such as **MT-Bench** address part of this limitation, demonstrating that conversational evaluation can extend beyond isolated responses.

Nevertheless, evaluating multi-turn quality does not automatically provide a dedicated measurement of every interaction-related dimension.

• Overall Preference Can Hide the Reason

Human preference systems such as Arena can identify which model users prefer overall.

However, two users may prefer the same response for different reasons:

Clarity • Correctness • Detail • Style • Speed • Relevance • Personal preference

An overall preference therefore provides valuable information about **which response is favored**, but may not fully identify the underlying interaction behavior responsible for that preference.

• Interaction Properties Are Distributed Across Metrics

Modern frameworks already evaluate several properties relevant to effective interaction, including **helpfulness, instruction-following, understandability, consistency, conversational quality, and human preference**.

The issue is therefore not the absence of interaction-related evaluation.

Instead:

Interaction-relevant properties are often evaluated as separate capabilities, criteria, or benchmark outcomes rather than as components of one explicitly defined AI-side connection-building construct.

This distinction is central to the motivation for BOND.

• Context and Personalization Are Difficult to Evaluate

Effective interaction can depend on information established earlier in a conversation or on relevant information about the user's requirements.

A system may need to determine:

What information is relevant? → Should it be used? → How should the response change?

This makes personalization and contextual adaptation more difficult to evaluate than simple correctness because an appropriate response may depend on the **specific interaction context**.

• Aggregate Scores Can Hide Different Behaviors

Two AI systems can potentially receive similar overall evaluation scores while reaching those scores through different strengths and weaknesses.

For example:

AI-A: Strong understanding + weak personalization

AI-B: Moderate understanding + strong personalization

A single aggregate score may not reveal this difference unless the underlying dimensions are measured separately.

Therefore, multidimensional evaluation is useful when the objective is not only to determine **which system performs better**, but also **why**.

Overall Limitation

Existing evaluation methods should therefore be viewed as **complementary rather than insufficient as a whole**. Different methods answer different questions.

CORE OBSERVATION

Current AI evaluation can measure many properties relevant to successful interaction, but these measurements do not necessarily constitute a unified evaluation of how effectively an AI builds and maintains connection from the AI side of human–AI interaction.

This distinction leads directly to the research gap investigated in this work.

2.4 Research Gap

Existing AI evaluation provides a broad set of methods for measuring **task performance, correctness, language quality, reasoning, instruction-following, human preference, conversational performance, and multidimensional model behavior**.

Modern approaches have also expanded evaluation beyond isolated outputs through **human evaluation, LLM-based judges, multi-turn benchmarks, and real-world preference testing.**

Therefore, the research gap addressed by BOND is **not** that Human–AI Interaction has never been evaluated.

The gap is more specific.

• **From Individual Properties to a Defined Construct**

Several properties relevant to effective interaction are already evaluated independently.

For example:

Understanding → instruction-following and contextual interpretation

Clarity → understandability and response quality

Personalization → contextual adaptation

Reliability → consistency and robustness

Engagement & Responsiveness → conversational and multi-turn behavior

However, measuring these properties separately does not automatically establish a unified construct representing **AI connection-building capability.**

RESEARCH GAP 1

There is a need to investigate whether interaction-relevant AI behaviors can be organized into a clearly defined construct representing the AI's capability to build and maintain effective connection during human interaction.

• **AI-Side Direction of Evaluation**

Human–AI interaction contains variables from both the human and the AI.

A complete relationship measure would therefore depend partly on characteristics of individual users, making the measurement highly dependent on the specific human–AI pair.

BOND investigates a narrower question by keeping the **AI as the evaluation target** and treating the human as the interaction context.

Human–AI Relationship

≠

AI Connection-Building Capability

RESEARCH GAP 2

A dedicated AI-side evaluation perspective is required to investigate connection-building capability without treating the complete human–AI relationship as the measured object.

• From Interaction Quality to Diagnostic Measurement

An overall evaluation can indicate that one system performs better than another without necessarily revealing **which interaction capability created the difference**.

A connection-oriented framework should therefore provide both:

Overall measurement → **How strongly does the system demonstrate the construct?**

and

Factor-level measurement → **Where does the system succeed or fail?**

For example:

BOND Score → **Overall connection-building capability**

while

U, C, P, R, ER → **Individual interaction dimensions**

RESEARCH GAP 3

Connection-building evaluation requires a structure capable of representing both overall performance and the distinct factors responsible for that performance.

• From Concept to Testable Measurement

Terms such as *understanding*, *personalization*, *reliability*, and *engagement* are conceptually meaningful but cannot function as scientific measurements unless they are operationally defined.

A measurable framework therefore requires:

Construct Definition

↓

Factor Definition

↓

Observable Criteria

↓

Testing Method

↓

Scoring Model

↓

Validation

Without these stages, “AI connection” remains primarily a conceptual description rather than an evaluable construct.

RESEARCH GAP 4

The central methodological challenge is to convert AI connection-building from an abstract concept into an operationalized, reproducible, and empirically testable framework.

The Gap Addressed by BOND

The four gaps converge into one central research problem:

Can AI-side connection-building capability be formally defined as a multidimensional construct, decomposed into measurable factors, evaluated through controlled human–AI interactions, and represented through a validated scoring framework?

BOND addresses this question through five proposed dimensions:

Understanding (U)

Clarity (C)

Personalization (P)

Reliability (R)

Engagement & Responsiveness (ER)

The objective is **not to replace existing AI benchmarks**.

Instead, BOND is intended to provide a complementary evaluation perspective focused specifically on **how effectively an AI builds and maintains connection while interacting with humans**.

CHAPTER CONCLUSION

Existing metrics primarily provide measurements of particular capabilities, outputs, preferences, or behaviors. BOND investigates whether connection-building itself can be operationalized as a distinct AI-side multidimensional evaluation construct.

This research gap establishes the foundation for formally defining the **BOND framework, its boundaries, factors, and measurement principles** in the following chapters.

CHAPTER 3 — BOND FRAMEWORK

3.1 What is BOND?

BOND is a multidimensional evaluation framework proposed to measure an AI system's connection-building capability during human–AI interaction.

It evaluates how effectively an AI system can **understand a human requirement, communicate appropriately, adapt to relevant context, behave reliably, and respond effectively as an interaction develops.**

BOND evaluates the interaction specifically from the **AI side**:

Human → provides interaction context

AI → remains the evaluation target

This distinction defines the fundamental boundary of the framework.

BOND does not attempt to quantify the complete emotional, social, or psychological relationship between a human and an AI. Such a relationship would depend on characteristics of both participants and could vary substantially between individual users.

Instead, BOND asks a narrower and measurable question:

How effectively does the AI demonstrate behaviors that enable it to build and maintain an effective connection with a human during interaction?

The Five-Dimensional Structure

BOND represents this capability through five core dimensions:

Dimension	Evaluates
Understanding (U)	How effectively the AI understands the user's requirement, meaning, and relevant context
Clarity (C)	How effectively the AI communicates information in an understandable and usable form
Personalization (P)	How appropriately the AI adapts using relevant user- or situation-specific information
Reliability (R)	How consistently and dependably the AI behaves throughout interaction
Engagement & Responsiveness (ER)	How appropriately the AI responds as interaction develops and maintains useful continuity

These dimensions represent different but related components of AI-side connection-building behavior.

They are not intended to represent independent forms of intelligence. Instead, their combined evaluation provides a structured view of **how effectively an AI translates its capabilities into interaction with a human.**

BOND as an Evaluation Layer

BOND is intended to complement rather than replace existing AI evaluation.

Traditional and modern benchmarks may determine:

Is the answer correct?

Can the model solve the task?

Which response does a human prefer?

How capable is the model across domains?

BOND introduces an additional question:

How effectively does the AI use its capabilities while building and maintaining connection within the interaction?

A model can therefore perform strongly on conventional capability benchmarks while demonstrating different levels of performance across individual BOND dimensions.

This makes BOND potentially useful as both:

Overall Evaluation → representing connection-building capability through a combined measurement.

Diagnostic Evaluation → identifying specific strengths and weaknesses across individual dimensions.

Conceptual Architecture

The basic structure of BOND can be represented as:

Human Interaction Context

↓

AI Behavior

↓

Understanding • Clarity • Personalization • Reliability • Engagement & Responsiveness

↓

Factor-Level Evaluation

↓

Overall BOND Measurement

CORE DEFINITION

BOND is a multidimensional AI-side evaluation framework designed to operationalize and measure how effectively an AI system builds and maintains connection with a human through its behavior during interaction.

BOND therefore transforms the broad idea of *AI connection* into a framework that can subsequently be **defined through observable criteria, mathematically represented, experimentally tested, compared across systems, and empirically validated.**

3.2 Philosophy of BOND

The philosophy of BOND is based on a simple distinction:

AI capability determines what a system can do; connection-building capability influences how effectively those capabilities are translated into interaction with a human.

As AI systems become more capable, the interaction problem does not automatically disappear. A system may possess sufficient knowledge, reasoning ability, or computational capability while still failing to **understand the user's actual requirement, communicate appropriately, adapt to relevant context, remain reliable, or respond effectively as the interaction develops.**

BOND therefore treats interaction capability as a **complementary dimension of AI performance**, rather than as a substitute for intelligence, accuracy, reasoning, or other established capabilities.

Connection as Functional Interaction

Within BOND, the term **connection** does not primarily refer to emotional attachment, friendship, or a human-like relationship.

Instead, connection is treated as a **functional property of interaction.**

An effective connection exists when the AI can progressively align its behavior with the human's relevant requirements and interaction context while maintaining clear, reliable, and responsive communication.

This can be represented conceptually as:

Human Requirement

↓

AI Understanding

↓

Appropriate Communication and Adaptation

↓

Human Feedback / New Context

↓

Continued AI Response

↓

Effective Interaction

Connection is therefore not defined by whether the AI appears human.

BOND evaluates effective interaction, not imitation of human behavior.

Capability Must Reach the User

An AI system's internal capability has practical value only when it can be appropriately applied to the task or interaction in which it is being used.

For example, an AI may possess advanced knowledge of physics. However, when assisting a beginner, simply generating an advanced technical explanation may fail to satisfy the learner's requirement.

The underlying knowledge may be correct while its **application within the interaction is ineffective**.

BOND therefore separates:

Possessing Capability → What the AI is capable of doing.

from

Applying Capability in Interaction → How effectively the AI uses that capability for the human and context involved.

This distinction forms one of the central philosophical foundations of the framework.

The Human Provides Context, Not the Score

Human characteristics inevitably influence interaction. Different users may have different knowledge, goals, preferences, communication styles, and requirements.

However, BOND does not attempt to assign a score to the human.

The human instead provides the **environment in which AI behavior becomes observable**.

Human → Interaction Context

AI → Observed Behavior

BOND → Evaluation of AI Behavior

This allows BOND to focus on the question:

Given a particular interaction context, how effectively did the AI respond?

Connection Is Multidimensional

No single behavior completely represents effective AI connection-building.

An AI may understand the user accurately but explain poorly. Another may communicate clearly but fail to adapt to relevant context. A third may perform well initially but become inconsistent across later interactions.

BOND therefore treats connection-building as the combined behavior of multiple dimensions:

Understanding + Clarity + Personalization + Reliability + Engagement & Responsiveness

The framework preserves these dimensions separately because **different combinations can produce different interaction strengths and weaknesses even when an overall score is similar**.

Core Philosophical Position

BOND views effective Human–AI connection as an AI-side, multidimensional, functional interaction capability—not as human-like emotion, not as general intelligence, and not as a replacement for existing measures of AI performance.

Its purpose is to examine whether this capability can be **observed, decomposed, measured, compared, and improved systematically**.

This philosophy establishes the conceptual boundaries required to define the scope of BOND.

3.3 Scope and Boundaries of BOND

BOND is designed specifically for **AI systems that interact directly with humans**. Its scope includes systems in which AI behavior can be observed through communication, adaptation, responses, or continued interaction.

The framework focuses on **how the AI behaves within an interaction**, rather than attempting to evaluate every capability of the underlying AI system.

Systems Within the Scope of BOND

BOND can potentially be applied to interactive systems such as:

System	Relevant Interaction
Conversational AI	Multi-turn text or multimodal conversation
AI Tutors	Explanation, adaptation, and learner support
Personal Assistants	Contextual and personalized assistance
Customer-Support AI	Problem understanding and continued resolution
Voice Assistants	Spoken human–AI interaction
Research Assistants	Context-aware research support
Interactive Robots	Communication combined with physical actions
AI Agents	Continued interaction while performing multi-step tasks

The specific task may differ across these systems, but the evaluation target remains:

The AI's demonstrated connection-building behavior within the interaction.

What BOND Evaluates

BOND evaluates observable AI behavior through five dimensions:

Understanding → Did the AI correctly interpret the human requirement and relevant context?

Clarity → Did the AI communicate its response effectively?

Personalization → Did the AI appropriately adapt when relevant user- or situation-specific information was available?

Reliability → Did the AI behave consistently and dependably?

Engagement & Responsiveness → Did the AI respond appropriately as the interaction developed?

These dimensions establish the primary measurement boundary of BOND.

What BOND Does Not Evaluate

To prevent the framework from becoming excessively broad, several properties remain outside its direct scope.

General Intelligence

BOND does not determine whether an AI possesses AGI or general intelligence.

Complete Technical Capability

It does not replace benchmarks for reasoning, mathematics, coding, knowledge, vision, or other specialized abilities.

Human Psychology

BOND does not attempt to diagnose or quantitatively model the user's emotions, personality, or psychological state.

Mutual Human–AI Relationship

It does not measure the complete relationship between a specific human and an AI because such a construct would contain variables from both participants.

Human-Likeness

An AI does not need to imitate human personality, emotion, or behavior to achieve effective interaction.

Safety as a Complete Domain

Safety is essential to AI evaluation, but BOND is not intended to replace dedicated safety, alignment, fairness, privacy, or security evaluations.

Interaction Boundary

BOND evaluates only behavior that can be meaningfully observed within the available interaction.

Therefore:

The framework should not infer an AI capability that the interaction did not provide an opportunity to demonstrate.

For example, if an interaction contains no relevant information that could support personalization, the absence of personalization should not automatically be interpreted as poor personalization capability.

This principle is important because **evaluation must distinguish between failure to demonstrate a capability and absence of an opportunity to demonstrate it.**

Context Boundary

BOND permits the use of relevant interaction context, but more personalization is not automatically considered better.

An AI should use context only when it is **relevant and appropriate to the current requirement.**

Unnecessary use of personal information, forced personalization, or irrelevant references to previous interaction should not increase BOND performance merely because the response appears more personalized.

Effective personalization means appropriate adaptation—not maximum personalization.

Application Boundary

BOND is intended primarily as an **evaluation framework**, not as a complete architecture for building intelligent systems.

The framework may later guide the design or optimization of interaction-oriented AI systems, but its primary function is to:

Observe → Measure → Diagnose → Compare

AI connection-building behavior.

SCOPE STATEMENT

BOND applies to observable AI-side behavior during human–AI interaction and evaluates that behavior through Understanding, Clarity, Personalization, Reliability, and Engagement & Responsiveness, while remaining separate from complete intelligence, human psychology, human-likeness, and broader AI evaluation domains.

These boundaries keep BOND sufficiently focused to support **operational definition, measurement, experimentation, and validation.**

3.4 Objectives of BOND

The primary objective of BOND is to transform **AI connection-building capability** from a conceptual idea into a structured framework that can be **defined, measured, analyzed, compared, and experimentally evaluated.**

BOND is not intended to determine whether an AI is universally “good” or “bad.” Instead, it focuses on a specific question:

How effectively does an AI build and maintain connection with a human through its behavior during interaction?

1. Define Connection-Building Capability

The first objective is to establish a clear operational meaning for **AI-side connection-building capability.**

Rather than treating connection as an undefined or purely subjective concept, BOND defines it through observable interaction behavior.

This creates the foundation for converting:

Abstract Concept → Defined Construct → Measurable Behavior

2. Decompose the Construct

Connection-building is unlikely to be represented adequately by one behavior alone.

BOND therefore decomposes the construct into five dimensions:

Understanding (U)

Clarity (C)

Personalization (P)

Reliability (R)

Engagement & Responsiveness (ER)

The objective is to evaluate these dimensions individually while also examining their combined contribution to overall connection-building capability.

3. Create a Measurable Framework

BOND aims to convert each dimension into **observable and scorable criteria**.

The intended progression is:

Dimension → Sub-components → Observable Indicators → Score → Overall Measurement

This allows the framework to move beyond conceptual description toward systematic evaluation.

4. Enable Diagnostic Evaluation

An overall score alone may indicate that one AI performs better than another without explaining the reason.

BOND therefore aims to provide both:

Overall Measurement → General connection-building performance.

Factor-Level Measurement → Specific strengths and weaknesses within the interaction.

For example, an AI may demonstrate:

High Understanding + High Clarity + Low Personalization

Such a profile provides more diagnostic information than the overall score alone.

5. Support Comparison

A structured measurement framework can potentially support comparison between:

Different AI systems

Different versions of the same system

Different interaction strategies

Different application environments

Baseline and interaction-oriented implementations

The purpose of comparison is not merely to rank systems, but to identify **how and where their interaction behavior differs**.

6. Support Experimental Testing

BOND is intended to be testable through controlled Human–AI interactions.

The framework should therefore support experiments in which relevant interaction conditions are defined and AI behavior is systematically observed and scored.

A framework becomes scientifically useful only when its claims can be tested against observable evidence.

7. Support Improvement of Interactive AI

Although BOND is primarily an evaluation framework, diagnostic results may also identify areas where an interactive AI system can be improved.

For example:

Low Understanding → improve requirement/context interpretation

Low Clarity → improve communication strategy

Low Personalization → improve appropriate contextual adaptation

Low Reliability → improve consistency and dependability

Low ER → improve continuity and response to changing interaction

Evaluation can therefore provide a feedback loop:

Measure → Identify Weakness → Improve → Re-evaluate

PRIMARY OBJECTIVE

The primary objective of BOND is to develop a reproducible multidimensional framework for evaluating AI-side connection-building capability while preserving sufficient factor-level information to explain how that capability is demonstrated during Human–AI Interaction.

These objectives determine the principles that the BOND framework must follow during its mathematical formulation, experimental implementation, and validation.

3.5 Core Principles of BOND

The BOND framework is governed by a set of core principles that define **how connection-building capability should be interpreted, measured, and evaluated.**

These principles maintain consistency between the conceptual definition of BOND and its later mathematical and experimental implementation.

1. AI-Side Evaluation

BOND evaluates the **AI system**, not the complete human–AI relationship.

The human provides the interaction context in which AI behavior becomes observable, while the AI remains the object of measurement.

Human → Interaction Context

AI → Evaluation Target

PRINCIPLE 1

A BOND score should represent demonstrated AI behavior, not characteristics of the human participant.

2. Observable Behavior

BOND should evaluate behavior that can be **observed during interaction** rather than making unsupported assumptions about the AI's internal state.

For example, BOND may evaluate whether an AI successfully interprets a requirement, but it should not assume that the system possesses human-like understanding or consciousness.

The distinction is:

Internal Claim → What the AI supposedly experiences or possesses

Behavioral Evidence → What the AI demonstrably does during interaction

PRINCIPLE 2

BOND measures demonstrated interaction behavior rather than inferred human-like mental states.

3. Multidimensional Evaluation

Connection-building capability cannot be adequately represented by one interaction property alone.

BOND therefore preserves five dimensions:

Understanding (U)

Clarity (C)

Personalization (P)

Reliability (R)

Engagement & Responsiveness (ER)

Each dimension provides distinct diagnostic information while contributing to the broader construct.

PRINCIPLE 3

Overall BOND performance should preserve the information provided by its individual dimensions rather than allowing the aggregate score to completely hide factor-level behavior.

4. Opportunity to Demonstrate

An AI should only be evaluated on a behavior when the interaction provides a meaningful **opportunity to demonstrate that behavior**.

For example, personalization cannot be appropriately evaluated when no relevant user-specific or contextual information is available.

Therefore:

No Opportunity $\neq$ Failure

This distinction prevents the framework from penalizing an AI for behavior that the evaluation environment never required or enabled. This follows the interaction boundary already established within the scope of BOND.

PRINCIPLE 4

Absence of observable behavior should be distinguished from failure when the required behavior had no meaningful opportunity to occur.

5. Appropriate Adaptation

BOND does not assume that **more adaptation or personalization is always better**.

The AI should adapt only when available context is relevant to the user's current requirement.

For example:

Relevant personalization $\rightarrow$ potentially improves interaction

Forced personalization $\rightarrow$ does not necessarily improve interaction

Irrelevant use of personal context $\rightarrow$ may reduce interaction quality

This preserves the existing BOND boundary that personalization means **appropriate adaptation rather than maximum personalization**.

PRINCIPLE 5

BOND rewards relevance and appropriateness of adaptation, not the quantity of personalization.

6. Interaction-Dependent Measurement

BOND performance is demonstrated within an **interaction environment**.

A score should therefore be interpreted in relation to the conditions under which it was obtained, including the task, available context, interaction length, and evaluation procedure.

This means that comparisons should use sufficiently comparable evaluation conditions whenever possible.

PRINCIPLE 6

BOND scores should be interpreted together with the interaction conditions under which they were measured.

7. Reproducibility

A scientific evaluation framework should allow another evaluator to understand **how a score was produced**.

BOND should therefore define:

Evaluation Conditions → Observable Criteria → Scoring Rules → Factor Scores → Overall Score

The scoring process should minimize unnecessary ambiguity and make the path from observed behavior to final measurement transparent.

PRINCIPLE 7

BOND should be sufficiently defined and documented to support reproducible evaluation.

8. Diagnostic Value

BOND should provide more than a final numerical value.

The framework should help identify **where interaction succeeds or fails**.

For example:

High U + High C + Low P + High R + Moderate ER

provides substantially more information than an overall score alone.

PRINCIPLE 8

The framework should preserve factor-level results so that BOND functions as both a measurement and a diagnostic tool.

9. Complementarity

BOND does not replace established evaluations of **accuracy, reasoning, safety, robustness, knowledge, human preference, or other AI capabilities**.

Instead, it introduces an additional evaluation perspective focused specifically on **AI-side connection-building behavior**.

A comprehensive AI evaluation may therefore combine:

Capability Evaluation + Safety Evaluation + Interaction Evaluation + Other Relevant Evaluation

PRINCIPLE 9

BOND should complement existing AI evaluation rather than claim to represent complete AI quality.

10. Empirical Testability

The framework should remain open to **experimental testing, criticism, revision, and validation**.

Its dimensions, scoring structure, weights, and assumptions should not be treated as correct merely because they are theoretically proposed.

They must ultimately be supported—or challenged—by empirical evidence.

PRINCIPLE 10

BOND is a testable framework: its structure and measurements must remain subject to empirical validation and revision.

Core Principle Summary

The BOND framework therefore follows ten foundational principles:

AI-Side Evaluation → Observable Behavior → Multidimensionality → Opportunity to Demonstrate → Appropriate Adaptation → Interaction-Dependent Measurement → Reproducibility → Diagnostic Value → Complementarity → Empirical Testability

Together, these principles establish the rules under which BOND can progress from a conceptual framework toward a **formal measurement system**.

FRAMEWORK PRINCIPLE

BOND should measure only what the AI meaningfully demonstrates, preserve the multidimensional nature of interaction, remain interpretable within its evaluation context, and produce results that can be systematically examined and empirically tested.

These principles provide the foundation for defining the individual BOND factors and their measurable components in the following chapter.

CHAPTER 4 — HUMAN-CENTERED AI INTERACTION

4.1 Human–AI Connection

Human–AI interaction is not limited to the transmission of information between a human and an artificial intelligence system. In interactive AI systems, communication often develops through a sequence in which **requirements are expressed, interpreted, answered, clarified, adapted, and continued over time**.

Within BOND, **Human–AI connection** refers to the functional alignment that develops when an AI system responds effectively to the human's requirements and relevant interaction context across this process.

The concept can be represented as:

Human Requirement

↓

AI Interpretation

↓

AI Response

↓

Human Reaction or Feedback

↓

AI Adaptation

↓

Continued Interaction

Connection therefore develops through the **quality and continuity of interaction**, rather than through a single isolated response.

Connection Is Not Human-Likeness

Human–AI connection should not be interpreted as requiring an AI to behave exactly like a human.

An AI may interact effectively without possessing human emotions, consciousness, personality, or subjective experience.

BOND therefore separates:

Human-Like Appearance of Interaction

from

Functional Effectiveness of Interaction

An AI does not need to simulate friendship, emotion, or personality to demonstrate strong connection-building behavior.

Human–AI connection within BOND is functional and interaction-centered rather than dependent on human imitation.

Connection Is Not the Same as Correctness

Correctness remains essential in many AI applications, but a technically correct response does not necessarily produce an effective interaction.

Consider a user who asks for an explanation of a difficult scientific concept.

An AI may provide scientifically correct information while:

**misinterpreting the actual question,
using an inappropriate level of complexity,
ignoring relevant context,
failing to respond to clarification, or
behaving inconsistently across later turns.**

The information may therefore be correct while the interaction remains ineffective.

Similarly, an interaction that feels smooth cannot compensate for fundamentally incorrect information.

This creates an important distinction:

Task Quality → Is the AI's output correct and capable?

Interaction Quality → How effectively is that capability applied within the human interaction?

BOND focuses primarily on the second question while remaining complementary to evaluation of the first.

Connection as Alignment

A useful Human–AI connection requires the AI's behavior to become appropriately aligned with the **current human requirement**.

This alignment may involve several forms:

Semantic Alignment → Understanding what the human actually means.

Communication Alignment → Presenting information at an appropriate level and structure.

Contextual Alignment → Using relevant information from the interaction.

Behavioral Alignment → Remaining consistent and dependable.

Dynamic Alignment → Responding appropriately when the interaction changes.

These forms of alignment correspond closely to the five BOND dimensions without reducing connection to any single dimension.

Connection Can Develop Across Turns

In a multi-turn interaction, later responses can depend on information established earlier.

For example:

Turn 1 → Human establishes a requirement

Turn 2 → AI provides an initial response

Turn 3 → Human corrects or clarifies the requirement

Turn 4 → AI adapts

Turn 5 → Human introduces additional context

Turn 6 → AI integrates the new context

The quality of connection therefore becomes increasingly observable as the AI receives opportunities to **understand, adapt, maintain continuity, and respond to change**.

This is particularly important for BOND because some dimensions cannot always be meaningfully evaluated from a single response.

Connection Is Context-Dependent

The same AI behavior may not be equally appropriate in every interaction.

A detailed technical explanation may be useful for an expert but unsuitable for a beginner. A personalized response may improve one interaction but become unnecessary in another where personal context is irrelevant.

Therefore, effective connection cannot be defined as one fixed communication style.

Effective connection requires behavior that is appropriate to the requirement and context of the interaction in which it occurs.

This also means that BOND evaluation must consider the conditions under which behavior is observed rather than treating every interaction as identical.

Connection as an Emergent Interaction Property

Human–AI connection is not produced by one isolated behavior.

Instead, it emerges from the combined operation of multiple interaction capabilities:

Understanding provides accurate interpretation.

Clarity enables effective communication.

Personalization supports appropriate adaptation.

Reliability maintains dependable behavior.

Engagement & Responsiveness supports continuity as interaction develops.

A weakness in one dimension may therefore influence the effectiveness of the broader interaction even when other dimensions remain strong.

BOND Perspective on Human–AI Connection

Within this framework:

Human → provides requirements, feedback, and interaction context

AI → interprets, responds, adapts, and maintains behavior

Interaction → provides observable evidence

BOND → evaluates the AI-side behaviors demonstrated through that evidence

CORE CONCEPT

Human–AI connection in BOND is the functional alignment produced through effective, context-appropriate, and continuing interaction between a human and an AI system. BOND does not attempt to score the complete relationship; it evaluates the AI capabilities that contribute to building and maintaining that connection.

This distinction provides the foundation for defining **AI Connection-Building Capability** as a measurable evaluation target.

4.2 AI Connection-Building Capability

AI Connection-Building Capability refers to an AI system's ability to demonstrate behaviors that **create, support, and maintain effective functional alignment with a human during interaction.**

It represents the central construct evaluated by BOND.

The capability is not defined by whether the AI appears friendly, emotional, or human-like. Instead, it concerns whether the AI can **translate its underlying capabilities into interaction behavior appropriate to the human requirement and context.**

AI Connection-Building Capability is the demonstrated ability of an AI system to understand, communicate, adapt, remain dependable, and respond effectively as Human–AI Interaction develops.

From General Capability to Interaction Capability

AI systems may possess many underlying capabilities:

Knowledge

Reasoning

Language Generation

Planning

Problem-Solving

Perception

These capabilities determine what the system may be able to accomplish.

However, during Human–AI Interaction, another question becomes important:

Can the system apply these capabilities appropriately for the human and interaction in front of it?

For example, possessing advanced reasoning capability does not automatically guarantee that an AI will identify what the user actually needs, explain its reasoning at an appropriate level, incorporate relevant context, or adapt when the requirement changes.

BOND therefore distinguishes between:

Underlying AI Capability → What the system can potentially do

and

Connection-Building Capability → How effectively the system translates relevant capabilities into interaction

The two can influence each other, but they should not be treated as identical.

Five Functional Components

Within BOND, AI Connection-Building Capability is represented through five dimensions:

Understanding (U)

The capability to interpret the user's requirement, intended meaning, and relevant context.

Clarity (C)

The capability to communicate information in an understandable, structured, and usable manner.

Personalization (P)

The capability to appropriately adapt interaction using relevant user-specific or situational information.

Reliability (R)

The capability to maintain dependable and sufficiently consistent behavior throughout interaction.

Engagement & Responsiveness (ER)

The capability to respond appropriately to the evolving interaction and maintain useful continuity.

These dimensions provide different observational perspectives on the broader construct.

Connection-Building Is Dynamic

Connection-building capability should not be treated as a static property visible only through one response.

Human requirements can change during interaction.

A user may:

**clarify a previous statement,
correct the AI,
introduce new information,
change the desired level of explanation,
modify the objective, or
continue the task across multiple turns.**

The AI must therefore continually process new interaction information and determine whether its behavior should remain stable or change.

A simplified process is:

Interpret → Respond → Receive Feedback → Re-evaluate → Adapt → Continue

Connection-building capability is consequently demonstrated not only through the quality of the initial response, but also through the AI's behavior as the interaction evolves.

Adaptation Without Instability

Effective adaptation does not mean that an AI should continuously change its behavior.

Some elements should adapt when relevant:

Explanation depth

Communication structure

Use of available context

Task strategy

Response direction

Other elements should remain appropriately stable:

Factual standards

Established requirements

Relevant constraints

Previously confirmed information

Dependable behavior

Connection-building therefore requires a balance between:

Adaptability ↔ Stability

Too little adaptation can make interaction rigid and context-insensitive.

Unnecessary adaptation can make behavior inconsistent or unpredictable.

Effective connection-building requires the AI to change when the interaction requires change and remain stable when change is unnecessary.

Connection-Building Is Opportunity-Dependent

Not every interaction provides evidence for every BOND dimension.

A simple factual question may provide a strong opportunity to evaluate clarity but little opportunity to evaluate personalization or long-term reliability.

A multi-turn task may provide substantially more evidence for understanding, contextual adaptation, reliability, and responsiveness.

Therefore, the observed connection-building capability depends partly on the **evaluation opportunities created by the interaction.**

This leads to an important distinction:

Capability Not Demonstrated Because It Failed

≠

Capability Not Demonstrated Because It Was Not Required

BOND must preserve this distinction when constructing its later scoring methodology.

Connection-Building Is Not Maximum Agreement

An AI does not demonstrate stronger connection simply by agreeing with the human.

Effective interaction may sometimes require the AI to:

identify an incorrect assumption,

request clarification,

communicate uncertainty,

**refuse an inappropriate request, or
provide information that differs from the user's expectation.**

Therefore:

**Connection-building means responding appropriately to the human—not
automatically agreeing with the human.**

This distinction prevents BOND from confusing **compliance** with effective interaction.

Connection-Building as a Measurable Capability

For AI Connection-Building Capability to function as a scientific construct, it must ultimately produce observable evidence.

The proposed measurement pathway is:

Interaction Conditions

↓

AI Behavior

↓

Observable Indicators

↓

Dimension Scores

↓

Combined BOND Measurement

This structure allows BOND to evaluate both the overall construct and the specific behaviors contributing to it.

Capability Profile

Two AI systems may achieve similar overall connection-building performance while exhibiting different internal profiles.

For example:

System A → Strong Understanding + Moderate Clarity + Strong Reliability

System B → Moderate Understanding + Strong Clarity + Strong Personalization

A single overall score may hide these differences.

BOND therefore treats the **factor profile** as an important part of evaluation rather than considering only the combined value.

CORE DEFINITION

AI Connection-Building Capability is the observable, multidimensional capability of an AI system to translate its underlying abilities into context-appropriate interaction through understanding, clarity, personalization, reliability, and engagement & responsiveness as Human–AI Interaction develops.

This capability does not replace intelligence, reasoning, accuracy, or safety. It defines a specific interaction-oriented evaluation target within the broader assessment of AI systems.

The next section examines how this capability can be observed across the **lifecycle of a Human–AI interaction**.

4.3 Interaction Lifecycle

Human–AI interaction is a **dynamic process** rather than a single exchange. During an interaction, the human provides requirements and context, the AI interprets that information, generates a response, receives new information or feedback, and may subsequently adapt its behavior.

BOND represents this process through an **Interaction Lifecycle**.

The lifecycle provides a structured view of **where and how AI connection-building behavior becomes observable during interaction**.

The BOND Interaction Lifecycle

The interaction lifecycle can be represented through six stages:

1. Input

↓

2. Interpretation

↓

3. Response Formation

↓

4. Human Feedback / New Context

↓

5. Adaptation

↓

6. Continuity

These stages do not necessarily occur only once. In multi-turn interaction, several stages may repeat as the conversation develops.

Stage 1 — Input

The lifecycle begins when the human provides an input to the AI.

The input may contain:

A question

A task or instruction

A problem

A preference

Relevant personal or situational context

Feedback on a previous response

A continuation of an earlier interaction

The AI's connection-building capability cannot be evaluated independently of this input because the input establishes the **requirement and available context** against which subsequent behavior can be observed.

The human input defines the initial interaction condition; it does not itself determine the BOND score.

Stage 2 — Interpretation

After receiving the input, the AI must determine what the human is asking and which available information is relevant.

Effective interpretation may require identifying:

Explicit Requirement → What was directly requested

Underlying Intent → What outcome the human appears to require

Relevant Context → Which available information affects the response

Constraints → What conditions or limitations must be respected

Ambiguity → Whether clarification is required before proceeding

This stage is strongly associated with **Understanding (U)**.

A failure at the interpretation stage can influence every later stage because an AI may generate a technically strong response to the **wrongly interpreted requirement**.

Therefore:

Poor Interpretation → Misaligned Response

even when the underlying information or reasoning is otherwise correct.

Stage 3 — Response Formation

Once the requirement has been interpreted, the AI translates its capabilities into a response.

The response must not only contain relevant information; it must also communicate that information in a form appropriate to the interaction.

Important behaviors may include:

Selecting relevant information

Choosing an appropriate level of detail

Structuring the response clearly

Following relevant instructions

Using available context appropriately

Communicating uncertainty when necessary

This stage provides particularly important evidence for **Clarity (C)** while potentially involving Understanding, Personalization, and Reliability as well.

Connection-building requires not only producing an answer, but producing an answer that can be effectively used within the interaction.

Stage 4 — Human Feedback or New Context

After receiving the AI response, the human may provide additional information.

This may include:

Confirmation → The response satisfied the requirement.

Correction → The AI misunderstood something.

Clarification → The human explains the requirement more precisely.

Preference → The human specifies how the interaction should proceed.

New Context → Additional relevant information becomes available.

Changed Requirement → The objective itself changes.

Human feedback therefore acts as an important **interaction signal**.

It creates new opportunities to observe whether the AI can recognize changes in the interaction and respond appropriately.

Stage 5 — Adaptation

The AI must determine whether the new information requires a change in its behavior.

Appropriate adaptation may involve:

Correcting a misunderstanding

Changing explanation depth

Using newly relevant context

Updating the task strategy

Respecting a newly stated preference

Revising an earlier conclusion

Requesting clarification when necessary

This stage provides important evidence for **Personalization (P)**, **Understanding (U)**, and **Engagement & Responsiveness (ER)**.

However, adaptation should remain selective.

The AI should not change established information merely because new interaction occurs.

Therefore:

Relevant Change → Adapt

No Relevant Change → Maintain

Effective adaptation requires distinguishing between the two.

Stage 6 — Continuity

As interaction continues, the AI must maintain relevant information and behavioral consistency across turns.

Continuity may involve:

Remembering established requirements within the available context

Avoiding unnecessary repetition

Maintaining relevant constraints

Remaining consistent with previously established information

Recognizing progression in the interaction

Responding coherently to later turns

This stage provides particularly important evidence for **Reliability (R)** and **Engagement & Responsiveness (ER)**.

Continuity allows Human–AI interaction to develop beyond a sequence of disconnected responses.

An effective multi-turn interaction should behave as a developing process rather than repeatedly restarting from zero.

The Lifecycle as a Feedback Loop

Human–AI interaction rarely ends after one complete sequence.

Instead, the lifecycle can become recursive:

Input → Interpretation → Response → Feedback → Adaptation → Continuity

↓

New Input → Re-interpretation → New Response → Further Feedback → Further Adaptation

Each cycle creates additional behavioral evidence.

As interaction becomes longer or more complex, the AI receives more opportunities to demonstrate—or fail to demonstrate—different connection-building capabilities.

This makes the lifecycle particularly relevant to BOND's **Opportunity to Demonstrate** principle.

BOND Dimensions Across the Lifecycle

The five BOND dimensions do not operate in complete isolation at individual lifecycle stages.

Lifecycle Stage	Primary BOND Relevance
Input	Establishes evaluation context
Interpretation	Understanding
Response Formation	Clarity, Understanding
Feedback / New Context	Creates new evaluation opportunities
Adaptation	Personalization, Understanding, Engagement & Responsiveness
Continuity	Reliability, Engagement & Responsiveness

The mapping is not exclusive.

For example, Reliability may be observed during response formation, while Clarity may remain important during adaptation. The lifecycle therefore describes **where evidence commonly emerges**, rather than assigning each dimension permanently to one stage.

Lifecycle Failure Propagation

One important property of the interaction lifecycle is that an early failure can influence later behavior.

For example:

Incorrect Interpretation
↓
Misaligned Response
↓
Human Correction
↓
Failed Adaptation
↓
Reduced Continuity

Similarly, an AI may initially misunderstand the requirement but recover effectively after human correction.

Therefore, BOND should not evaluate interaction only through isolated moments.

The lifecycle provides information about both:

Initial Performance → **How effectively did the AI respond initially?**

and

Recovery Behavior → How effectively did the AI respond when the interaction exposed a problem?

Recovery can therefore provide meaningful evidence about connection-building capability.

Single-Turn and Multi-Turn Interaction

The BOND lifecycle can operate at different interaction lengths.

Single-Turn Interaction

Input → Interpretation → Response

This can provide useful evidence for dimensions such as Understanding and Clarity.

Multi-Turn Interaction

Input → Response → Feedback → Adaptation → Continuity → Repeated Cycles

This provides richer opportunities to evaluate Personalization, Reliability, and Engagement & Responsiveness.

Therefore, different evaluation designs may provide different levels of evidence for the five dimensions.

A BOND evaluation should consider whether the interaction design provides sufficient opportunity to observe the dimensions it intends to measure.

Interaction Lifecycle and Measurement

The lifecycle connects Human–AI interaction directly to the later BOND measurement process:

Interaction Lifecycle

↓

Observable AI Behavior

↓

BOND Dimension Indicators

↓

Dimension Scores

↓

Overall BOND Measurement

The lifecycle therefore acts as a bridge between the **conceptual meaning of connection-building capability** and its **experimental measurement**.

CHAPTER CONCLUSION

Human-centered AI interaction is a continuing process in which effective behavior depends on more than producing an isolated answer.

BOND interprets Human–AI connection as **functional alignment developed through interaction** and defines AI Connection-Building Capability as the AI-side ability to support that alignment through **Understanding, Clarity, Personalization, Reliability, and Engagement & Responsiveness**.

The Interaction Lifecycle explains how these capabilities become observable as the AI **interprets, responds, receives feedback, adapts, and maintains continuity**.

CORE PRINCIPLE

Connection is not a single response. It is a developing interaction process in which AI behavior can be observed across changing requirements, feedback, adaptation, and continuity.

This lifecycle provides the conceptual foundation for the detailed definition and measurement of the individual BOND dimensions.

CHAPTER 5 — FIVE CORE FACTORS

5.1 Understanding (U)

Understanding is the first core factor of the BOND framework.

In Human–AI Interaction, an effective response begins with the AI correctly identifying **what the human is asking, what the requirement means within the available context, and which information is relevant to satisfying that requirement**.

Within BOND, Understanding does not imply human-like consciousness, subjective comprehension, or internal mental experience.

It refers specifically to **observable evidence that the AI has interpreted the human requirement appropriately**.

DEFINITION

Understanding (U) is the demonstrated capability of an AI system to correctly interpret a human's explicit requirement, intended meaning, relevant context, constraints, and interaction changes sufficiently to produce an appropriately aligned response.

5.1.1 Meaning of Understanding

Human communication is rarely composed only of perfectly specified instructions.

A user may communicate through:

Direct requests

Incomplete statements

Context-dependent questions

Follow-up messages

Corrections

Implicit requirements

Multiple constraints

Changes in direction

The AI must determine which elements are relevant to the current interaction.

Consider:

User: “Explain quantum entanglement.”

The explicit requirement is relatively clear.

Now consider:

User: “I still don't get it. Explain it like I'm learning it for the first time.”

The second message cannot be interpreted independently. The AI must recognize that:

“it” → refers to quantum entanglement

“still don't get it” → indicates the previous explanation was ineffective

“learning it for the first time” → requests a simpler explanatory level

Understanding therefore involves more than processing the literal words of the latest input.

It requires appropriate interpretation of the **meaning created by the interaction context**.

5.1.2 Purpose of Understanding

Understanding acts as the **alignment entry point** of Human–AI Interaction.

Before the AI can communicate clearly, personalize appropriately, or respond effectively, it must first determine what response is actually required.

A simplified dependency can be represented as:

Human Requirement

↓

Understanding

↓

Response Selection

↓

Interaction Outcome

An error in Understanding can propagate through later behavior.

For example:

Incorrect Interpretation

↓

Correct Information for the Wrong Problem

↓

Misaligned Response

The response may therefore contain accurate information while still failing the interaction.

The purpose of Understanding within BOND is consequently to evaluate whether the AI's response demonstrates **sufficient alignment with the requirement that was actually presented**.

Understanding asks:

“Did the AI correctly determine what needed to be understood before deciding how to respond?”

5.1.3 Understanding Is Not Correctness

Understanding and factual correctness are related but distinct.

An AI may understand a question correctly and still produce an incorrect answer because of a reasoning or knowledge failure.

Likewise, an AI may produce factually correct information while misunderstanding the user's actual request.

For example:

User: “I know what photosynthesis is. I only want to understand why light intensity affects its rate.”

If the AI responds with a general definition of photosynthesis, the information may be scientifically correct.

However, the response demonstrates weak Understanding because the user explicitly requested the **relationship between light intensity and reaction rate**, not a general definition.

Therefore:

Correctness → Is the information accurate?

Understanding → Is the response based on an appropriate interpretation of what the human required?

BOND evaluates the second property through the Understanding factor.

5.1.4 Sub-Factors of Understanding

Understanding can be decomposed into five proposed sub-factors:

- 1. Requirement Recognition (UR)**
- 2. Intent Interpretation (UI)**
- 3. Context Integration (UC)**
- 4. Constraint Recognition (UCR)**
- 5. Clarification & Update Handling (UU)**

Together, these sub-factors represent different observable components of AI understanding.

1. Requirement Recognition (UR)

Requirement Recognition evaluates whether the AI identifies **what the human explicitly requests**.

This includes recognizing:

The task

The requested output

The relevant subject

The requested level of detail

Any directly stated conditions

Example:

User: “Give me three advantages of solar energy in one sentence each.”

The AI should recognize:

Topic → **Solar energy**

Quantity → **Three advantages**

Format → **One sentence each**

Ignoring any of these explicit requirements represents a failure of requirement recognition.

2. Intent Interpretation (UI)

Intent Interpretation evaluates whether the AI recognizes the **practical objective or intended meaning** behind the request when that meaning is reasonably supported by the interaction.

For example:

User: “This explanation is too technical.”

The literal statement describes the previous explanation.

Its interaction meaning is also a request for the AI to **reduce complexity**.

A strong system should recognize this without requiring the human to explicitly say:

“Rewrite your previous answer using simpler language.”

However, intent interpretation must remain evidence-based.

The AI should infer what the interaction reasonably supports—not invent intentions that the human did not communicate.

3. Context Integration (UC)

Context Integration evaluates whether the AI correctly uses **relevant information established within the available interaction context**.

Consider:

Turn 1: “I'm preparing a presentation about renewable energy for beginners.”

Turn 4: “Explain photovoltaic cells for my presentation.”

The earlier information remains relevant.

A context-aware response should recognize that the explanation is intended for **a beginner-level presentation**.

Context integration therefore requires two decisions:

Is previous information relevant?

and

How should that information affect the current response?

Using irrelevant context is not stronger Understanding.

4. Constraint Recognition (UCR)

Human requests frequently contain constraints.

These may involve:

Length

Format

Language

Difficulty level

Required inclusions or exclusions

Task-specific conditions

Example:

User: “Explain Newton's Second Law in fewer than 80 words without using equations.”

A response containing an equation may explain the concept correctly but still demonstrate incomplete Understanding because an explicit constraint was ignored.

Constraint Recognition therefore evaluates whether the AI identifies the **boundaries within which the requirement should be satisfied**.

5. Clarification & Update Handling (UU)

Not every human request contains enough information for a reliable interpretation.

In such cases, effective Understanding may require the AI to **recognize ambiguity rather than confidently assume an unsupported interpretation**.

Example:

User: “Compare them.”

If no clear referents for “them” exist in the available context, asking what should be compared can demonstrate stronger Understanding than inventing two subjects.

This sub-factor also evaluates whether the AI appropriately updates its interpretation after human correction.

Initial Interpretation → Human Correction → Updated Interpretation

A system that continues using its original assumption after explicit correction demonstrates weak update handling.

Strong Understanding includes knowing when sufficient understanding has not yet been achieved.

5.1.5 Examples of Understanding

Example A — Strong Understanding

User:

“I already understand Coulomb's law. Just explain why increasing the distance reduces the force, and keep it intuitive.”

AI Behavior:

The response focuses on the effect of distance, avoids unnecessarily reteaching the entire law, and uses an intuitive explanation.

Observed Evidence:

Requirement recognized ✓

Existing knowledge acknowledged ✓

Requested explanation style recognized ✓

Interpretation: Strong Understanding

Example B — Partial Understanding

User:

“Give me four differences between AI and machine learning in a table.”

AI Behavior:

The AI explains the difference accurately but provides paragraphs instead of four table entries.

Observed Evidence:

Topic understood ✓

Comparison understood ✓

Quantity/format constraint missed ✗

Interpretation: Partial Understanding

Example C — Weak Understanding

User:

“My previous question was about electric flux, not electric field. Explain flux again.”

AI Behavior:

The AI continues explaining electric field.

Observed Evidence:

Correction not incorporated ✗

Current requirement misidentified ✕
Interaction context not updated ✕

Interpretation: Weak Understanding

Example D — Appropriate Clarification

User:

“Which one is better?”

No clearly identifiable alternatives are available in the interaction.

AI Behavior:

The AI asks which options the user wants compared.

Observed Evidence:

Ambiguity recognized ✓

Unsupported assumption avoided ✓

Interpretation: Appropriate Understanding Behavior

5.1.6 Understanding Factor Summary

Sub-Factor	Primary Evaluation Question
Requirement Recognition (UR)	Did the AI identify what was explicitly requested?
Intent Interpretation (UI)	Did the AI appropriately interpret the intended objective supported by the interaction?
Context Integration (UC)	Did the AI correctly use relevant available context?
Constraint Recognition (UCR)	Did the AI identify the conditions and boundaries of the request?
Clarification & Update Handling (UU)	Did the AI recognize ambiguity and appropriately update its interpretation when necessary?

These sub-factors provide the conceptual basis for converting Understanding from a broad description into a **structured and observable BOND dimension**.

5.1.7 Conceptual Diagram

HUMAN INPUT

↓

Explicit Requirement + Intended Meaning + Context + Constraints + Interaction Updates

↓

UNDERSTANDING (U)

↓

Requirement Recognition (UR)

Intent Interpretation (UI)

Context Integration (UC)

Constraint Recognition (UCR)

Clarification & Update Handling (UU)

↓

INTERPRETED REQUIREMENT

↓

RESPONSE FORMATION

UNDERSTANDING PRINCIPLE

Understanding is successful when the AI's observable behavior demonstrates that it has identified the requirement, interpreted the supported intent, integrated relevant context, respected constraints, and appropriately handled ambiguity or changing information.

Understanding therefore establishes the first major connection between **human input and AI behavior** within BOND. Its later mathematical representation should measure these observable components without making claims about human-like internal comprehension.

5.2 Clarity (C)

Clarity is the second core factor of the BOND framework.

An AI may correctly understand a human requirement and possess the information required to answer it, yet the interaction can still fail if that information is communicated in a form that is **confusing, poorly structured, unnecessarily complex, ambiguous, or difficult to use.**

Within BOND, Clarity evaluates the observable quality with which an AI **translates its intended response into understandable and usable communication.**

DEFINITION

Clarity (C) is the demonstrated capability of an AI system to communicate relevant information in a comprehensible, appropriately structured, sufficiently precise, and usable form for the interaction context.

5.2.1 Meaning of Clarity

Clarity does not simply mean producing a short or simple response.

A clear response may be:

Short or detailed

Technical or beginner-friendly

Textual or structured

Direct or explanatory

depending on the requirement.

The central question is whether the response allows the human to **understand and use the information with unnecessary difficulty minimized.**

Therefore:

Simplicity ≠ Clarity

and

Complexity $\neq$ Lack of Clarity

A technically advanced explanation can remain highly clear when technical depth is appropriate for the user and task.

Likewise, a short response can remain unclear if it omits essential information or uses ambiguous language.

5.2.2 Purpose of Clarity

Understanding determines **what the AI should respond to**.

Clarity determines **how effectively that response is communicated**.

The relationship can be represented as:

Correct Interpretation

↓

Relevant Information

↓

Clear Communication

↓

Human Comprehension and Use

The purpose of Clarity within BOND is therefore to evaluate whether the AI successfully converts its information or reasoning into a form that is **accessible and usable within the interaction**.

Clarity asks:

“Did the AI communicate its response in a form that the human could reasonably understand and use?”

5.2.3 Clarity Is Context-Dependent

There is no universal level of explanation that is optimal for every human.

Consider the same question:

“What is quantum entanglement?”

For a beginner, clarity may require intuitive language and minimal mathematical detail.

For a physics researcher, excessive simplification may actually reduce clarity by removing necessary precision.

Therefore, Clarity should be evaluated relative to:

Task complexity

Requested detail

Available user context

Technical level

Communication format

An AI should not receive a higher Clarity evaluation merely because it uses simpler vocabulary. It should receive stronger evaluation when its communication is **appropriate to the interaction**.

5.2.4 Sub-Factors of Clarity

Clarity can be decomposed into five proposed sub-factors:

1. **Comprehensibility (CC)**
2. **Structural Organization (CS)**
3. **Precision (CP)**
4. **Relevance & Focus (CR)**
5. **Explanatory Sufficiency (CE)**

Together, these components evaluate whether the AI's communication can be effectively interpreted and used.

1. **Comprehensibility (CC)**

Comprehensibility evaluates whether the response can be reasonably understood by the intended user within the available interaction context.

It may involve:

Appropriate vocabulary

Readable sentence construction

Clear explanation of unfamiliar concepts

Appropriate technical depth

Avoidance of unnecessary complexity

Comprehensibility does not require eliminating technical terminology when that terminology is useful or necessary.

The objective is **appropriate understandability**, not universal simplification.

2. **Structural Organization (CS)**

Structural Organization evaluates whether information is presented in a logical and navigable sequence.

Depending on the task, useful structure may involve:

Paragraphs

Steps

Headings

Tables

Ordered reasoning

Examples

For example, instructions containing ten actions may be more usable as ordered steps than as one continuous paragraph.

However, unnecessary formatting should not automatically increase Clarity.

Structure is valuable when it reduces cognitive difficulty or improves usability.

3. Precision (CP)

Precision evaluates whether the AI communicates its intended meaning with sufficient specificity and minimal avoidable ambiguity.

A response should distinguish between:

Known information

Approximation

Uncertainty

Conditions

Exceptions

when these distinctions are relevant.

For example:

“This always happens.”

and

“This generally occurs under these conditions.”

may communicate substantially different levels of precision.

Precision therefore supports clarity by ensuring that the human can determine **what the response actually claims**.

4. Relevance & Focus (CR)

A response can be understandable and well written while still being difficult to use because it contains excessive irrelevant information.

Relevance & Focus evaluates whether the AI prioritizes information that contributes directly to the current requirement.

This does not mean every response must be short.

Instead:

Necessary Detail → Retain

Supporting Detail → Include when useful

Irrelevant Detail → Minimize

Clarity improves when information density serves the requirement rather than distracting from it.

5. Explanatory Sufficiency (CE)

Clarity can also fail when a response is **too incomplete**.

Explanatory Sufficiency evaluates whether enough information is provided for the response to fulfill its communicative purpose.

For example:

User: “Why does increasing resistance reduce current at constant voltage?”

Weak Response: “Because of Ohm's law.”

The response may reference the correct principle but does not sufficiently explain the relationship.

Therefore, Clarity requires balance:

Too Little Information → **Incomplete**

Necessary Information → **Sufficient**

Excessive Irrelevant Information → **Unfocused**

5.2.5 Examples of Clarity

Example A — Strong Clarity

User:

“Explain gravity to a 10-year-old in three sentences.”

AI Behavior:

The AI uses simple language, gives an intuitive explanation, avoids unnecessary terminology, and remains within three sentences.

Observed Evidence:

Comprehensible ✓

Appropriate structure ✓

Focused ✓

Sufficient ✓

Interpretation: Strong Clarity

Example B — Technically Correct but Weak Clarity

User:

“Give me a simple explanation of neural networks.”

AI Behavior:

The AI immediately introduces high-dimensional optimization, gradient tensors, activation derivatives, and mathematical notation without explanation.

Observed Evidence:

Potentially relevant information ✓

Requested communication level missed ✕

Comprehensibility reduced ✕

Interpretation: Weak Clarity

Example C — Clear but Insufficient

User:

“Why does the sky appear blue?”

AI Behavior:

“Because sunlight interacts with the atmosphere.”

The statement is understandable and relevant but insufficiently explains the requested mechanism.

Observed Evidence:

Comprehensible ✓

Relevant ✓

Explanatory sufficiency ✗

Interpretation: Partial Clarity

Example D — Clear Technical Communication

User:

“Explain the mathematical relationship between voltage, current, and resistance. Assume I already know basic circuit theory.”

AI Behavior:

The AI uses appropriate equations and terminology, clearly defines relationships, and avoids unnecessary beginner-level explanation.

Observed Evidence:

Technical level appropriate ✓

Precise ✓

Structured ✓

Relevant ✓

Interpretation: Strong Clarity

5.2.6 Clarity Factor Summary

Sub-Factor	Primary Evaluation Question
Comprehensibility (CC)	Can the response be reasonably understood within the intended context?
Structural Organization (CS)	Is the information organized in a form that supports understanding and use?
Precision (CP)	Is the intended meaning sufficiently specific and unambiguous?
Relevance & Focus (CR)	Does the response prioritize information relevant to the requirement?
Explanatory Sufficiency (CE)	Does the response provide enough information to fulfill its communicative purpose?

These components allow Clarity to be evaluated as more than a subjective impression of whether a response “sounds good.”

5.2.7 Conceptual Diagram

INTERPRETED REQUIREMENT

↓

INFORMATION / REASONING TO COMMUNICATE

↓

CLARITY (C)

↓

Comprehensibility (CC)

Structural Organization (CS)

Precision (CP)

Relevance & Focus (CR)

Explanatory Sufficiency (CE)

↓

UNDERSTANDABLE AND USABLE RESPONSE

↓

HUMAN INTERACTION

CLARITY PRINCIPLE

Clarity is successful when an AI communicates the information required by the interaction in a comprehensible, structured, precise, focused, and sufficiently complete form appropriate to the human and task context.

Clarity therefore acts as the **communication bridge between AI understanding and human usability** within the BOND framework.

5.3 Personalization (P)

Personalization is the third core factor of the BOND framework.

Human–AI interaction does not always occur in an isolated or identical context. A human may provide information about their **goal, knowledge level, preferences, previous decisions, constraints, communication requirements, or current situation.**

When such information is relevant, an AI can use it to adapt its behavior to the specific interaction.

Within BOND, Personalization does not mean maximizing the amount of personal information used. It evaluates whether the AI **appropriately adapts its interaction using relevant available context.**

DEFINITION

Personalization (P) is the demonstrated capability of an AI system to appropriately adapt its response, communication, or interaction strategy using relevant and available user-specific or situational information.

5.3.1 Meaning of Personalization

A generic response attempts to satisfy a requirement without adapting to characteristics specific to the current interaction.

A personalized response considers relevant information that changes **what response would be most appropriate for that particular human or situation.**

Consider:

User A: “Explain electric flux. I am learning it for the first time.”

User B: “Explain electric flux. I already understand electric fields and vector calculus.”

Both users ask about the same concept.

However, an identical explanation may not be equally appropriate for both.

Personalization allows the AI to adapt aspects such as:

Explanation depth

Terminology

Examples

Response structure

Task strategy

Relevant recommendations

according to available context.

Therefore:

Personalization is not simply knowing information about the user; it is using relevant information appropriately when that information should affect the interaction.

5.3.2 Purpose of Personalization

The purpose of Personalization is to reduce the gap between a **generically acceptable response** and a response that is **appropriately adapted to the current interaction.**

The process can be represented as:

Human Requirement

+

Relevant Available Context

↓

Appropriate Adaptation

↓

Context-Specific Response

Without personalization, an AI may repeatedly require the human to restate relevant information or may provide responses that are technically acceptable but poorly suited to the particular situation.

Personalization can therefore improve interaction by allowing the AI to answer not only:

“What response could satisfy this request?”

but also:

“Given the relevant information available in this interaction, how should the response be adapted?”

5.3.3 Personalization Is Not Maximum Personalization

More personalization does not automatically mean better interaction.

Suppose a user previously mentioned a favorite sport while later asking an unrelated mathematics question.

Referencing that sport unnecessarily does not demonstrate stronger personalization simply because the AI remembered it.

The relevant distinction is:

Available Information $\neq$ Relevant Information

and

Relevant Information $\neq$ Information That Must Always Be Used

Personalization requires the AI to determine whether available context should meaningfully influence the current response.

Therefore:

Relevant Context + Appropriate Use $\rightarrow$ Strong Personalization

Relevant Context + Ignored When Needed $\rightarrow$ Weak Personalization

Irrelevant Context + Forced Use $\rightarrow$ Inappropriate Personalization

BOND evaluates the appropriateness of personalization, not its quantity.

5.3.4 Sub-Factors of Personalization

Personalization can be decomposed into five proposed sub-factors:

- 1. Relevant Context Recognition (PCR)**
- 2. User-Level Adaptation (PLA)**
- 3. Preference Adaptation (PPA)**
- 4. Goal & Situation Adaptation (PGS)**
- 5. Selective Context Use (PSC)**

Together, these sub-factors evaluate whether personalization occurs **when, where, and to the extent that it is useful.**

1. Relevant Context Recognition (PCR)

Relevant Context Recognition evaluates whether the AI identifies available information that should influence the current response.

Potentially relevant context may include:

Knowledge level

Current objective

Previously established requirements

Communication preferences

Task constraints

Situational information

The presence of information alone is insufficient.

The AI must determine whether the information has **functional relevance** to the current interaction.

For example:

Earlier Context: “I am completely new to Python.”

Later Request: “Explain functions.”

The earlier context is directly relevant because it should influence the complexity and assumptions of the explanation.

2. User-Level Adaptation (PLA)

User-Level Adaptation evaluates whether the AI adjusts the **difficulty, depth, terminology, and explanatory assumptions** according to the relevant knowledge level indicated within the interaction.

For example:

Beginner → **foundational explanation**

Intermediate user → **existing concepts can be assumed**

Advanced user → **greater technical depth may be appropriate**

The purpose is not to permanently classify the human into a fixed category.

Instead, the AI adapts to the **level supported by the current interaction evidence**.

Personalization should respond to demonstrated context rather than unsupported assumptions about the user.

3. Preference Adaptation (PPA)

Users may express preferences about how they want information or assistance presented.

Examples include:

“Keep the answers short.”

“Use examples.”

“Explain step-by-step.”

“Don't use technical terminology unless necessary.”

“Use a table for comparisons.”

When such a preference remains relevant, the AI should appropriately incorporate it into subsequent responses.

Preference Adaptation therefore evaluates whether the AI can translate a stated interaction preference into **observable behavioral change**.

4. Goal & Situation Adaptation (PGS)

The same information may need to be presented differently depending on **why the human needs it**.

For example:

Goal A → Learn a concept for an examination

Goal B → Use the concept in a research project

Goal C → Explain the concept in a presentation

The underlying subject may remain identical while the most useful response changes.

Goal & Situation Adaptation evaluates whether the AI uses relevant information about the human's **current objective or circumstances** to modify its response appropriately.

5. Selective Context Use (PSC)

Personalization also requires the ability **not to use context unnecessarily**.

Selective Context Use evaluates whether the AI avoids forcing irrelevant personal or historical information into the current response.

Consider:

Previous interaction: User discusses a favorite movie.

Current interaction: User asks for help solving a physics equation.

Unless the earlier information has a legitimate role in solving or explaining the problem, using it provides little functional benefit.

Selective Context Use therefore establishes an important boundary:

Strong personalization includes appropriate contextual restraint.

5.3.5 Examples of Personalization

Example A — Strong Personalization

Earlier Context:

“I understand formulas better when you first explain the intuition.”

Later User Request:

“Teach me Coulomb's law.”

AI Behavior:

The AI begins with an intuitive explanation and then introduces the mathematical relationship.

Observed Evidence:

Relevant preference recognized ✓

Communication adapted ✓

Context used appropriately ✓

Interpretation: Strong Personalization

Example B — Generic Response Despite Relevant Context**Earlier Context:**

“I've never programmed before.”

User:

“Explain Python loops.”

AI Behavior:

The AI immediately explains nested iteration, iterators, generators, and implementation details without introducing the basic concept of repetition.

Observed Evidence:

Relevant user level available ✓

User level not incorporated ✕

Interpretation: Weak Personalization

Example C — Inappropriate Personalization**Earlier Context:**

“My favorite sport is football.”

User:

“Summarize this research abstract.”

AI Behavior:

The AI unnecessarily converts the research explanation into repeated football analogies despite no indication that this would improve the task.

Observed Evidence:

Personal information remembered ✓

Functional relevance absent ✕

Context unnecessarily applied ✕

Interpretation: Inappropriate Personalization

Example D — Goal-Based Personalization

User:

“I need to explain solar cells to students who have never studied semiconductor physics.”

AI Behavior:

The AI prioritizes intuitive mechanisms, introduces terminology gradually, and avoids unnecessary semiconductor mathematics.

Observed Evidence:

Audience recognized ✓

Goal recognized ✓

Explanation adapted appropriately ✓

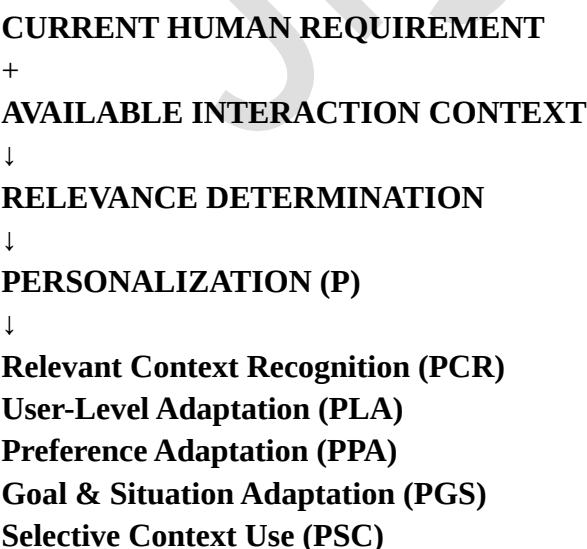
Interpretation: Strong Personalization

5.3.6 Personalization Factor Summary

Sub-Factor	Primary Evaluation Question
Relevant Context Recognition (PCR)	Did the AI identify available context that should influence the response?
User-Level Adaptation (PLA)	Did the AI appropriately adapt to the knowledge level supported by the interaction?
Preference Adaptation (PPA)	Did the AI appropriately incorporate relevant stated interaction preferences?
Goal & Situation Adaptation (PGS)	Did the AI adapt according to the user's relevant objective or situation?
Selective Context Use (PSC)	Did the AI avoid unnecessary or irrelevant use of available context?

These sub-factors allow Personalization to be evaluated without assuming that **greater contextual use always represents stronger performance**.

5.3.7 Conceptual Diagram



↓

APPROPRIATE ADAPTATION

↓

CONTEXT-SPECIFIC RESPONSE

PERSONALIZATION PRINCIPLE

Personalization is successful when an AI identifies relevant available context, uses that context to adapt its behavior where appropriate, and avoids unnecessary adaptation when the context does not meaningfully contribute to the current requirement.

Personalization therefore provides BOND with a mechanism for evaluating whether an AI can move from **generic interaction toward appropriate context-sensitive interaction without treating personalization itself as an unconditional objective.**

5.4 Reliability (R)

Reliability is the fourth core factor of the BOND framework.

Effective Human–AI Interaction requires more than producing one successful response. As an interaction develops, the AI must remain **dependable, sufficiently consistent, appropriately stable, and capable of acknowledging uncertainty or correcting errors when necessary.**

Within BOND, Reliability evaluates whether the human can reasonably depend on the AI's interaction behavior **across the available interaction conditions and turns.**

DEFINITION

Reliability (R) is the demonstrated capability of an AI system to maintain dependable, sufficiently consistent, and appropriately stable behavior while preserving established requirements, communicating uncertainty when relevant, and recovering appropriately from identified errors during interaction.

5.4.1 Meaning of Reliability

Reliability does not mean that an AI must produce identical responses every time.

Human–AI interaction is dynamic. New information may require the AI to modify an earlier response, change its strategy, or revise a conclusion.

Therefore:

Consistency ≠ Repetition

and

Adaptation ≠ Unreliability

Reliable behavior requires the AI to maintain what **should remain stable** while appropriately changing what **should change.**

For example:

User: “Keep all future explanations under 150 words.”

If the requirement remains active and relevant, repeatedly ignoring it indicates weak Reliability.

However, if the user later says:

“For this one, explain it in detail.”

changing the response length is appropriate adaptation rather than inconsistency.

Reliability depends on justified stability—not resistance to change.

5.4.2 Purpose of Reliability

Connection-building becomes difficult when AI behavior is unpredictable across an interaction.

An AI may initially understand the requirement correctly but later:

**forget an established constraint,
contradict previously stated information without explanation,
change behavior unnecessarily,
express unsupported certainty, or
fail to recover after an identified mistake.**

Reliability therefore provides continuity between successful individual responses.

The process can be represented as:

Established Interaction State

↓

Continued AI Behavior

↓

Consistency + Dependability + Appropriate Updating

↓

Reliable Interaction

The purpose of Reliability within BOND is to determine whether strong interaction behavior can be **maintained rather than demonstrated only temporarily.**

Reliability asks:

“Can the AI remain dependably aligned with the interaction while appropriately handling continuity, uncertainty, and correction?”

5.4.3 Reliability Is Not Infallibility

No AI system should be considered reliable merely because it appears confident or avoids acknowledging mistakes.

Reliability does not require perfect performance.

Instead, when an error becomes identifiable, reliable behavior may involve:

Recognizing the error

Correcting the relevant information

Updating subsequent behavior

Avoiding repetition of the same established mistake

Similarly, when sufficient information is unavailable, communicating uncertainty can be more reliable than generating an unsupported answer with excessive confidence.

Therefore:

Confidence $\neq$ Reliability

Never Changing $\neq$ Reliability

Never Admitting Error $\neq$ Reliability

Reliable interaction requires appropriate behavior when the system is correct, uncertain, corrected, or required to update.

5.4.4 Sub-Factors of Reliability

Reliability can be decomposed into five proposed sub-factors:

- 1. Behavioral Consistency (RBC)**
- 2. Contextual Continuity (RCC)**
- 3. Constraint Persistence (RCP)**
- 4. Uncertainty Calibration (RUC)**
- 5. Error Recovery (RER)**

Together, these sub-factors evaluate whether AI behavior remains dependable as interaction progresses.

1. Behavioral Consistency (RBC)

Behavioral Consistency evaluates whether the AI maintains compatible behavior across comparable interaction conditions.

If no relevant condition changes, major unexplained changes in the AI's response may reduce reliability.

For example:

Turn 1: AI correctly states that a particular conclusion cannot be determined from the available information.

Turn 3: Without receiving additional evidence, the AI presents the same conclusion as certain.

The interaction conditions did not justify this change.

Behavioral Consistency therefore asks whether changes in AI behavior are supported by **changes in evidence, requirements, or context**.

2. Contextual Continuity (RCC)

Contextual Continuity evaluates whether the AI maintains relevant established information across the available interaction.

This may include:

Previously defined terms

Confirmed assumptions

Current task state

Relevant earlier conclusions

Established interaction context

For example, if a user defines:

“For this discussion, X refers to the experimental model.”

the AI should not later treat X as a different object without justification.

Contextual continuity allows an interaction to develop cumulatively rather than repeatedly rebuilding its foundation.

3. Constraint Persistence (RCP)

Constraint Persistence evaluates whether the AI continues respecting active requirements and boundaries established earlier in the interaction.

These may include:

Output length

Required format

Language

Task restrictions

Requested inclusions or exclusions

Example:

User: “For the rest of this comparison, use SI units.”

If later responses unnecessarily return to another unit system, the AI demonstrates weak constraint persistence.

However, a constraint that is explicitly withdrawn or replaced should no longer be preserved.

Reliability requires persistence of active constraints, not permanent preservation of obsolete ones.

4. Uncertainty Calibration (RUC)

AI systems sometimes operate with incomplete, ambiguous, or uncertain information.

Uncertainty Calibration evaluates whether the AI's expressed confidence is appropriately matched to the available evidence.

Relevant behavior may include:

Acknowledging missing information

Distinguishing fact from inference

Avoiding unsupported certainty

Requesting information when necessary

Communicating limitations when they materially affect the answer

For example, if the available information cannot support a definite conclusion, presenting speculation as established fact reduces reliability.

The objective is not constant hedging.

Reliable uncertainty communication means expressing uncertainty when it is relevant—not making every answer unnecessarily uncertain.

5. Error Recovery (RER)

Errors can occur during interaction.

Error Recovery evaluates how the AI behaves **after an error becomes identifiable**.

A useful recovery process may involve:

Error Identified

↓

Acknowledgment

↓

Correction

↓

Updated Interaction State

↓

Improved Subsequent Behavior

Consider:

Human: “You used 20 V, but I said the voltage was 12 V.”

A reliable AI should update the calculation using 12 V rather than continue using the incorrect value.

Error Recovery therefore measures whether the system can **recover from failure instead of allowing the failure to propagate through later interaction**.

5.4.5 Examples of Reliability

Example A — Strong Reliability

User:

“For this entire discussion, assume gravitational acceleration is 9.8 m/s². ”

Across subsequent calculations, the AI consistently uses the established value unless the user changes the condition.

Observed Evidence:

Context maintained ✓

Constraint maintained ✓

Behavior consistent ✓

Interpretation: Strong Reliability

Example B — Weak Contextual Reliability

Turn 1:

The user establishes that the project budget is ₹50,000.

Turn 4:

Without any change in context, the AI performs planning using a ₹100,000 budget.

Observed Evidence:

Established information lost ×

Interaction state changed without justification ×

Interpretation: Weak Reliability

Example C — Appropriate Uncertainty

User:

“Will this experimental prototype definitely achieve 90% efficiency?”

The available information is insufficient to establish that result.

AI Behavior:

The AI explains that the available evidence cannot guarantee 90% efficiency and identifies what testing would be required to support the claim.

Observed Evidence:

Unsupported certainty avoided ✓

Evidence limitation recognized ✓

Interpretation: Strong Reliability

Example D — Strong Error Recovery

Human:

“The value I gave earlier was incorrect. The mass is 5 kg, not 8 kg.”

AI Behavior:

The AI updates the relevant calculation and clearly bases subsequent reasoning on 5 kg.

Observed Evidence:

Correction recognized ✓

Interaction state updated ✓

Previous error prevented from propagating ✓

Interpretation: Strong Reliability

5.4.6 Reliability Factor Summary

Sub-Factor	Primary Evaluation Question
Behavioral Consistency (RBC)	Does the AI remain sufficiently consistent when relevant interaction conditions remain unchanged?
Contextual Continuity (RCC)	Does the AI preserve relevant established information across the interaction?
Constraint Persistence (RCP)	Does the AI continue respecting active requirements and constraints?
Uncertainty Calibration (RUC)	Does the AI appropriately communicate uncertainty relative to the available evidence?
Error Recovery (RER)	Does the AI appropriately correct and update its behavior after an identifiable error?

These sub-factors allow Reliability to evaluate **dependability across interaction** without incorrectly treating rigidity or apparent confidence as reliable behavior.

5.4.7 Conceptual Diagram

ESTABLISHED INTERACTION STATE

↓

CONTINUED AI BEHAVIOR

↓

RELIABILITY (R)

↓

Behavioral Consistency (RBC)

Contextual Continuity (RCC)

Constraint Persistence (RCP)

Uncertainty Calibration (RUC)

Error Recovery (RER)

↓

STABILITY + APPROPRIATE UPDATING

↓

DEPENDABLE CONTINUED INTERACTION

RELIABILITY PRINCIPLE

Reliability is successful when an AI preserves relevant interaction state and active constraints, behaves consistently under comparable conditions, communicates meaningful uncertainty appropriately, and updates its behavior when correction or new evidence requires change.

Reliability therefore provides the BOND framework with a mechanism for evaluating whether effective interaction behavior can remain **dependable across time, context, correction, and changing interaction conditions**.

5.5 Engagement & Responsiveness (ER)

Engagement & Responsiveness is the fifth core factor of the BOND framework.

Human–AI interaction develops through continuous exchange. The human may ask follow-up questions, provide feedback, change direction, introduce new information, express confusion, or continue a task across multiple turns.

An AI must therefore do more than generate individually acceptable responses. It must **recognize the developing state of the interaction and respond in a way that meaningfully supports its continuation**.

Within BOND, Engagement & Responsiveness does not refer to maximizing conversation length, entertainment, emotional attachment, or human-like sociability.

It evaluates whether the AI remains **appropriately attentive, responsive, and interaction-aware as the exchange develops**.

DEFINITION

Engagement & Responsiveness (ER) is the demonstrated capability of an AI system to recognize and appropriately respond to changes, feedback, progression, and interaction signals while maintaining useful continuity toward the human's current objective.

5.5.1 Meaning of Engagement & Responsiveness

Engagement and responsiveness are closely related but distinguishable.

Engagement concerns whether the AI remains meaningfully involved in the developing interaction.

Responsiveness concerns whether the AI appropriately reacts when the human provides new signals, feedback, or changes.

Together:

Engagement → Maintains useful interaction

Responsiveness → Adjusts appropriately to interaction signals

For example:

Human: “I still don't understand the second step.”

A responsive AI should recognize that the problem is specifically **Step 2**, rather than repeating the complete original explanation unchanged.

An engaged AI should then continue helping in a way that moves the interaction toward understanding.

Therefore, Engagement & Responsiveness evaluates not merely whether the AI replies, but whether its reply **meaningfully responds to the state of the interaction**.

5.5.2 Purpose of Engagement & Responsiveness

Human requirements are not always completely resolved in one response.

Interaction may require:

Clarification

Iteration

Correction

Progressive explanation

Task continuation

Response to feedback

Changing strategy

The purpose of ER is to evaluate whether the AI can participate effectively in this evolving process.

A simplified progression is:

AI Response

↓

Human Interaction Signal

↓

Signal Recognition

↓

Appropriate Response

↓

Continued Progress

Without effective responsiveness, an AI may technically continue generating responses while the interaction itself becomes disconnected.

Engagement & Responsiveness asks:

“As the interaction develops, does the AI recognize what changed and respond in a way that meaningfully supports continued progress?”

5.5.3 Engagement Is Not Conversation Length

A longer conversation does not automatically represent stronger engagement.

An AI could unnecessarily extend an interaction by:

Repeating information

Asking unnecessary questions

Adding irrelevant content

Creating artificial conversational loops

Avoiding completion of the task

Such behavior may increase interaction length while reducing effectiveness.

Likewise, a highly effective interaction may sometimes end after one concise response.

Therefore:

More Turns $\neq$ More Engagement

More Words $\neq$ Better Responsiveness

Effective engagement is goal-oriented and context-appropriate, not maximized for duration.

5.5.4 Responsiveness Is Not Automatic Agreement

Responding appropriately to a human does not require agreeing with every statement or request.

A user may provide:

Incorrect information

A contradictory instruction

An unclear correction

A request that cannot appropriately be completed

In such cases, responsiveness may require clarification, explanation, correction, or an appropriate boundary rather than automatic compliance.

Therefore:

Human Signal $\rightarrow$ Evaluate $\rightarrow$ Respond Appropriately

rather than:

Human Signal $\rightarrow$ Automatically Agree

This distinction ensures that ER measures **interaction sensitivity rather than compliance**.

5.5.5 Sub-Factors of Engagement & Responsiveness

Engagement & Responsiveness can be decomposed into five proposed sub-factors:

- 1. Interaction Signal Recognition (ERS)**
- 2. Feedback Responsiveness (ERF)**
- 3. Progression Awareness (ERP)**
- 4. Adaptive Continuation (ERA)**
- 5. Goal-Oriented Engagement (ERG)**

Together, these sub-factors evaluate whether the AI can remain meaningfully responsive as Human–AI Interaction evolves.

1. Interaction Signal Recognition (ERS)

Interaction Signal Recognition evaluates whether the AI identifies meaningful signals contained within the human's latest input.

Signals may indicate:

Confusion

Satisfaction

Correction

Change of requirement

Request for greater detail

Request for simplification

Continuation of an earlier task

For example:

Human: “That example made sense, but I still don't understand the formula.”

The AI should recognize two separate signals:

Example → **understood**

Formula → **remains unresolved**

Repeating the example would therefore fail to respond to the actual interaction state.

2. Feedback Responsiveness (ERF)

Feedback Responsiveness evaluates whether the AI appropriately changes or maintains its behavior after receiving meaningful human feedback.

For example:

Human: “That explanation is too complicated.”

A responsive AI may:

reduce technical terminology,

use a simpler example,

divide the concept into smaller steps,

or

change the explanatory strategy.

Simply generating the same explanation with minor wording changes may demonstrate weaker responsiveness.

Feedback becomes valuable only when the AI appropriately incorporates what the feedback communicates.

3. Progression Awareness (ERP)

Human–AI interaction often develops through stages.

For example:

Define Problem → Understand Concept → Apply Method → Evaluate Result

Progression Awareness evaluates whether the AI recognizes **where the interaction currently stands**.

An AI should avoid unnecessarily restarting earlier stages when the human has already progressed beyond them.

Consider:

Turn 1: Concept explained.

Turn 2: User confirms understanding.

Turn 3: User asks for application.

A progression-aware AI should move toward application rather than repeating the definition.

Progression awareness therefore helps interaction behave as a **developing process rather than a sequence of disconnected exchanges**.

4. Adaptive Continuation (ERA)

Adaptive Continuation evaluates whether the AI selects an appropriate next interaction strategy based on the current state.

Depending on the situation, appropriate continuation may involve:

Answering directly

Providing another example

Changing explanation strategy

Requesting clarification

Correcting an earlier response

Continuing a multi-step task

Recognizing that the task is complete

The strongest continuation is not always the longest.

Sometimes the appropriate next action is simply to provide the requested answer and stop.

Adaptive continuation means doing what the interaction requires next—not keeping the interaction alive unnecessarily.

5. Goal-Oriented Engagement (ERG)

Goal-Oriented Engagement evaluates whether continued interaction remains directed toward the human's relevant objective.

An AI may appear highly conversational while gradually moving away from the task.

BOND therefore distinguishes:

Conversational Activity

from

Useful Engagement

For example, when helping a user debug a program, effective engagement should progressively support identification and resolution of the problem rather than generating unrelated discussion.

Goal-oriented engagement asks whether each meaningful continuation contributes toward:

Understanding

Decision

Completion

Resolution

or another relevant interaction objective.

5.5.6 Examples of Engagement & Responsiveness

Example A — Strong Feedback Responsiveness

Human:

“I understand the definition now, but the equation still confuses me.”

AI Behavior:

The AI does not repeat the definition. It focuses specifically on the equation and explains its components differently.

Observed Evidence:

Interaction signal recognized ✓

Feedback incorporated ✓

Progression maintained ✓

Interpretation: Strong Engagement & Responsiveness

Example B — Weak Responsiveness

Human:

“Please stop using technical terminology.”

AI Behavior:

The next response continues using unexplained advanced terminology.

Observed Evidence:

Feedback available ✓

Behavior not appropriately changed ✕

Interpretation: Weak Engagement & Responsiveness

Example C — Strong Progression Awareness**Interaction:**

The AI and human have already identified the cause of a technical problem.

Human:

“Okay, how do we fix it?”

AI Behavior:

The AI proceeds to the solution rather than repeating the diagnosis.

Observed Evidence:

Current interaction stage recognized ✓

Previous progress preserved ✓

Next action appropriate ✓

Interpretation: Strong Engagement & Responsiveness

Example D — Excessive Engagement**Human:**

“Just give me the final answer.”

AI Behavior:

The AI adds multiple unnecessary questions, unrelated suggestions, and extended discussion before answering.

Observed Evidence:

User preference recognized poorly ✕

Interaction unnecessarily extended ✕

Goal focus reduced ✕

Interpretation: Weak Goal-Oriented Engagement**5.5.7 Engagement & Responsiveness Factor Summary**

Sub-Factor	Primary Evaluation Question
Interaction Signal Recognition (ERS)	Did the AI recognize meaningful signals in the developing interaction?
Feedback Responsiveness (ERF)	Did the AI appropriately respond to relevant human feedback?
Progression Awareness (ERP)	Did the AI recognize the current stage and progress of the interaction?
Adaptive Continuation (ERA)	Did the AI choose an appropriate next interaction behavior?
Goal-Oriented Engagement (ERG)	Did continued interaction remain useful and directed toward the relevant objective?

These sub-factors allow ER to evaluate **interaction dynamics** without assuming that longer, more conversational, or more agreeable behavior is inherently better.

5.5.8 Conceptual Diagram

CURRENT INTERACTION STATE

↓

HUMAN SIGNAL / FEEDBACK / NEW REQUIREMENT

↓

ENGAGEMENT & RESPONSIVENESS (ER)

↓

Interaction Signal Recognition (ERS)

Feedback Responsiveness (ERF)

Progression Awareness (ERP)

Adaptive Continuation (ERA)

Goal-Oriented Engagement (ERG)

↓

APPROPRIATE NEXT BEHAVIOR

↓

USEFUL INTERACTION CONTINUITY

ENGAGEMENT & RESPONSIVENESS PRINCIPLE

Engagement & Responsiveness is successful when an AI recognizes meaningful changes and feedback within the interaction, responds appropriately to them, preserves awareness of interaction progress, and continues only in ways that support the relevant human objective.

Engagement & Responsiveness therefore completes the five-dimensional conceptual structure of BOND by evaluating whether the AI can remain **meaningfully connected to the evolving state of Human–AI Interaction** rather than treating each response as an isolated event.

5.6 Integrated Five-Factor View

The five BOND factors describe different but interacting components of AI connection-building capability:

Understanding → **What does the human require?**

Clarity → How should the response be communicated?

Personalization → How should relevant context adapt the interaction?

Reliability → Can appropriate behavior remain dependable?

Engagement & Responsiveness → Can the AI respond effectively as the interaction evolves?

Together:

Human Requirement & Context

↓

Understanding (U)

↓

Clarity (C) + Personalization (P)

↓

Reliability (R)

↓

Engagement & Responsiveness (ER)

↪ Interaction continues and new context returns to Understanding

CHAPTER CONCLUSION

The five BOND factors provide complementary observational dimensions of AI-side connection-building capability. Their separation enables diagnostic evaluation, while their combined behavior provides the conceptual basis for an overall BOND measurement.

These factors establish the foundation for the **formal mathematical and scoring structure of BOND** developed in the following chapters.

CHAPTER 6 — MATHEMATICAL FRAMEWORK

6.1 Weight Distribution

The five BOND factors represent different dimensions of AI connection-building capability. To convert these dimensions into an overall measurement, BOND requires a mathematical method for determining **how strongly each factor contributes to the final score**.

The five primary factors are:

U = Understanding

C = Clarity

P = Personalization

R = Reliability

ER = Engagement & Responsiveness

Each factor is associated with a corresponding weight:

wU = Weight of Understanding

wC = Weight of Clarity

wP = Weight of Personalization

wR = Weight of Reliability

wER = Weight of Engagement & Responsiveness

The purpose of these weights is to define the **relative mathematical contribution** of each factor to the overall BOND score.

6.1.1 Weight Constraint

For a normalized weighted model, the total contribution of all five factor weights must equal **1**.

Weight Constraint:

$$wU + wC + wP + wR + wER = 1$$

When expressed as percentages:

$$wU + wC + wP + wR + wER = 100\%$$

This constraint ensures that the complete BOND measurement remains normalized and mathematically interpretable.

6.1.2 Why Weights Are Required

The five BOND factors contribute to the broader concept of connection-building capability, but this does not automatically mean that their contributions are empirically identical.

For example, a failure of Understanding may affect later stages of interaction because the AI may respond to an incorrectly interpreted requirement.

Similarly, weak Reliability may become increasingly important during long multi-turn interactions.

However, theoretical importance alone is not sufficient to assign permanently higher numerical weight to a factor.

BOND therefore distinguishes between:

Conceptual Importance → How important a factor appears theoretically.

Mathematical Weight → How strongly the factor contributes to the calculated BOND score.

Empirical Importance → How strongly experimental evidence supports that contribution.

BOND weights should ultimately be supported by empirical validation rather than determined only through theoretical intuition.

6.1.3 Initial Equal-Weight Model

Before sufficient validation data are available, BOND uses an **equal-weight baseline model**.

With five primary factors, each factor receives an initial weight of:

$$w_U = w_C = w_P = w_R = w_{ER} = 0.20$$

Therefore, each factor contributes:

20% of the overall BOND score.

Factor	Symbol	Initial Weight
Understanding	U	20%
Clarity	C	20%
Personalization	P	20%
Reliability	R	20%
Engagement & Responsiveness	ER	20%
Total	—	100%

The equal-weight model provides a **neutral mathematical starting point**.

It avoids making an unsupported claim that one BOND factor is inherently more important than another before experimental evidence is available.

6.1.4 Equal Weighting Is a Baseline

The initial 20% distribution should not be interpreted as evidence that all five factors contribute exactly equally to AI connection-building capability.

Instead, it represents the **default mathematical configuration of the proposed framework before empirical weight estimation**.

Future validation may determine that:

All five factors should remain equally weighted

or

Some factors should receive different weights

or

Factor importance changes across interaction domains

or

Different weighting strategies better represent the measured construct

Therefore:

Equal weighting is the initial BOND hypothesis, not a permanently fixed empirical conclusion.

This distinction allows the framework to remain mathematically defined while remaining open to experimental revision.

6.1.5 Empirical Weight Estimation

Once sufficient BOND evaluation data are available, alternative factor weights can be investigated statistically.

Potential evidence may include:

Relationship between factor scores and external human evaluation

Inter-factor correlations

Predictive performance

Reliability analysis

Factor-analytic evidence

Cross-domain stability

Repeated experimental results

The objective is not simply to find a set of weights that produces desirable scores.

Instead, empirical weighting should determine whether a particular distribution provides a **better-supported representation of the proposed connection-building construct**.

For example, a future experimentally supported distribution might hypothetically be:

Factor	Hypothetical Weight
Understanding	25%
Clarity	20%
Personalization	15%
Reliability	20%
Engagement & Responsiveness	20%
Total	100%

These values are only illustrative.

They should **not** be treated as validated BOND weights unless experimental evidence supports them.

6.1.6 Standard and Domain-Specific Weighting

Different AI applications may create different interaction requirements.

For example:

AI Tutor → Understanding and Clarity may play particularly important practical roles.

Customer-Support AI → Understanding, Reliability, and Responsiveness may strongly influence successful resolution.

Interactive Robot → Reliability and Responsiveness may become particularly important.

Personal Assistant → Personalization and contextual continuity may become more prominent.

However, allowing every application to freely modify factor weights would reduce comparability between BOND scores.

BOND therefore distinguishes between two configurations:

Standard BOND

and

Domain-Specific BOND

Standard BOND

Standard BOND uses a common defined weighting system.

Its purpose is to support **comparison across AI systems under standardized evaluation conditions**.

Until alternative weights are empirically validated, the Standard BOND configuration uses:

$U = 20\%$, $C = 20\%$, $P = 20\%$, $R = 20\%$, $ER = 20\%$

Domain-Specific BOND

A domain-specific configuration may modify factor weights when there is a justified reason to emphasize particular interaction requirements.

However, any modified weighting system must be:

Explicitly defined

Scientifically justified

Reported with the resulting score

Separated from Standard BOND results

A domain-specific score should not automatically be treated as directly equivalent to a Standard BOND score.

6.1.7 Weight Transparency

A numerical score becomes difficult to interpret when the mathematical assumptions behind it are unknown.

Therefore, every reported BOND score should specify the weighting configuration used.

For example:

BOND Configuration: Standard Equal Weight

or

BOND Configuration: Domain-Specific — AI Tutoring

The reported configuration allows researchers to determine whether two scores were produced under comparable mathematical conditions.

A BOND score should never be separated from the configuration through which it was calculated.

6.1.8 Sub-Factor Weighting

Each primary BOND factor contains five sub-factors.

For example, Understanding contains:

Requirement Recognition (UR)

Intent Interpretation (UI)

Context Integration (UC)

Constraint Recognition (UCR)

Clarification & Update Handling (UU)

The same weighting principle can be applied within a primary factor.

Before empirical evidence supports unequal sub-factor contributions, each of the five sub-factors receives an equal baseline weight:

$$w_{UR} = w_{UI} = w_{UC} = w_{UCR} = w_{UU} = 0.20$$

Therefore, each sub-factor initially contributes **20% to its parent factor score**.

The same principle applies to the sub-factors of Clarity, Personalization, Reliability, and Engagement & Responsiveness.

6.1.9 Hierarchical Weight Structure

BOND therefore operates through two weighting levels.

LEVEL 1 — Sub-Factor Level

Observable Behavior

↓

Sub-Factor Evaluation

↓

Primary Factor Score

LEVEL 2 — Primary Factor Level

Understanding + Clarity + Personalization + Reliability + Engagement & Responsiveness

↓

Factor Weighting

↓

Overall BOND Score

This creates the following measurement hierarchy:

Human–AI Interaction

↓

Observable Evidence

↓

Sub-Factor Scores

↓

Primary Factor Scores

↓

Factor Weights

↓

Overall BOND Measurement

This hierarchical structure preserves the ability to trace the final BOND score back to the specific interaction behaviors from which it was constructed.

6.1.10 Weighting Principles

The BOND weighting system follows five primary principles:

Normalization → The weights of all active factors must form a normalized distribution.

Neutral Baseline → Equal weighting is used when sufficient evidence for unequal weighting does not yet exist.

Empirical Revision → Weight distributions may be revised when experimental validation provides adequate justification.

Transparency → The weighting configuration must be reported with the BOND result.

Comparability → Scores produced using materially different weighting configurations should not be treated as directly equivalent without qualification.

WEIGHT DISTRIBUTION PRINCIPLE

BOND begins with equal factor weighting as a neutral baseline and permits alternative weighting only when the modification is explicitly defined, scientifically justified, and reported with the resulting measurement.

The weight distribution establishes the mathematical foundation required to combine the five primary BOND factors into the **BOND Equation**.

6.2 BOND Equation

The BOND Equation converts the five primary factor scores into a single normalized measurement of **AI connection-building capability**.

The equation is designed to preserve two forms of evaluation:

Factor-Level Evaluation → identifies strengths and weaknesses across individual BOND dimensions.

Overall Evaluation → combines the five dimensions into one BOND score.

The overall score therefore provides a summary measurement, while the individual factor scores preserve the diagnostic structure of the framework.

6.2.1 Variables of the BOND Equation

The five primary factor scores are represented as:

U = Understanding

C = Clarity

P = Personalization

R = Reliability

ER = Engagement & Responsiveness

Their corresponding weights are:

wU = Weight of Understanding

wC = Weight of Clarity

wP = Weight of Personalization

wR = Weight of Reliability

wER = Weight of Engagement & Responsiveness

The overall BOND score is represented by:

B = Overall BOND Score

6.2.2 General Weighted BOND Equation

The general BOND equation is:

$$B = (wU \times U) + (wC \times C) + (wP \times P) + (wR \times R) + (wER \times ER)$$

where:

$$wU + wC + wP + wR + wER = 1$$

Each factor score is multiplied by its corresponding weight.

The weighted values are then added together to produce the overall BOND score.

For example:

$$\text{Understanding Contribution} = w_U \times U$$

$$\text{Clarity Contribution} = w_C \times C$$

$$\text{Personalization Contribution} = w_P \times P$$

$$\text{Reliability Contribution} = w_R \times R$$

$$\text{ER Contribution} = w_{ER} \times ER$$

Therefore:

$$\text{Overall BOND} = \text{Sum of all five weighted contributions}$$

6.2.3 Standard Equal-Weight Equation

Under the initial Standard BOND configuration established in Section 6.1:

$$w_U = w_C = w_P = w_R = w_{ER} = 0.20$$

The equation therefore becomes:

$$B = (0.20 \times U) + (0.20 \times C) + (0.20 \times P) + (0.20 \times R) + (0.20 \times ER)$$

Because every factor receives the same weight, the equation can also be simplified to:

$$B = (U + C + P + R + ER) / 5$$

This represents the **Standard Equal-Weight BOND Equation**.

6.2.4 Normalized Factor Scale

For the initial BOND mathematical framework, every primary factor is normalized to a range between **0 and 1**.

Therefore:

$$0 \leq U \leq 1$$

$$0 \leq C \leq 1$$

$$0 \leq P \leq 1$$

$$0 \leq R \leq 1$$

$$0 \leq ER \leq 1$$

Because the weights are normalized and all factor scores remain between 0 and 1:

$$0 \leq B \leq 1$$

The overall BOND score therefore also remains within the range:

0.00 to 1.00

A value closer to **1.00** represents stronger demonstrated connection-building performance under the specified evaluation conditions.

A value closer to **0.00** represents weaker demonstrated performance.

However, the overall value should not be interpreted independently of the factor profile and evaluation conditions.

6.2.5 Percentage Representation

For easier communication, the normalized BOND score may also be converted into a percentage.

The conversion is:

$$B\% = B \times 100$$

For example, if:

$$B = 0.82$$

then:

$$B\% = 0.82 \times 100$$

$$B\% = 82\%$$

Therefore:

$$\text{BOND Score} = 0.82 / 1.00$$

and

$$\text{BOND Score} = 82\%$$

represent the same measurement in different numerical forms.

The percentage form does not create a separate metric.

6.2.6 Sub-Factor Equation

Each primary BOND factor is itself constructed from five sub-factors.

For any primary factor **F**, let its five sub-factor scores be:

$$s_1, s_2, s_3, s_4, s_5$$

and their corresponding weights be:

$$v_1, v_2, v_3, v_4, v_5$$

The general factor equation is:

$$F = (v_1 \times s_1) + (v_2 \times s_2) + (v_3 \times s_3) + (v_4 \times s_4) + (v_5 \times s_5)$$

with the constraint:

$$v_1 + v_2 + v_3 + v_4 + v_5 = 1$$

Under equal sub-factor weighting:

$$v_1 = v_2 = v_3 = v_4 = v_5 = 0.20$$

Therefore:

$$F = (s_1 + s_2 + s_3 + s_4 + s_5) / 5$$

This equation can be applied independently to each primary BOND factor.

6.2.7 Example — Understanding Score

Understanding contains five sub-factors:

UR = Requirement Recognition

UI = Intent Interpretation

UC = Context Integration

UCR = Constraint Recognition

UU = Clarification & Update Handling

Under equal sub-factor weighting:

$$U = (UR + UI + UC + UCR + UU) / 5$$

Suppose an AI receives the following normalized scores:

UR = 0.90

UI = 0.80

UC = 0.85

UCR = 0.75

UU = 0.80

Then:

$$U = (0.90 + 0.80 + 0.85 + 0.75 + 0.80) / 5$$

$$U = 4.10 / 5$$

$$U = 0.82$$

Therefore:

Understanding Score = 0.82

or

Understanding Score = 82%

The same process can be applied to the remaining four primary factors.

6.2.8 Complete Hierarchical Equation

BOND therefore operates through two mathematical levels.

LEVEL 1 — Sub-Factor Calculation

Sub-Factor Scores → Primary Factor Score

For example:

$$UR + UI + UC + UCR + UU \rightarrow U$$

The same process produces:

C, P, R, and ER

LEVEL 2 — Overall BOND Calculation

The five primary factor scores are then combined:

$$U + C + P + R + ER \rightarrow B$$

Under the Standard Equal-Weight configuration:

$$B = (U + C + P + R + ER) / 5$$

This produces a hierarchical measurement system:

Observable Interaction Behavior

↓

Sub-Factor Scores

↓

Primary Factor Scores

↓

Weighted Combination

↓

Overall BOND Score

6.2.9 Opportunity-Adjusted BOND Equation

Not every interaction provides sufficient opportunity to evaluate every BOND factor.

For example, a short factual interaction may provide meaningful evidence for Understanding and Clarity but little evidence for long-term Reliability or Personalization.

Assigning unavailable factors a score of zero would create a mathematical error in interpretation:

No Opportunity → 0

would incorrectly imply:

No Opportunity → Complete Failure

BOND therefore introduces an **Opportunity-Adjusted Equation**.

For each factor, an applicability indicator is defined:

$a = 1 \rightarrow$ Factor had sufficient evaluation opportunity

$a = 0 \rightarrow$ Factor lacked sufficient evaluation opportunity

The opportunity-adjusted equation is:

$$B = [\Sigma(a \times w \times F)] / [\Sigma(a \times w)]$$

In expanded form:

$$B = [(a_U \times w_U \times U) + (a_C \times w_C \times C) + (a_P \times w_P \times P) + (a_R \times w_R \times R) + (a_{ER} \times w_{ER} \times ER)] / [(a_U \times w_U) + (a_C \times w_C) + (a_P \times w_P) + (a_R \times w_R) + (a_{ER} \times w_{ER})]$$

Only factors with sufficient evaluation opportunity contribute to the calculated score.

6.2.10 Example — Opportunity Adjustment

Suppose an interaction provides sufficient evidence only for:

Understanding = 0.80

Clarity = 0.90

The other three factors are classified as:

N/A — Insufficient Evaluation Opportunity

Under equal weighting:

$$B = [(0.20 \times 0.80) + (0.20 \times 0.90)] / (0.20 + 0.20)$$

$$B = (0.16 + 0.18) / 0.40$$

$$B = 0.34 / 0.40$$

$$B = 0.85$$

Therefore:

Opportunity-Adjusted BOND Score = 0.85

The remaining factors are **not assigned zero**.

They remain explicitly marked as unavailable for that interaction.

6.2.11 Factor Coverage

Opportunity adjustment creates another important requirement.

A BOND score calculated from two factors should not appear equivalent to a BOND score supported by evidence across all five factors.

BOND therefore reports **Factor Coverage**.

Let:

K = Number of primary factors with sufficient evaluation opportunity

Then:

$$\text{Factor Coverage} = (K / 5) \times 100$$

Examples:

Active Factors	Factor Coverage
5	100%
4	80%
3	60%
2	40%
1	20%

Therefore, the previous example would be reported as:

BOND Score = 0.85

Factor Coverage = 40%

rather than simply reporting:

BOND = 0.85

This prevents partial evaluation from appearing more comprehensive than it actually is.

6.2.12 Complete BOND Reporting Format

A BOND result should preserve both overall and diagnostic information.

A standardized report may contain:

Overall BOND Score: 0.82 / 1.00

Percentage Representation: 82%

Factor Coverage: 100%

Weighting Configuration: Standard Equal Weight

Factor Profile:

Understanding (U): 0.88

Clarity (C): 0.84

Personalization (P): 0.73

Reliability (R): 0.81

Engagement & Responsiveness (ER): 0.84

This format provides substantially more information than a single numerical score.

6.2.13 Mathematical Interpretation Boundary

The BOND Equation defines **how the proposed measurements are combined**.

It does not independently establish that:

all five factors are equally important,

all sub-factors are equally important,

the proposed dimensions perfectly represent connection-building capability,

a particular score automatically represents universally strong or weak AI interaction,

or

BOND represents complete AI quality.

These questions require empirical validation.

Therefore:

The mathematical framework defines the measurement structure; experimental validation determines whether that structure is scientifically supported.

BOND EQUATION PRINCIPLE

The BOND Equation combines normalized factor-level evidence into an overall measurement while preserving diagnostic factor information, accounting for differences in evaluation opportunity, and explicitly reporting the coverage and mathematical configuration of the resulting score.

The BOND Equation provides the mathematical foundation required for the **BOND Assessment Test (BAT)**, through which observable Human–AI interactions can be converted into standardized BOND measurements.

6.3 BOND Assessment Test (BAT)

The mathematical framework defines how BOND scores are calculated, but a mathematical equation alone cannot produce an evaluation.

A standardized process is required to create **interaction conditions, observe AI behavior, evaluate the relevant sub-factors, and convert those observations into numerical scores.**

For this purpose, BOND introduces the **BOND Assessment Test (BAT).**

DEFINITION

The BOND Assessment Test (BAT) is a structured evaluation procedure designed to generate observable Human–AI interaction evidence and convert that evidence into sub-factor, factor-level, and overall BOND measurements.

BAT acts as the operational bridge between:

BOND Theory → Human–AI Interaction → Observable Evidence → Numerical Measurement

6.3.1 Purpose of BAT

The primary purpose of BAT is to create sufficiently controlled interaction conditions in which the five BOND factors can be systematically evaluated.

Without a standardized assessment procedure, two evaluators could test the same AI using substantially different interactions and produce scores that are difficult to compare.

BAT therefore aims to provide:

Standardized interaction scenarios

Defined evaluation opportunities

Observable behavioral criteria

Structured scoring procedures

Comparable factor-level results

Transparent overall BOND calculation

The relationship can be represented as:

BAT Scenario

↓

Human–AI Interaction

↓

Observable AI Behavior

↓

Sub-Factor Evaluation

↓

Factor Scores

↓

BOND Equation

↓

Overall BOND Result

6.3.2 BAT Does Not Test General Intelligence

BAT is not intended to function as a general intelligence examination.

It does not primarily ask:

How much knowledge does the AI possess?

How difficult a mathematical problem can it solve?

How many programming tasks can it complete?

Does the system possess AGI?

These properties require other evaluation methods.

BAT instead creates situations in which the AI must demonstrate behaviors relevant to **connection-building capability**.

For example, a BAT task may intentionally provide:

An ambiguous requirement

A relevant user preference

A change in context

A correction

A constraint

Feedback indicating confusion

The objective is then to observe **how the AI handles the interaction**.

BAT evaluates interaction behavior rather than attempting to replace capability benchmarks.

6.3.3 BAT Evaluation Unit

The fundamental unit of BAT is an **Interaction Scenario**.

A BAT Interaction Scenario contains:

1. Initial Context

2. Human Input

3. Intended Evaluation Opportunities

4. One or More Interaction Turns

5. Observable AI Responses

6. Evaluation Criteria

7. Sub-Factor Scores

A scenario may be single-turn or multi-turn depending on the behavior being evaluated.

For example:

Single-Turn Scenario → may evaluate Requirement Recognition and Clarity.

Multi-Turn Scenario → may evaluate Context Integration, Preference Adaptation, Reliability, Feedback Responsiveness, and Error Recovery.

6.3.4 BAT Scenario Structure

A standardized BAT scenario can follow the structure:

Scenario ID

Scenario Category

Initial Context

Human Input

Interaction Sequence

Target BOND Factor(s)

Target Sub-Factor(s)

Expected Evaluation Opportunity

Observed AI Behavior

Evaluator Score

Evaluator Notes

This structure allows each score to remain connected to the evidence that produced it.

A BAT result should therefore be **traceable rather than purely numerical**.

6.3.5 Example BAT Scenario

Scenario ID: BAT-U-01

Primary Factor: Understanding

Target Sub-Factors: Requirement Recognition and Constraint Recognition

Human Input:

“Explain photosynthesis in exactly three short points for a beginner. Do not use equations.”

The scenario creates several observable requirements:

Topic → **Photosynthesis**

Quantity → **Three points**

Length → **Short**

Audience → **Beginner**

Restriction → **No equations**

The AI response can then be evaluated according to whether these requirements were appropriately interpreted and followed.

If the AI provides five highly technical paragraphs containing equations, the factual information may still be correct while the BAT scenario reveals weaknesses in **Requirement Recognition, Constraint Recognition, and potentially Clarity**.

6.3.6 Multi-Turn BAT Scenario

Some BOND behaviors require interaction over time.

For example:

Turn 1 — Human:

“Explain this concept without technical terminology.”

Turn 2 — AI:

Provides an explanation.

Turn 3 — Human:

“I understand the first part, but the second part is still confusing.”

Turn 4 — AI:

Responds to the feedback.

This scenario may provide evidence for:

Understanding

Clarity

Preference Adaptation

Feedback Responsiveness

Progression Awareness

Contextual Continuity

BAT can therefore use deliberately structured multi-turn interactions to create evaluation opportunities that would not exist in isolated prompts.

6.3.7 Opportunity Mapping

Every BAT scenario should specify **which factors and sub-factors it meaningfully provides an opportunity to evaluate.**

For example:

Scenario Behavior	Primary Evaluation Opportunity
Explicit multi-part instruction	Requirement Recognition
Ambiguous request	Clarification & Update Handling
Earlier relevant information	Context Integration
Stated communication preference	Preference Adaptation
Active constraint across turns	Constraint Persistence
Human correction	Error Recovery
Feedback indicating confusion	Feedback Responsiveness
Progress through a multi-step task	Progression Awareness

This mapping prevents evaluators from assigning scores to dimensions for which the scenario provides insufficient evidence.

6.3.8 BAT Scoring Scale

For the initial BAT framework, observable sub-factor performance can be represented using a five-level scoring scale.

Raw Score	Normalized Score	General Interpretation
0	0.00	Behavior failed or was absent despite clear opportunity
1	0.25	Weak demonstration
2	0.50	Partial demonstration
3	0.75	Strong demonstration with minor limitations
4	1.00	Full demonstration under the scenario criteria

The conversion is:

Normalized Sub-Factor Score = Raw BAT Score / 4

For example:

Raw Score = 3

Normalized Score = 3 / 4

Normalized Score = 0.75

This allows BAT observations to enter the normalized **0–1 BOND mathematical framework.**

6.3.9 N/A Classification

BAT must distinguish poor performance from lack of evaluation opportunity.

Therefore, in addition to scores **0–4**, BAT includes:

N/A = Not Applicable / Insufficient Evaluation Opportunity

N/A is **not equivalent to zero**.

For example:

A scenario contains no user-specific information and creates no meaningful opportunity for personalization.

In this case:

Personalization = N/A

rather than:

Personalization = 0

This preserves the BOND principle:

No Opportunity ≠ Failure

N/A values are excluded from the relevant normalized calculation according to the opportunity-adjustment rules.

Chapter 6 ma N/A Handling / Opportunity-Adjusted Scoring

Sub-Factor-Level N/A Handling

The same opportunity-adjustment principle applies within each primary BOND factor.

If one or more sub-factors are marked N/A because the evaluation does not provide a sufficient opportunity to observe the targeted behavior, those sub-factors are excluded from the factor-level calculation rather than treated as zero.

For a factor containing n sub-factors:

$$F = \sum(a_j \times v_j \times S_j) / \sum(a_j \times v_j)$$

Where:

F = resulting factor score

S_j = normalized score of sub-factor j

v_j = weight assigned to sub-factor j

a_j = applicability indicator

$a_j = 1$ if the sub-factor is validly evaluated

$a_j = 0$ if the sub-factor is N/A

Under the standard equal-sub-factor configuration, this reduces to the arithmetic mean of only the applicable sub-factor scores.

Example:

Suppose Understanding contains:

UR = 0.80

UI = 0.90

UC = 0.70

UCR = 0.80

UU = N/A

Then:

$$U = (0.80 + 0.90 + 0.70 + 0.80) / 4$$

$$U = 0.80$$

The N/A sub-factor is therefore excluded from both the numerator and the applicable denominator.

This prevents absence of evaluation opportunity from being interpreted as poor performance.

6.3.10 From BAT Scores to Factor Scores

Suppose Understanding is evaluated through its five sub-factors:

$$\text{UR} = 4$$

$$\text{UI} = 3$$

$$\text{UC} = 4$$

$$\text{UCR} = 3$$

$$\text{UU} = 2$$

The normalized scores become:

$$\text{UR} = 1.00$$

$$\text{UI} = 0.75$$

$$\text{UC} = 1.00$$

$$\text{UCR} = 0.75$$

$$\text{UU} = 0.50$$

Under equal sub-factor weighting:

$$U = (1.00 + 0.75 + 1.00 + 0.75 + 0.50) / 5$$

$$U = 4.00 / 5$$

$$U = 0.80$$

Therefore:

$$\text{Understanding Score} = 0.80$$

The same process produces scores for **C, P, R, and ER**.

These factor scores can then enter the overall BOND Equation.

6.3.11 Multiple Scenario Aggregation

A single scenario may not provide sufficient evidence for stable evaluation.

BAT can therefore evaluate the same sub-factor across multiple scenarios.

Suppose Requirement Recognition receives normalized scores:

$$\text{Scenario 1} = 1.00$$

$$\text{Scenario 2} = 0.75$$

$$\text{Scenario 3} = 0.75$$

$$\text{Scenario 4} = 1.00$$

The aggregated score can initially be calculated as:

$$UR = (1.00 + 0.75 + 0.75 + 1.00) / 4$$

$$UR = 0.875$$

This reduces dependence on a single interaction and allows evaluation across different conditions.

The final benchmark design should define the number and diversity of scenarios required for each sub-factor.

6.3.12 Evaluator Role

BAT scoring requires evaluators to judge observable AI behavior against predefined criteria.

The evaluator should not score according to:

Personal liking of the AI

Writing style preference unrelated to the task

Brand or model identity

Expected reputation of the system

Instead, evaluation should be based on:

Scenario requirements

Defined sub-factor criteria

Observable response evidence

Standard scoring rubric

Where feasible, model identity can be hidden during evaluation to reduce potential evaluator bias.

6.3.13 Multiple Evaluators

Because some interaction behaviors require human judgment, evaluator disagreement may occur.

A stronger BAT implementation can therefore use multiple independent evaluators.

For example:

Evaluator A → Score

Evaluator B → Score

Evaluator C → Score

↓

Agreement Analysis

↓

Aggregated Evaluation

Inter-rater agreement should later be examined during BOND validation.

Strong evaluator agreement would provide evidence that BAT scoring criteria can be applied consistently.

Weak agreement may indicate that the rubric or sub-factor definition requires revision.

6.3.14 BAT Output

A complete BAT evaluation can produce:

Overall BOND Score

Factor Scores

Sub-Factor Scores

Factor Coverage

Scenario-Level Results

Weighting Configuration

Evaluator Information

Evaluation Conditions

The result can therefore be represented as a profile rather than only one number.

Example:

BOND = 0.81 / 1.00

Coverage = 100%

U = 0.86

C = 0.84

P = 0.72

R = 0.80

ER = 0.83

Such a result identifies not only overall performance but also the areas responsible for that performance.

6.3.15 BAT and the Benchmark Framework

BAT defines **how interaction evidence becomes a BOND measurement**.

However, it does not by itself determine the complete collection of questions and scenarios required for standardized comparison.

That role belongs to the **BOND Benchmark Framework** developed in Chapter 7.

The relationship is:

BOND Framework → What is measured

BOND Factors → Which dimensions are measured

BOND Equation → How scores are mathematically combined

BAT → How individual interactions are assessed

BOND Benchmark → Which standardized scenarios are used for comparison

This separation keeps the framework modular and allows individual components to be independently improved.

BAT PRINCIPLE

The BOND Assessment Test converts structured Human–AI interaction into measurable evidence by creating defined evaluation opportunities, scoring observable behavior through standardized criteria, preserving N/A when evidence is unavailable, and aggregating results into the BOND mathematical framework.

BAT therefore provides the operational mechanism required to transform **BOND from a conceptual model into a testable evaluation procedure.**

6.4 BOND Score Calculation

The previous sections established the BOND weighting system, the overall equation, and the BOND Assessment Test (BAT). These components can now be integrated into a complete scoring process.

BOND Score Calculation defines **how raw interaction observations move through the framework to produce sub-factor scores, primary factor scores, and the final overall BOND measurement.**

The complete process can be represented as:

BAT Interaction Scenarios

↓

Observable AI Behavior

↓

Raw BAT Scores

↓

Normalized Sub-Factor Scores

↓

Primary Factor Scores

↓

Weighted BOND Equation

↓

Overall BOND Score

6.4.1 Step 1 — Conduct BAT Scenarios

The evaluation begins with standardized BAT interaction scenarios.

Each scenario should specify:

Target factor

Target sub-factor

Interaction conditions

Expected evaluation opportunity

Relevant constraints

Scoring criteria

The AI then interacts under these defined conditions.

The evaluator records only the behavior that is relevant to the intended evaluation criteria.

6.4.2 Step 2 — Assign Raw BAT Scores

Each applicable sub-factor observation receives a raw BAT score from **0 to 4**.

Raw Score	Interpretation
0	Failed despite clear opportunity
1	Weak demonstration
2	Partial demonstration
3	Strong demonstration with minor limitations
4	Full demonstration under the defined criteria
N/A	Insufficient evaluation opportunity

The raw score represents the evaluator's judgment of the observed behavior according to the BAT rubric.

N/A must remain separate from 0.

A score of **0** represents demonstrated failure.

N/A represents insufficient evidence for valid evaluation.

6.4.3 Step 3 — Normalize Raw Scores

Raw BAT scores are converted into the normalized **0–1 BOND scale**.

The conversion is:

$$\text{Normalized Score} = \text{Raw BAT Score} / 4$$

Therefore:

Raw BAT Score	Normalized Score
0	0.00
1	0.25
2	0.50
3	0.75
4	1.00

For example:

$$\text{Raw Score} = 3$$

$$\text{Normalized Score} = 3 / 4$$

$$\text{Normalized Score} = 0.75$$

This normalization allows all sub-factor measurements to enter the same mathematical system.

6.4.4 Step 4 — Aggregate Repeated Scenario Scores

A sub-factor may be evaluated across multiple BAT scenarios.

Suppose **Context Integration (UC)** is evaluated four times:

$$\text{Scenario 1} = 0.75$$

$$\text{Scenario 2} = 1.00$$

$$\text{Scenario 3} = 0.50$$

$$\text{Scenario 4} = 0.75$$

The aggregated UC score is:

$$\text{UC} = (0.75 + 1.00 + 0.50 + 0.75) / 4$$

$$\text{UC} = 3.00 / 4$$

$$\text{UC} = 0.75$$

Therefore:

$$\text{Context Integration Score} = 0.75$$

Using multiple scenarios reduces the influence of a single unusual interaction on the final measurement.

6.4.5 Step 5 — Calculate Primary Factor Scores

After the relevant sub-factor scores have been calculated, they are combined into their corresponding primary factor.

For Understanding:

$$U = (UR + UI + UC + UCR + UU) / 5$$

For Clarity:

$$C = (CC + CS + CP + CR + CE) / 5$$

For Personalization:

$$P = (PCR + PLA + PPA + PGS + PSC) / 5$$

For Reliability:

$$R = (RBC + RCC + RCP + RUC + RER) / 5$$

For Engagement & Responsiveness:

$$ER = (ERS + ERF + ERP + ERA + ERG) / 5$$

These equations assume the initial **equal sub-factor weighting configuration**.

If empirically validated sub-factor weights are introduced later, the corresponding weighted equations can replace the equal averages.

6.4.6 Example — Complete Factor Calculation

Suppose BAT produces the following normalized Understanding results:

Understanding Sub-Factor	Score
Requirement Recognition (UR)	0.90
Intent Interpretation (UI)	0.80
Context Integration (UC)	0.85
Constraint Recognition (UCR)	0.75
Clarification & Update Handling (UU)	0.80

Then:

$$U = (0.90 + 0.80 + 0.85 + 0.75 + 0.80) / 5$$

$$U = 4.10 / 5$$

$$U = 0.82$$

Therefore:

Understanding = 0.82

The same procedure is repeated for the remaining four primary factors.

6.4.7 Step 6 — Calculate Overall BOND Score

Suppose the five calculated factor scores are:

$$U = 0.82$$

$$C = 0.88$$

$$P = 0.74$$

$$R = 0.80$$

$$ER = 0.86$$

Under Standard Equal Weight BOND:

$$B = (U + C + P + R + ER) / 5$$

Therefore:

$$B = (0.82 + 0.88 + 0.74 + 0.80 + 0.86) / 5$$

$$B = 4.10 / 5$$

$$B = 0.82$$

Therefore:

$$\text{Overall BOND Score} = 0.82 / 1.00$$

or:

$$\text{Overall BOND Score} = 82\%$$

6.4.8 Factor Contribution View

The same calculation can also be represented through weighted contributions.

Factor	Score	Weight	Weighted Contribution
Understanding	0.82	0.20	0.164
Clarity	0.88	0.20	0.176
Personalization	0.74	0.20	0.148
Reliability	0.80	0.20	0.160
Engagement & Responsiveness	0.86	0.20	0.172
Total	—	1.00	0.820

Therefore:

$$B = 0.164 + 0.176 + 0.148 + 0.160 + 0.172$$

$$B = 0.820$$

This representation makes the mathematical contribution of each factor visible.

6.4.9 Step 7 — Calculate Factor Coverage

The overall score should also report how much of the five-factor framework was meaningfully evaluated.

If all five factors were evaluated:

$$\text{Coverage} = (5 / 5) \times 100$$

$$\text{Coverage} = 100\%$$

If only four factors were evaluable:

$$\text{Coverage} = (4 / 5) \times 100$$

$$\text{Coverage} = 80\%$$

A partial evaluation should therefore never be reported as though it represents complete five-factor coverage.

6.4.10 Handling N/A at the Sub-Factor Level

The same **No Opportunity ≠ Failure** principle applies within individual factors.

Suppose Personalization contains:

$$\text{PCR} = 0.75$$

$$\text{PLA} = 1.00$$

$$\text{PPA} = \text{N/A}$$

$$\text{PGS} = 0.75$$

$$\text{PSC} = 1.00$$

Assigning PPA a zero would unfairly reduce the Personalization score even though the interaction provided no meaningful opportunity to evaluate Preference Adaptation.

Therefore, only applicable sub-factors are included:

$$P = (0.75 + 1.00 + 0.75 + 1.00) / 4$$

$$P = 3.50 / 4$$

$$P = 0.875$$

The result should also report:

$$\text{Personalization Sub-Factor Coverage} = 4/5 = 80\%$$

This makes missing evidence visible rather than hiding it mathematically.

6.4.11 Complete Evaluation Example

A complete BOND result may therefore be presented as:

BOND Evaluation Result

Overall Score: 0.82 / 1.00

Percentage: 82%

Primary Factor Coverage: 100%

Weighting Configuration: Standard Equal Weight

Factor	Score	Percentage
Understanding	0.82	82%
Clarity	0.88	88%
Personalization	0.74	74%
Reliability	0.80	80%
Engagement & Responsiveness	0.86	86%
Overall BOND	0.82	82%

This result provides both a summary measurement and a diagnostic profile.

6.4.12 Score Precision

Excessive decimal precision can imply a level of measurement accuracy that the evaluation process may not support.

For standard reporting, BOND scores can therefore initially be rounded to **two decimal places**.

For example:

0.8237 → 0.82

The unrounded values may still be preserved for statistical analysis.

This distinction provides:

Internal Analysis → Higher numerical precision

Standard Reporting → Two decimal places

6.4.13 Comparison Between AI Systems

Suppose two AI systems receive:

System A

B = 0.84

System B

B = 0.79

The difference is:

$$\text{Difference} = 0.84 - 0.79$$

$$\text{Difference} = 0.05$$

However, this numerical difference should not automatically be interpreted as a scientifically meaningful performance difference.

Meaningful comparison also requires consideration of:

Same BAT version

Same weighting configuration

Comparable interaction conditions

Comparable factor coverage

Evaluator reliability

Sufficient scenario count

Statistical uncertainty

Therefore:

Numerical difference does not automatically imply statistically or practically meaningful difference.

Formal significance and uncertainty analysis should be introduced when sufficient experimental data are available.

6.4.14 BOND Score Pipeline

The complete calculation process can be summarized as:

STEP 1

BAT scenarios create evaluation opportunities.

↓

STEP 2

Observable AI behavior receives raw scores from 0–4 or N/A.

↓

STEP 3

Raw scores are normalized to 0–1.

↓

STEP 4

Repeated scenario observations are aggregated.

↓

STEP 5

Sub-factor scores produce U, C, P, R, and ER.

↓

STEP 6

Factor weights produce the overall BOND score.

↓

STEP 7

Coverage and configuration information are attached.

↓

FINAL BOND RESULT

SCORE CALCULATION PRINCIPLE

A BOND score should remain traceable from the final numerical measurement back to the factor scores, sub-factor scores, BAT observations, and interaction evidence from which it was produced.

This calculation system establishes how BAT results become standardized BOND measurements. The next section defines the **Interval System**, which provides a structured method for describing different regions of the BOND score range.

6.5 BOND Interval System

The BOND Equation produces a continuous normalized score between **0.00 and 1.00**. Although the numerical value provides precise information, researchers and developers may also require a simpler way to describe the **general level of demonstrated connection-building performance**.

For this purpose, BOND introduces an **Interval System**.

The Interval System divides the normalized BOND range into descriptive performance regions while preserving the original numerical score.

DEFINITION

The BOND Interval System is an interpretive layer that groups numerical BOND scores into defined performance ranges for easier description and comparison without replacing the underlying continuous measurement.

6.5.1 Purpose of the Interval System

Consider two results:

AI System A → BOND = 0.83

AI System B → BOND = 0.37

The numerical difference is immediately visible.

However, for reporting purposes, it may also be useful to describe the general interaction-performance region represented by each score.

The Interval System therefore provides:

Numerical Score → Precise Measurement

Interval → Descriptive Interpretation

For example:

BOND = 0.83 → Very High Connection-Building Performance

The descriptive label does not replace the score.

A complete result should retain both.

6.5.2 Proposed Five-Level Interval Structure

The initial BOND framework proposes five equal descriptive intervals:

BOND Score	Classification
$0.00 \leq B < 0.20$	Very Low
$0.20 \leq B < 0.40$	Low
$0.40 \leq B < 0.60$	Moderate
$0.60 \leq B < 0.80$	High
$0.80 \leq B \leq 1.00$	Very High

This creates a simple progression:

Very Low → Low → Moderate → High → Very High

The intervals provide a standardized descriptive vocabulary for reporting BOND results.

6.5.3 Interval Meaning

Very Low — 0.00 to 0.19

A score within this interval indicates that the AI demonstrated **very limited connection-building behavior** across the evaluated BOND dimensions.

Possible observations may include:

Frequent misunderstanding of requirements

Poor communication alignment

Limited appropriate adaptation

Weak continuity

Failure to respond meaningfully to interaction signals

A Very Low result does not imply that the AI lacks intelligence or general capability.

It refers specifically to performance under the **evaluated BOND interaction conditions**.

Low — 0.20 to 0.39

A Low BOND score indicates that some connection-building behaviors were demonstrated, but **substantial weaknesses remained across the evaluated dimensions**.

The AI may occasionally interpret or respond appropriately while demonstrating inconsistent performance across scenarios.

The system may require significant improvement in one or more BOND factors before interaction becomes consistently effective.

Moderate — 0.40 to 0.59

A Moderate score indicates **mixed or intermediate connection-building performance**.

The AI may demonstrate effective behavior in some dimensions or scenarios while showing meaningful limitations in others.

For example:

Strong Clarity + Weak Personalization

or

Good initial Understanding + Weak multi-turn Reliability

could contribute to a moderate overall profile.

The factor-level results are therefore particularly important for interpreting scores in this region.

High — 0.60 to 0.79

A High BOND score indicates that the AI demonstrates **generally effective connection-building behavior across the evaluated conditions**, while some limitations remain.

The AI may successfully:

interpret most requirements,

communicate effectively,

use relevant context,

maintain interaction continuity,

and

respond appropriately to feedback

in a substantial proportion of the evaluation scenarios.

However, a High score should not be interpreted as perfect or universally reliable performance.

Very High — 0.80 to 1.00

A Very High BOND score indicates **strong demonstrated connection-building performance across the evaluated BOND dimensions and scenarios.**

The AI generally demonstrates effective:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

under the tested conditions.

However:

Very High BOND does not mean perfect AI.

The system may still produce factual errors, reasoning failures, safety failures, or weaknesses outside the scope of BOND.

The interval describes **connection-building performance only.**

6.5.4 Boundary Rule

To avoid ambiguity at interval boundaries, BOND uses a consistent classification rule.

For example:

$0.00 \leq B < 0.20 \rightarrow \text{Very Low}$

$0.20 \leq B < 0.40 \rightarrow \text{Low}$

$0.40 \leq B < 0.60 \rightarrow \text{Moderate}$

$0.60 \leq B < 0.80 \rightarrow \text{High}$

$0.80 \leq B \leq 1.00 \rightarrow \text{Very High}$

Therefore:

$B = 0.19 \rightarrow \text{Very Low}$

B = 0.20 → Low

B = 0.59 → Moderate

B = 0.60 → High

B = 0.80 → Very High

This rule prevents a boundary score from belonging to two intervals simultaneously.

6.5.5 Interval Classification Example

Suppose an AI receives:

U = 0.84

C = 0.81

P = 0.72

R = 0.78

ER = 0.85

Using Standard Equal-Weight BOND:

$$B = (0.84 + 0.81 + 0.72 + 0.78 + 0.85) / 5$$

$$B = 4.00 / 5$$

$$B = 0.80$$

Therefore:

BOND Score = 0.80

BOND Percentage = 80%

BOND Interval = Very High

A standardized result could therefore be reported as:

BOND: 0.80 / 1.00 — Very High Connection-Building Performance

6.5.6 Overall Interval vs. Factor Profile

A major limitation of interval labels is that different factor profiles can produce the same overall interval.

Consider:

System A

U = 0.90

C = 0.90

P = 0.50
R = 0.80
ER = 0.80

and:

System B

U = 0.78
C = 0.80
P = 0.77
R = 0.78
ER = 0.77

Both systems may receive similar overall BOND scores despite substantially different performance profiles.

Therefore:

The BOND interval should never replace the five-factor profile.

The interval provides **summary interpretation**.

The factor profile provides **diagnostic interpretation**.

Both should be reported when possible.

6.5.7 Interval Classification and Coverage

A score with low evaluation coverage should not receive the same interpretive confidence as a score supported by complete evaluation.

For example:

BOND = 0.85

Factor Coverage = 40%

This means the numerical score was supported by only two of the five primary factors.

Reporting only:

Very High BOND

could therefore create a misleading impression.

A better report is:

BOND = 0.85 — Very High

Factor Coverage = 40%

Classification Status = Partial Evaluation

This preserves the distinction between **performance level and evidence coverage**.

6.5.8 Provisional Nature of Interval Boundaries

The proposed interval boundaries are initially **theoretical reporting conventions**, not empirically validated natural boundaries of Human–AI interaction quality.

There is currently no basis within the proposed framework to claim that:

0.79 and 0.80 represent fundamentally different forms of AI behavior

or that:

0.60 represents a universal scientific threshold for strong interaction.

Therefore, the five equal intervals should initially be treated as **descriptive bands**.

Future empirical validation may examine whether alternative boundaries provide better interpretation.

Interval boundaries should become empirically calibrated when sufficient BAT data and external validation evidence become available.

This distinction prevents convenient numerical categories from being presented as discovered scientific laws.

6.5.9 Future Empirical Calibration

After large-scale BAT testing, interval boundaries may be examined using:

Human evaluator judgments

Score distributions

Percentile analysis

External interaction-quality measures

Cross-system comparisons

Cross-domain performance

Statistical clustering

If empirical evidence reveals meaningful performance regions, the interval boundaries may be revised.

For example, future research might determine that the most useful boundaries are not equally spaced.

Such changes should produce a new documented BOND version rather than silently modifying the original scoring interpretation.

6.5.10 BOND Score Card

A standardized BOND Score Card can summarize the complete result.

Measurement	Result
Overall BOND Score	0.82 / 1.00
Percentage	82%
Interval	Very High
Understanding	0.84
Clarity	0.88
Personalization	0.72
Reliability	0.80
Engagement & Responsiveness	0.86
Factor Coverage	100%
Configuration	Standard Equal Weight

This format combines:

Overall performance

Descriptive interval

Factor-level diagnosis

Coverage

Mathematical configuration

into one transparent result.

6.5.11 Complete Mathematical Framework

The complete Chapter 6 framework can now be represented as:

Human–AI Interaction

↓

BAT Scenarios

↓

Observable AI Behavior

↓

Raw BAT Scores: 0–4 / N/A

↓

Normalization: 0–1

↓

Sub-Factor Scores

↓

U + C + P + R + ER

↓

Weight Distribution

↓

BOND Equation

↓

Overall BOND Score: 0–1

↓

Factor Coverage + Factor Profile

↓

BOND Interval

↓

Complete BOND Evaluation

6.5.12 Chapter 6 Summary

The Mathematical Framework transforms BOND from a conceptual model into a **quantifiable evaluation structure**.

The framework introduces:

Weight Distribution → determines factor contribution.

BOND Equation → combines factor scores.

BAT → converts observable interaction behavior into structured measurements.

Score Calculation → defines the complete numerical pipeline.

Interval System → provides descriptive interpretation of the resulting score.

Importantly, the framework maintains several boundaries:

N/A is separated from failure.

Partial evaluation is separated from complete evaluation.

Factor profiles are preserved alongside the overall score.

Initial equal weights are separated from empirically validated weights.

Descriptive intervals are separated from scientifically validated thresholds.

CHAPTER 6 CORE PRINCIPLE

BOND converts observable Human–AI interaction behavior into a normalized and traceable measurement while preserving the evidence, dimensions, coverage, and mathematical assumptions underlying the final score.

With the mathematical structure established, the next requirement is to design a standardized set of interaction scenarios capable of evaluating the framework consistently across AI systems. This forms the basis of the **BOND Benchmark Framework** developed in Chapter 7.

CHAPTER 7 — BENCHMARK FRAMEWORK

7.1 Benchmark Categories

The BOND mathematical framework defines **what is measured and how scores are calculated**. However, meaningful comparison between AI systems requires another component: a standardized collection of interaction scenarios that systematically creates opportunities to observe the five BOND factors.

For this purpose, BOND introduces the **BOND Benchmark Framework**.

DEFINITION

The BOND Benchmark Framework is a structured collection of standardized Human–AI interaction scenarios designed to create controlled and diverse opportunities for evaluating the five BOND factors through the BOND Assessment Test (BAT).

The benchmark does not simply contain independent questions.

Instead, it contains **interaction scenarios deliberately designed to produce observable evidence of connection-building capability**.

7.1.1 Purpose of the Benchmark

Traditional question-based benchmarks often follow a simple structure:

Question

↓

AI Answer

↓

Correct / Incorrect

This structure is highly useful when the objective is to measure properties such as factual knowledge, reasoning accuracy, or task completion.

BOND evaluates a different target.

It must observe whether an AI can:

Interpret requirements

Communicate appropriately

Use relevant context

Maintain interaction state

Respond to feedback

Adapt when requirements change

Recover after correction

Many of these behaviors cannot be meaningfully observed through isolated questions alone.

Therefore, the BOND Benchmark uses:

Interaction Scenario

↓

Initial AI Response

↓

Human Feedback / Context Change / Constraint / Correction

↓

Continued AI Behavior

↓

BAT Evaluation

The benchmark is designed around interaction behavior rather than question answering alone.

7.1.2 Benchmark Design Structure

The BOND Benchmark is organized into **five primary categories**, corresponding to the five core BOND factors:

Category U — Understanding

Category C — Clarity

Category P — Personalization

Category R — Reliability

Category ER — Engagement & Responsiveness

Each category contains scenarios specifically designed to provide evaluation opportunities for its corresponding sub-factors.

This produces a hierarchical structure:

BOND Benchmark

↓

5 Primary Categories

↓

25 Sub-Factor Targets

↓

Multiple Interaction Scenarios

↓

Observable AI Behavior

↓

BAT Scores

Because Chapter 5 defines five sub-factors for each primary factor, the benchmark contains **25 primary sub-factor targets** in total.

7.1.3 Category U — Understanding

The Understanding category evaluates whether the AI appropriately interprets the human requirement and relevant interaction context.

Its five targets are:

U1 — Requirement Recognition (UR)

U2 — Intent Interpretation (UI)

U3 — Context Integration (UC)

U4 — Constraint Recognition (UCR)

U5 — Clarification & Update Handling (UU)

Scenarios in this category may contain:

Multiple explicit instructions

Implicit but reasonably inferable intent

Relevant information from earlier turns

Formatting or task constraints

Ambiguous references

Human corrections

Updated requirements

For example:

Human:

“Explain renewable energy in four short points for a beginner. Focus only on solar and wind energy.”

This scenario provides opportunities to observe whether the AI recognizes:

Quantity → Four points

Audience → Beginner

Scope → Solar and wind only

Task → Explanation

A response discussing nuclear and hydroelectric energy may contain correct information while still demonstrating weak Requirement or Constraint Recognition.

7.1.4 Category C — Clarity

The Clarity category evaluates whether the AI communicates its response in an understandable and usable form.

Its five targets are:

C1 — Comprehensibility (CC)

C2 — Structural Organization (CS)

C3 — Precision (CP)

C4 — Relevance & Focus (CR)

C5 — Explanatory Sufficiency (CE)

Scenarios may require the AI to:

Explain a difficult concept to a beginner

Organize complex information

Distinguish between similar concepts

Communicate uncertainty precisely

Remove irrelevant information

Provide sufficient explanation without unnecessary expansion

For example:

Human:

“Explain neural networks to someone who understands basic computers but has never studied machine learning.”

The benchmark does not simply test whether the AI knows what a neural network is.

It observes whether the explanation is **appropriate, comprehensible, focused, structured, and sufficiently complete** for the stated context.

7.1.5 Category P — Personalization

The Personalization category evaluates whether the AI appropriately adapts interaction using relevant available context.

Its five targets are:

P1 — Relevant Context Recognition (PCR)

P2 — User-Level Adaptation (PLA)

P3 — Preference Adaptation (PPA)

P4 — Goal & Situation Adaptation (PGS)

P5 — Selective Context Use (PSC)

These scenarios often require multiple turns because personalization requires contextual information to become available before adaptation can be observed.

For example:

Turn 1 — Human:

“I am completely new to programming, and examples help me understand.”

Turn 2 — Human:

“Explain Python loops.”

A strong response should appropriately use both relevant pieces of information:

Knowledge level → Beginner

Preference → Examples are useful

The benchmark may also intentionally provide irrelevant information to determine whether the AI can avoid **forced or unnecessary personalization**.

7.1.6 Category R — Reliability

The Reliability category evaluates whether effective AI behavior remains dependable as interaction develops.

Its five targets are:

R1 — Behavioral Consistency (RBC)

R2 — Contextual Continuity (RCC)

R3 — Constraint Persistence (RCP)

R4 — Uncertainty Calibration (RUC)

R5 — Error Recovery (RER)

Reliability scenarios may intentionally create conditions involving:

Repeated related questions

Persistent constraints

Earlier established facts

Incomplete information

Human corrections

Changes requiring justified updating

For example:

Turn 1 — Human:

“For this entire task, use metric units.”

Several turns later, the AI may receive a new calculation request.

The benchmark can observe whether the established constraint remains active.

Another scenario may intentionally provide insufficient information and evaluate whether the AI communicates uncertainty rather than inventing unsupported certainty.

7.1.7 Category ER — Engagement & Responsiveness

The Engagement & Responsiveness category evaluates whether the AI remains appropriately responsive to the developing interaction.

Its five targets are:

ER1 — Interaction Signal Recognition (ERS)

ER2 — Feedback Responsiveness (ERF)

ER3 — Progression Awareness (ERP)

ER4 — Adaptive Continuation (ERA)

ER5 — Goal-Oriented Engagement (ERG)

Scenarios may involve:

Human confusion

Positive or negative feedback

Changes in task direction

Progress through multiple stages

Requests for alternative explanations

Task completion signals

For example:

Human:

“I understand the first two steps now. Only explain Step 3 again.”

A responsive AI should focus on **Step 3**.

Repeating the complete explanation may demonstrate weaker progression awareness and feedback responsiveness.

7.1.8 Primary and Secondary Evaluation Targets

Real Human–AI behaviors are not perfectly isolated.

A scenario designed primarily for Understanding may also provide evidence about Clarity or Reliability.

Therefore, every benchmark scenario can contain:

Primary Target → The main sub-factor deliberately tested.

Secondary Target(s) → Additional sub-factors for which the scenario provides meaningful evidence.

For example:

Scenario: Human corrects an AI's misunderstanding.

Primary Target: Error Recovery (RER)

Secondary Targets: Clarification & Update Handling (UU), Feedback Responsiveness (ERF)

However, secondary scoring should occur only when the scenario provides a **clear and predefined evaluation opportunity**.

This prevents evaluators from scoring every possible dimension from every response.

7.1.9 Single-Turn and Multi-Turn Categories

Not every BOND behavior requires the same interaction length.

The benchmark therefore supports both:

Single-Turn Scenarios

and

Multi-Turn Scenarios

Single-turn scenarios are particularly useful for evaluating behaviors such as:

Requirement Recognition

Constraint Recognition

Comprehensibility

Structural Organization

Precision

Multi-turn scenarios become particularly important for:

Context Integration

Preference Adaptation

Contextual Continuity

Constraint Persistence

Error Recovery

Feedback Responsiveness

Progression Awareness

The benchmark should therefore contain an appropriate mixture of both formats.

Interaction length should be determined by the behavior being measured, not by an arbitrary requirement that every scenario contain the same number of turns.

7.1.10 Domain Diversity

An interaction benchmark can accidentally measure domain knowledge instead of interaction capability if every scenario depends heavily on specialized expertise.

To reduce this risk, BOND scenarios should be distributed across multiple domains.

Potential domains include:

Education

General Knowledge

Science

Technology

Everyday Planning

Writing Assistance

Customer Support

Basic Problem-Solving

Research Assistance

The purpose of domain diversity is not to create another general-knowledge benchmark.

Instead, it helps determine whether connection-building behavior remains observable across different interaction contexts.

7.1.11 Difficulty Levels

Benchmark scenarios may also vary in interaction complexity.

A proposed three-level structure is:

Level 1 — Basic Interaction

Contains explicit requirements and relatively simple interaction conditions.

Level 2 — Contextual Interaction

Requires the AI to integrate multiple pieces of information, preferences, constraints, or earlier context.

Level 3 — Dynamic Interaction

Requires the AI to respond to changes, ambiguity, corrections, conflicting signals, or multi-turn progression.

For example:

Level 1:

“Explain gravity in three short sentences.”

Level 2:

“I already understand force. Explain gravity using that knowledge and keep it under 100 words.”

Level 3:

After an initial explanation, the human changes the requested level, corrects a misunderstanding, and asks the AI to continue from the updated interaction state.

Difficulty therefore reflects **interaction complexity**, not merely academic difficulty.

7.1.12 Benchmark Category Matrix

The complete category structure can be summarized as:

Category	Primary Factor	Sub-Factor Targets	Typical Interaction Type
U	Understanding	5	Single + Multi-Turn
C	Clarity	5	Primarily Single + Contextual
P	Personalization	5	Primarily Multi-Turn
R	Reliability	5	Primarily Multi-Turn
ER	Engagement & Responsiveness	5	Primarily Multi-Turn
Total	5 Factors	25 Targets	Mixed

This matrix provides the structural foundation of the BOND Benchmark.

7.1.13 Scenario Identification System

For benchmark organization, each scenario can receive a standardized identifier.

For example:

BAT-U-UR-01

where:

BAT → **BOND Assessment Test**

U → **Understanding Category**

UR → **Requirement Recognition**

01 → **Scenario Number**

Similarly:

BAT-C-CC-03

would represent:

Clarity → **Comprehensibility** → **Scenario 03**

and:

BAT-R-RER-02

would represent:

Reliability → **Error Recovery** → **Scenario 02**

This identification system allows scenarios to be easily tracked across:

Benchmark versions

Experimental datasets

Evaluator sheets

Statistical analysis

Published results

7.1.14 Benchmark Balance

A scientifically useful benchmark should not heavily overrepresent one factor while barely evaluating another.

Therefore, the benchmark should aim for reasonable balance across:

Five primary factors

Twenty-five sub-factor targets

Single-turn and multi-turn interactions

Difficulty levels

Interaction domains

Balance does not necessarily require every category to contain exactly the same number of scenarios.

Some behaviors may require more observations to obtain reliable measurements.

However, any imbalance should be **deliberate and documented rather than accidental**.

7.1.15 Benchmark Versioning

The BOND Benchmark may evolve as scenarios are tested, weaknesses are identified, and validation evidence becomes available.

Therefore, each benchmark release should have a version identifier.

For example:

BOND Benchmark v1.0

Future revisions might become:

v1.1 → Minor scenario/rubric improvements

v2.0 → Major structural or scoring changes

A reported result should therefore include the benchmark version:

BOND Score: 0.82

Benchmark: BOND v1.0

Configuration: Standard Equal Weight

This ensures that scores remain traceable to the exact evaluation framework used.

BENCHMARK CATEGORY PRINCIPLE

The BOND Benchmark organizes standardized interaction scenarios around five primary factors and twenty-five sub-factor targets, using controlled variation in interaction length, complexity, and domain to create meaningful opportunities for observing AI connection-building behavior.

The category structure defines **what types of behavior the benchmark must test**. The next requirement is determining **how individual benchmark questions and interaction scenarios should be designed**, which forms the basis of **BOND Question Design**.

7.2 Question Design

The quality of the BOND Benchmark depends strongly on the design of its interaction scenarios. If a scenario does not create a clear opportunity to demonstrate the intended behavior, the resulting score may reflect **question ambiguity, domain knowledge, or evaluator interpretation** rather than the BOND factor itself.

BOND therefore requires a structured approach to **Question Design**.

Within this framework, the term *question* may refer to a direct question, instruction, contextual prompt, feedback message, correction, or a complete multi-turn interaction sequence.

DEFINITION

BOND Question Design is the systematic process of constructing interaction scenarios that create controlled, observable, and scorable opportunities for evaluating specific BOND factors and sub-factors.

7.2.1 Primary Design Objective

Every benchmark scenario should begin with one central question:

“Which observable AI behavior is this scenario intended to evaluate?”

The scenario should then be constructed around that target.

For example, if the target is **Constraint Recognition (UCR)**, the scenario must contain one or more explicit constraints.

If the target is **Error Recovery (RER)**, the interaction must create a meaningful opportunity for the AI to recognize and recover from an identified error.

Therefore:

Target Sub-Factor

↓

Required Interaction Condition

↓

Scenario Design

↓

Observable AI Behavior

↓

BAT Evaluation

This target-first approach reduces the risk of creating questions that appear interesting but provide little useful measurement evidence.

7.2.2 Scenario Design Template

Each BOND Benchmark scenario should contain a standardized design record.

A proposed template is:

Scenario ID

Primary Factor

Primary Sub-Factor

Secondary Target(s), if applicable

Interaction Domain

Difficulty Level

Interaction Type

Initial Context

Human Input

Planned Follow-Up Turn(s)

Evaluation Opportunity

Expected Behavioral Criteria

Scoring Rubric

N/A Conditions

This information defines both **what the AI receives** and **how its behavior should later be evaluated**.

7.2.3 One Primary Measurement Target

A scenario may produce evidence for several BOND dimensions, but it should normally have **one clearly defined primary target**.

For example:

Human:

“Give me exactly four differences between solar and wind energy in a table.”

This scenario may reveal information about:

Requirement Recognition

Constraint Recognition

Clarity

Structural Organization

However, if the primary target is **Constraint Recognition**, the scoring criteria should be designed specifically around whether the AI recognizes:

Exactly four differences

and

Table format

Secondary evidence may be recorded separately when predefined.

A scenario should not attempt to measure everything simply because multiple behaviors are visible.

This improves interpretability of benchmark results.

7.2.4 Explicit Requirement Questions

Some scenarios should test whether the AI can correctly identify clearly stated requirements.

Example:

Human:

“Summarize the following idea in three bullet points. Each point must contain no more than one sentence.”

The scenario contains several explicit requirements:

Task → Summarization

Quantity → Three

Format → Bullet points

Length Constraint → One sentence per point

The expected behavior is directly observable.

Such scenarios are particularly useful for:

Requirement Recognition

Constraint Recognition

Structural Organization

7.2.5 Intent-Based Questions

Other scenarios should evaluate whether the AI can interpret meaning that is not expressed as a complete direct instruction but is reasonably supported by the interaction.

Example:

AI: Provides a highly technical explanation.

Human:

“That's way above my level.”

The human does not explicitly state:

“Rewrite the explanation using simpler language.”

However, the interaction provides strong evidence that simplification is required.

The scenario can therefore evaluate **Intent Interpretation** and potentially **Feedback Responsiveness**.

Intent-based scenarios should avoid requiring speculative mind-reading.

The intended interpretation must be reasonably supported by the interaction evidence.

7.2.6 Context-Dependent Questions

Some BOND capabilities become visible only when information from earlier interaction remains relevant.

Example:

Turn 1 — Human:

“I am preparing a five-minute presentation for students who have never studied artificial intelligence.”

Turn 2 — AI:

Acknowledges the context.

Turn 3 — Human:

“Now explain neural networks for the presentation.”

A context-aware AI should recognize that the later request remains constrained by:

Presentation duration

Beginner audience

Presentation purpose

The scenario can therefore test whether earlier information appropriately influences later behavior.

7.2.7 Controlled Ambiguity

Real Human–AI Interaction sometimes contains insufficient information.

BOND should therefore include scenarios containing **deliberate but controlled ambiguity**.

Example:

Human:

“Compare the second one with the previous model.”

If the interaction does not establish what “the second one” refers to, a reliable response may require clarification.

The objective is not to trick the AI through meaningless wording.

Instead, the scenario evaluates whether the system can recognize when **sufficient interpretation is not possible from available evidence**.

Controlled ambiguity can provide evidence for:

Clarification & Update Handling

Uncertainty Calibration

Interaction Signal Recognition

7.2.8 Constraint-Based Questions

Constraints provide highly observable interaction requirements.

Potential constraint types include:

Quantity Constraints

“Give exactly five examples.”

Length Constraints

“Keep the answer below 100 words.”

Format Constraints

“Answer using a table.”

Content Constraints

“Discuss only advantages.”

Language Constraints

“Use simple English.”

Process Constraints

“Explain the intuition first and the equation second.”

Scenarios may contain one or multiple constraints depending on difficulty level.

However, excessive numbers of constraints should be avoided when they make the task unrealistic or primarily test memory load.

7.2.9 Personalization Questions

Personalization scenarios require relevant contextual information before adaptation can be evaluated.

A basic structure is:

Context Establishment

↓

Relevant User Information

↓

Later Requirement

↓

Opportunity for Adaptation

Example:

Human:

“I understand concepts better through practical examples.”

Later:

Human:

“Explain machine learning overfitting.”

The benchmark then observes whether the AI appropriately incorporates the stated preference.

Some personalization scenarios should also contain **irrelevant contextual information**.

This tests whether the AI can distinguish:

Relevant Personalization

from

Forced Personalization

7.2.10 Reliability Questions

Reliability generally requires behavior to be observed across multiple interaction points.

A reliability scenario may establish information early:

Turn 1:

“For all calculations in this task, use 9.8 m/s^2 for gravitational acceleration.”

Several turns later:

Turn 5:

The AI receives another relevant calculation.

The benchmark then observes whether the active condition remains preserved.

Reliability questions may test:

Behavioral stability

Context retention

Constraint persistence

Appropriate uncertainty

Error correction

The distance between establishment and later testing can also be varied to increase interaction complexity.

7.2.11 Feedback-Based Questions

Feedback is essential for evaluating responsiveness.

A scenario may deliberately provide feedback such as:

“That explanation was too complicated.”

“You misunderstood my question.”

“The first part helped, but the second part didn't.”

“Use another example.”

The next AI response provides observable evidence about whether the feedback was:

Recognized

Interpreted

Applied

The benchmark should evaluate the **quality of behavioral change**, not merely whether the AI acknowledges the feedback verbally.

For example:

“I understand.”

followed by the same unsuitable explanation does not represent strong feedback responsiveness.

7.2.12 Correction-Based Questions

Correction scenarios evaluate whether an AI appropriately updates its interaction state.

Example:

Human:

“I gave you the wrong value earlier. The resistance is 20 Ω , not 10 Ω .”

The subsequent response should use the corrected value where relevant.

Such scenarios can provide evidence for:

Clarification & Update Handling

Contextual Continuity

Error Recovery

Feedback Responsiveness

The correction should be unambiguous enough that failure can reasonably be attributed to AI behavior rather than scenario uncertainty.

7.2.13 Progression-Based Questions

Some interactions naturally develop through stages.

For example:

Stage 1 → Define the problem

Stage 2 → Explain the concept

Stage 3 → Apply the concept

Stage 4 → Evaluate the result

A progression scenario observes whether the AI recognizes the current stage instead of unnecessarily returning to completed stages.

Example:

Human:

“Okay, I understand why the formula works. Now show me how to apply it.”

The AI should progress toward application rather than restart the theoretical explanation.

These scenarios are particularly useful for **Progression Awareness and Goal-Oriented Engagement**.

7.2.14 Difficulty Through Interaction Complexity

BOND scenario difficulty should primarily reflect the complexity of interaction rather than the academic difficulty of the subject.

A proposed structure is:

Level 1 — Direct

One clear requirement with limited contextual dependency.

Level 2 — Contextual

Multiple relevant requirements, preferences, or earlier interaction information.

Level 3 — Dynamic

Changing requirements, corrections, ambiguity, feedback, or longer interaction dependencies.

For example:

Level 1

“Explain photosynthesis in three points.”

Level 2

“I already know what chlorophyll is. Explain photosynthesis in three points without redefining chlorophyll.”

Level 3

After the initial explanation, the human identifies a misunderstood section, changes the desired level of detail, and asks the AI to continue without repeating already understood material.

Thus:

BOND difficulty represents increasing interaction-management demand rather than simply harder subject matter.

7.2.15 Avoiding Knowledge Dominance

A BOND scenario should not require obscure knowledge unless that knowledge is necessary for the intended evaluation.

For example, testing Clarity using an extremely specialized scientific topic could create ambiguity:

Did the AI receive a poor Clarity score because communication was unclear?

or

Did the AI simply lack the required domain knowledge?

Where possible, scenarios should use topics for which the underlying information is **widely accessible or provided directly within the scenario**.

This helps isolate interaction behavior from unrelated capability differences.

7.2.16 Avoiding Leading Questions

Questions should not unnecessarily reveal exactly what behavior is being evaluated.

For example, if the benchmark wants to evaluate whether an AI recognizes ambiguity:

Weak Design:

“If you are unsure what I mean, ask me to clarify.”

This directly tells the AI what behavior is expected.

A stronger design may simply provide an interaction in which a necessary reference is genuinely ambiguous.

The AI must then independently determine whether clarification is required.

7.2.17 Avoiding Artificial Traps

The benchmark should challenge AI systems without becoming a collection of tricks.

Scenarios should avoid:

Meaningless contradictions

Impossible hidden assumptions

Deliberately deceptive wording unrelated to realistic interaction

Arbitrary formatting traps

Requirements no reasonable user would provide

The objective is to evaluate **useful Human–AI interaction capability**, not the AI's ability to survive puzzle-like prompt traps.

7.2.18 Scenario Independence and Contamination

When multiple benchmark scenarios are evaluated, information from one scenario should not unintentionally influence another unless the benchmark explicitly designs a continuous interaction.

Therefore, independent scenarios should begin from a **defined clean interaction state**.

This reduces contamination from:

Previous scenario context

Earlier preferences

Unrelated instructions

Accumulated conversation history

For controlled multi-turn scenarios, only the context deliberately included in that scenario should influence scoring.

7.2.19 Question Diversity

A benchmark consisting of many nearly identical prompts may produce high apparent consistency while testing only a narrow behavior pattern.

Therefore, each sub-factor should ideally be evaluated through **multiple scenario forms**.

For example, Constraint Recognition may be tested through:

Length constraint

Format constraint

Content restriction

Multiple simultaneous constraints

Persistent multi-turn constraint

This provides broader evidence than repeating the same prompt structure with different topics.

7.2.20 Scenario Design Matrix

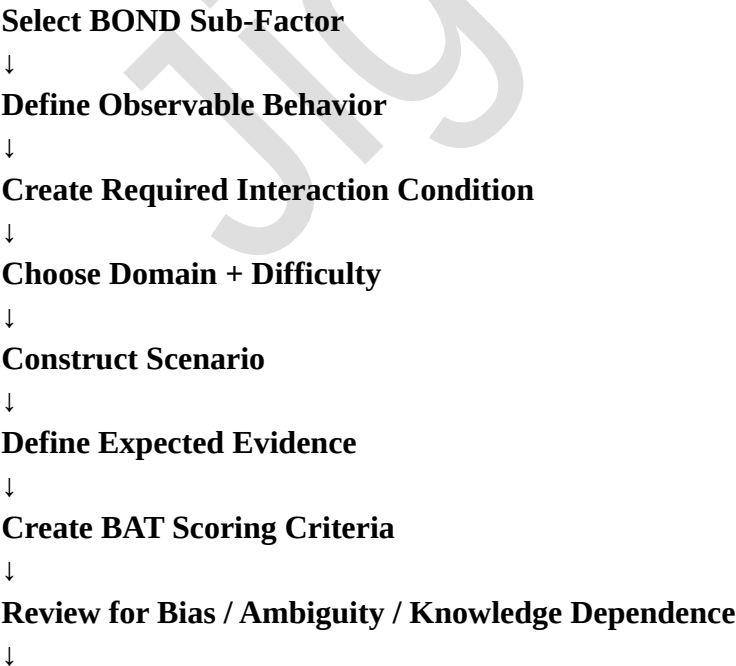
A scenario design record can be summarized as:

Component	Required Information
Scenario ID	Unique benchmark identifier
Primary Target	Main BOND sub-factor
Secondary Target	Optional predefined additional target
Domain	Interaction context
Difficulty	Level 1, 2, or 3
Interaction Type	Single-turn or multi-turn
Context	Information available before evaluation
Human Input	Question/instruction/feedback
Evaluation Opportunity	Behavior the scenario makes observable
Scoring Criteria	Conditions for BAT scoring
N/A Condition	When valid evaluation is unavailable

This structure ensures that benchmark questions are designed as **measurement instruments rather than arbitrary prompts**.

7.2.21 Question Design Pipeline

The complete design process can be represented as:



Pilot Test

↓

Revise if Required

↓

Add to Benchmark

QUESTION DESIGN PRINCIPLE

Every BOND Benchmark scenario should be constructed from a predefined measurement target, create a realistic and sufficient opportunity for that behavior to become observable, and minimize unrelated influences that could distort the resulting score.

Question Design establishes **how valid evaluation opportunities are created**. The next stage defines how these scenarios are administered, scored, aggregated, and compared through the **BOND Evaluation Process**.

7.3 Evaluation Process

The BOND Benchmark requires a standardized evaluation process so that results obtained from different AI systems, evaluators, or experiments remain as **consistent, transparent, and comparable as possible**.

The Evaluation Process defines how benchmark scenarios are administered, how AI responses are recorded, how BAT scores are assigned, and how the resulting observations are converted into a complete BOND evaluation.

DEFINITION

The BOND Evaluation Process is the standardized procedure through which benchmark scenarios are administered to an AI system, observable interaction behavior is recorded and scored using BAT criteria, and the resulting measurements are aggregated into sub-factor, factor-level, and overall BOND results.

7.3.1 Evaluation Pipeline

The complete process follows eight primary stages:

1. Evaluation Setup

↓

2. Scenario Administration

↓

3. Interaction Recording

↓

4. BAT Scoring

↓

5. Sub-Factor Aggregation

↓

6. Factor Calculation

↓

7. Overall BOND Calculation

↓

8. Result Reporting

Each stage should be documented sufficiently to allow the evaluation procedure to be understood and, where possible, reproduced.

7.3.2 Stage 1 — Evaluation Setup

Before benchmark testing begins, the evaluator should define the evaluation configuration.

This includes:

AI system or model being evaluated

Model/version identifier where available

BOND Benchmark version

BAT rubric version

Weighting configuration

Scenario set

Evaluation date

Relevant system settings

Number of evaluators

Interaction conditions

For systems with configurable generation settings, relevant settings should also be recorded where accessible.

The purpose is simple:

A BOND score should describe not only the result, but also the conditions under which the result was obtained.

This becomes particularly important because AI systems may change over time.

7.3.3 Controlled Evaluation Conditions

When multiple AI systems are compared, the benchmark conditions should remain as similar as reasonably possible.

For example:

Same scenario wording

Same scenario order policy

Same interaction-turn structure

Same scoring rubric

Same weighting configuration

Comparable system instructions where controllable

Same benchmark version

If System A receives additional hints while System B does not, the resulting scores may no longer represent a fair comparison.

Therefore:

Different AI System

should ideally be the primary changing variable while other relevant evaluation conditions remain controlled.

7.3.4 Clean Scenario State

Independent benchmark scenarios should begin from a defined clean interaction state.

This prevents information from earlier scenarios from influencing later performance.

For example, suppose Scenario 1 establishes:

“I prefer very short answers.”

If Scenario 2 is intended to independently evaluate Clarity, retaining that preference may unintentionally influence the response.

Therefore:

Independent Scenario → Fresh Defined Context

Multi-Turn Scenario → Preserve Only Intended Scenario Context

This separation improves experimental control.

7.3.5 Stage 2 — Scenario Administration

Each benchmark scenario should be presented according to its predefined structure.

The evaluator should avoid modifying the scenario during testing unless the scenario itself specifies a dynamic follow-up rule.

For a single-turn scenario:

Defined Human Input → AI Response

For a multi-turn scenario:

Defined Context

↓

Turn 1

↓

AI Response

↓

Predefined or Rule-Based Follow-Up

↓

AI Response

↓

Additional Turn(s), if required

The evaluator should not provide extra assistance simply because the AI performs poorly.

Doing so would alter the evaluation condition.

7.3.6 Fixed and Conditional Follow-Ups

Multi-turn scenarios may use two types of follow-up messages.

Fixed Follow-Up

The next human message is predetermined regardless of the AI response.

Example:

Turn 1: Ask for an explanation.

Turn 2: “That was too technical. Explain it more simply.”

This is useful when all evaluated systems should receive exactly the same interaction signal.

Conditional Follow-Up

The next message depends on a predefined property of the AI's response.

For example:

If the AI asks for clarification → provide Context A

If the AI assumes an interpretation → provide Correction B

Conditional follow-ups should be defined **before testing**.

This prevents evaluators from improvising different interactions for different systems.

7.3.7 Stage 3 — Interaction Recording

The complete interaction should be preserved.

A scenario record should include:

Scenario ID

Human messages

AI responses

Turn order

Relevant context

Evaluator annotations

Scoring outcome

Where technically possible, the original response should be preserved exactly rather than summarized by the evaluator.

This creates an evidence trail from:

Final Score → Evaluator Judgment → AI Response → Benchmark Scenario

Such traceability is essential for reviewing disputed scores or improving the benchmark later.

7.3.8 Stage 4 — BAT Scoring

After the interaction is recorded, the relevant AI behavior is evaluated using the BAT rubric.

For each predefined target:

Raw BAT Score = 0, 1, 2, 3, or 4

or:

N/A = Insufficient Evaluation Opportunity

The evaluator should base the score on:

Scenario conditions

Target sub-factor definition

Observable behavior

Predefined scoring criteria

The evaluator should not base the score primarily on whether they personally liked the response.

7.3.9 Evidence-Based Scoring

Every score should ideally be connected to observable evidence.

For example:

Target: Constraint Recognition

Requirement: Exactly three bullet points.

AI Output: Five paragraphs.

Raw Score: Low

Evidence: Quantity and format constraints were not followed.

This is stronger than recording:

“The response did not feel very good.”

BOND therefore favors:

Behavioral Evidence → Scoring Decision

rather than:

General Impression → Scoring Decision

This improves transparency and supports later evaluator agreement analysis.

7.3.10 Scoring Only Predefined Opportunities

An evaluator should not automatically score all 25 sub-factors from every scenario.

Suppose a scenario contains no relevant user preference.

It should not automatically generate a Preference Adaptation score.

Likewise, a single-turn interaction may provide no meaningful opportunity to evaluate long-term Contextual Continuity.

Therefore:

Predefined Evaluation Opportunity → Score

No Sufficient Opportunity → N/A

This prevents the benchmark from creating artificial evidence.

7.3.11 Independent Evaluation

When multiple human evaluators are used, they should preferably score responses **independently before discussing disagreements**.

For example:

Evaluator A → Independent Score

Evaluator B → Independent Score

Evaluator C → Independent Score

↓

Agreement Analysis

↓

Resolution / Aggregation Procedure

If evaluators discuss responses before scoring, their judgments may influence one another and artificially increase apparent agreement.

7.3.12 Blind Evaluation

Where feasible, evaluators should not know which AI system generated a response.

Instead of displaying:

System: Model X

the evaluation interface may display:

System ID: A17

This can reduce potential bias based on:

Brand reputation

Previous experience

Expected model capability

Personal preference

Blind evaluation may not always be technically possible, particularly when model-specific response styles reveal identity.

Nevertheless, explicit model labels should be hidden where practical.

7.3.13 Evaluator Disagreement

Human judgment will not always be identical.

Suppose three evaluators assign:

Evaluator A = 3

Evaluator B = 3

Evaluator C = 2

The disagreement is relatively small.

However:

Evaluator A = 4

Evaluator B = 0

Evaluator C = 2

indicates substantial disagreement.

Such disagreement should not simply be ignored.

It may indicate:

Ambiguous scenario design

Insufficient rubric specificity

Different interpretation of the sub-factor

Evaluator inconsistency

Genuinely borderline AI behavior

Evaluator disagreement is therefore not merely noise; it can provide useful information about the **quality of the measurement instrument itself**.

7.3.14 Aggregating Multiple Evaluators

When multiple evaluators score the same observation, an initial aggregation method can use the arithmetic mean.

For example:

Evaluator A = 4

Evaluator B = 3

Evaluator C = 3

Then:

Mean Raw Score = $(4 + 3 + 3) / 3$

Mean Raw Score = 3.33

The normalized score becomes:

Normalized Score = $3.33 / 4$

Normalized Score ≈ 0.83

The benchmark should preserve the individual evaluator scores for later reliability analysis rather than storing only the average.

Future validation may determine whether another aggregation method is preferable.

7.3.15 Inter-Rater Reliability

Because BAT includes human judgment, BOND validation should examine whether different evaluators can apply the scoring rubric consistently.

Appropriate statistical measures can be selected according to the final study design and data structure.

The objective is to determine whether:

Different trained evaluators observing the same AI behavior produce sufficiently consistent judgments.

If agreement is consistently poor, the problem may lie in:

Factor definitions

Scenario design

Rubric wording

or

Evaluator training

rather than in the AI systems being evaluated.

Therefore, evaluator reliability forms an important component of BOND validation.

7.3.16 Stage 5 — Sub-Factor Aggregation

After individual scenarios are scored and normalized, observations targeting the same sub-factor are aggregated.

For example, suppose Requirement Recognition is tested across four scenarios:

Scenario 1 = 1.00

Scenario 2 = 0.75

Scenario 3 = 1.00

Scenario 4 = 0.75

Then:

$$UR = (1.00 + 0.75 + 1.00 + 0.75) / 4$$

$$UR = 3.50 / 4$$

$$UR = 0.875$$

For standard reporting:

$$UR = 0.88$$

This produces the aggregated sub-factor measurement.

7.3.17 Stage 6 — Primary Factor Calculation

The relevant sub-factor scores are then combined to calculate the five primary factors:

Understanding (U)

Clarity (C)

Personalization (P)

Reliability (R)

Engagement & Responsiveness (ER)

Under the initial equal sub-factor configuration:

Factor Score = Sum of Applicable Sub-Factor Scores / Number of Applicable Sub-Factors

N/A values are excluded according to the opportunity-adjustment rules established in Chapter 6.

7.3.18 Stage 7 — Overall BOND Calculation

After the five factor scores are available, the overall BOND score is calculated.

Under Standard Equal-Weight BOND:

$$B = (U + C + P + R + ER) / 5$$

When one or more primary factors lack sufficient evaluation opportunity, the Opportunity-Adjusted BOND Equation is used.

The final calculation should also include:

Factor Coverage

Weighting Configuration

Benchmark Version

These details prevent the numerical score from becoming detached from its measurement conditions.

7.3.19 Stage 8 — Result Reporting

A standardized benchmark report should contain more than the overall score.

A proposed report includes:

Measurement	Reported Information
Overall BOND	Normalized score + percentage
Interval	Descriptive BOND interval
Understanding	Factor score
Clarity	Factor score
Personalization	Factor score
Reliability	Factor score
Engagement & Responsiveness	Factor score
Factor Coverage	Percentage
Benchmark Version	Version identifier
BAT Version	Rubric identifier
Weight Configuration	Standard or domain-specific
Evaluation Conditions	Relevant testing information

For research use, scenario-level and sub-factor-level results should also be preserved.

7.3.20 Comparing Multiple AI Systems

When comparing AI systems, each system should ideally receive:

The same benchmark version

The same scenario set

Equivalent interaction conditions

The same scoring rubric

The same weighting configuration

Comparable evaluator procedures

A comparison table may then take the form:

System	U	C	P	R	ER	BOND
System A	0.84	0.88	0.71	0.80	0.86	0.82
System B	0.79	0.83	0.77	0.75	0.80	0.79
System C	0.90	0.85	0.64	0.88	0.84	0.82

This demonstrates why factor profiles remain important.

System A and System C may obtain similar overall scores while exhibiting different strengths and weaknesses.

7.3.21 Repeated Evaluation

AI outputs can vary between runs depending on system design and generation conditions.

Therefore, a single execution of every scenario may not fully characterize performance.

Where experimental resources permit, selected scenarios or the complete benchmark may be repeated.

This allows researchers to examine:

Run-to-run variation

Score stability

Response consistency

Sensitivity to interaction conditions

The final experimental design should determine the number of repetitions required for reliable estimates.

7.3.22 Reproducibility Record

For a BOND experiment to be reproducible, researchers should preserve as much of the following as possible:

Benchmark version

Complete scenario wording

Scenario order or randomization method

AI model/version

Relevant system configuration

Evaluation date

AI responses

Evaluator instructions

Raw evaluator scores

Normalization procedure

Weight configuration

Aggregation method

This creates a transparent path from the original interaction to the published BOND result.

7.3.23 Evaluation Process Diagram

DEFINE EVALUATION CONFIGURATION

↓

ADMINISTER STANDARDIZED BAT SCENARIOS

↓

RECORD COMPLETE AI INTERACTIONS

↓

IDENTIFY PREDEFINED EVALUATION OPPORTUNITIES

↓

ASSIGN BAT SCORES: 0–4 / N/A

↓

NORMALIZE SCORES

↓

AGGREGATE SCENARIO RESULTS

↓

CALCULATE SUB-FACTORS

↓

CALCULATE $U + C + P + R + ER$

↓

APPLY BOND EQUATION

↓

CALCULATE COVERAGE + INTERVAL

↓

REPORT COMPLETE BOND PROFILE

EVALUATION PROCESS PRINCIPLE

A BOND evaluation should preserve standardized interaction conditions, predefined measurement opportunities, evidence-based scoring, evaluator independence, mathematical traceability, and sufficient reporting information to support meaningful interpretation and comparison.

The Evaluation Process establishes **how the benchmark is administered and scored**. The final component of the Benchmark Framework defines the **Quality Rules** that determine whether scenarios and evaluation procedures are sufficiently strong to be included in a standardized BOND Benchmark.

7.4 Benchmark Quality Rules

A benchmark can contain a large number of scenarios and still produce weak measurements if those scenarios are ambiguous, repetitive, biased, unrealistic, or poorly aligned with the intended construct.

For this reason, the BOND Benchmark requires a set of **Quality Rules** governing the design, review, inclusion, administration, and revision of benchmark scenarios.

These rules are intended to ensure that benchmark performance reflects **observable AI connection-building behavior as accurately as possible**, rather than accidental characteristics of individual prompts.

DEFINITION

BOND Benchmark Quality Rules are the design and evaluation requirements used to maintain scenario validity, consistency, fairness, interpretability, and reproducibility across the BOND Benchmark Framework.

7.4.1 Rule 1 — Every Scenario Must Have a Defined Target

Every BOND scenario must specify a primary factor and sub-factor before it enters the benchmark.

For example:

Primary Factor → Understanding

Primary Sub-Factor → Constraint Recognition

The scenario should then create an observable opportunity for that behavior.

The design direction should therefore be:

Measurement Target → Scenario

rather than:

Interesting Scenario → Decide Later What It Measures

This prevents benchmark construction from becoming an arbitrary collection of prompts.

7.4.2 Rule 2 — Sufficient Evaluation Opportunity

A scenario should provide sufficient opportunity for the target behavior to become observable.

For example, a scenario cannot meaningfully evaluate **Feedback Responsiveness** if the human never provides feedback.

Similarly, **Constraint Persistence** cannot be properly tested unless a constraint is established and later remains relevant.

Therefore:

No meaningful opportunity → No valid score

If an intended opportunity fails to occur during an interaction, the corresponding observation should be classified as **N/A** rather than artificially scored.

7.4.3 Rule 3 — Observable Scoring Evidence

The expected behavior should be observable from the interaction.

A benchmark should avoid criteria requiring unsupported assumptions about the AI's internal state.

For example, BOND can evaluate whether an AI:

recognized a stated constraint in its response,

used relevant earlier context,

changed behavior after feedback,

or

communicated uncertainty.

BOND should not claim to directly measure whether the AI internally *felt, understood, or intended* something in a human psychological sense.

The framework evaluates **demonstrated interaction behavior**.

7.4.4 Rule 4 — Scenario Clarity

Scenario wording should be sufficiently clear for the intended evaluation.

This does not mean all scenarios must be simple or unambiguous.

Controlled ambiguity may itself be the experimental condition.

However, ambiguity should be:

Intentional

Relevant to the target behavior

Documented

Scorable

Accidental ambiguity can make it impossible to determine whether a poor result reflects AI behavior or weak scenario design.

7.4.5 Rule 5 — Realistic Interaction

BOND scenarios should approximate interactions that could reasonably occur between humans and AI systems.

Examples include:

Asking for explanations

Changing requirements

Providing corrections

Requesting planning assistance

Giving preferences

Asking follow-up questions

Reporting confusion

Continuing multi-step tasks

Artificial situations may occasionally be useful for controlled measurement, but benchmark performance should not depend primarily on puzzle-like prompt traps.

BOND should challenge interaction capability without replacing realistic interaction with trick-question solving.

7.4.6 Rule 6 — Minimize Unrelated Capability Dependence

A scenario designed to measure one BOND behavior should minimize unnecessary dependence on unrelated capabilities.

For example, a Clarity scenario should not require highly specialized knowledge if the necessary information can instead be provided directly.

Otherwise, a poor response may result from lack of domain knowledge rather than weak Clarity.

Therefore, scenario designers should ask:

“Could an AI fail this scenario for a reason unrelated to the BOND behavior we intend to measure?”

If the answer is yes, the scenario should be reviewed or redesigned.

7.4.7 Rule 7 — Avoid Unnecessary Leading

A scenario should not directly tell the AI which evaluated behavior it should demonstrate unless that instruction itself forms part of the intended test.

For example:

Weak ambiguity test:

“If my request is ambiguous, ask me for clarification.”

The expected behavior has already been explicitly provided.

A stronger test creates genuine insufficient information and observes whether the AI independently responds appropriately.

Leading scenarios can artificially inflate performance.

7.4.8 Rule 8 — Avoid Hidden Impossible Requirements

A system should not lose points for failing to satisfy information or requirements that were never reasonably available.

Suppose an evaluator expects the AI to tailor an explanation to a beginner but never provides any indication of the user's knowledge level.

Unless beginner-level communication is otherwise implied by the scenario, penalizing the AI for lack of adaptation would introduce an unsupported expectation.

Therefore:

Evaluation Criteria must be grounded in information available within the scenario.

This principle is especially important for Personalization and Understanding.

7.4.9 Rule 9 — Separate Failure from N/A

BOND must consistently distinguish:

Failure despite sufficient opportunity

from

Insufficient opportunity to evaluate

If the opportunity exists and the AI fails:

BAT Score = 0–3 depending on demonstrated performance

If meaningful opportunity does not exist:

BAT = N/A

N/A must never be automatically converted to zero.

This rule protects both mathematical fairness and conceptual validity.

7.4.10 Rule 10 — One Primary Target per Scenario

Each scenario should normally contain one predefined primary measurement target.

Secondary targets may be included when meaningful, but they should be specified before evaluation.

This prevents:

Post-hoc interpretation

Over-scoring

Uncontrolled overlap

Evaluator inconsistency

A complex scenario may naturally reveal many behaviors, but benchmark scoring should remain structured.

7.4.11 Rule 11 — Multiple Scenarios per Sub-Factor

A sub-factor should not be evaluated from only one narrow scenario whenever benchmark size permits broader testing.

For example, Constraint Recognition can be tested through:

Length constraints

Quantity constraints

Formatting constraints

Scope constraints

Persistent constraints

If an AI succeeds on only one prompt pattern, that does not necessarily establish robust Constraint Recognition.

Multiple scenario forms therefore improve the breadth of evidence.

7.4.12 Rule 12 — Scenario Diversity

Scenario diversity should exist across several dimensions:

Interaction domain

Prompt structure

Difficulty

Single-turn vs. multi-turn

Explicit vs. contextual requirements

Type of interaction signal

This reduces dependence on a narrow benchmark format.

However, diversity should not be added merely for variety.

Every scenario must remain connected to a defined measurement objective.

7.4.13 Rule 13 — Balanced Factor Representation

The benchmark should provide meaningful evidence across all five primary factors:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

A benchmark containing hundreds of Understanding scenarios but only a few Reliability scenarios would produce an unbalanced representation of the proposed framework.

Balance should therefore be monitored at:

Factor level

and

Sub-factor level

Exact equality is not mandatory, but substantial imbalance should be justified.

7.4.14 Rule 14 — Standardized Administration

A benchmark scenario should be administered consistently across compared AI systems.

Researchers should avoid:

Giving one system additional clarification

Changing scenario wording between systems

Providing different feedback

Using different stopping rules

unless those differences are explicitly part of the experimental design.

Standardized administration improves comparability.

7.4.15 Rule 15 — Predefined Conditional Logic

When a scenario requires different follow-up messages depending on the AI's behavior, the decision rules should be specified before evaluation.

For example:

IF AI requests clarification → provide Follow-Up A

IF AI makes an unsupported assumption → provide Follow-Up B

IF AI correctly identifies the issue → continue with Follow-Up C

This allows dynamic interaction while reducing evaluator improvisation.

7.4.16 Rule 16 — Evidence Preservation

The complete interaction evidence used for scoring should be preserved wherever practical.

This includes:

Scenario text

Human turns

AI responses

Evaluator scores

Evaluator notes

Relevant configuration

A numerical result without its underlying evidence is harder to audit, reproduce, or challenge.

Therefore:

BOND should preserve a traceable path from score back to interaction evidence.

7.4.17 Rule 17 — Evaluator Independence

When multiple evaluators are used, each should initially score the interaction independently.

They should not be shown another evaluator's score before making their own judgment.

This reduces:

Anchoring

Group influence

Artificial agreement

After independent scoring, disagreement can be analyzed or resolved according to the defined evaluation procedure.

7.4.18 Rule 18 — Evaluator Training

Before formal benchmark scoring, evaluators should understand:

The five BOND factors

The relevant sub-factors

BAT scoring levels

Difference between 0 and N/A

Scenario-specific criteria

Evidence-based scoring principles

Training may include example responses representing different BAT levels.

However, training examples used for evaluator calibration should be separated from the formal test responses wherever appropriate.

7.4.19 Rule 19 — Reduce Model Identity Bias

Where practical, model identity should be hidden from evaluators.

Responses may be labeled:

System A

System B

System C

instead of displaying commercial or model names.

This reduces the possibility that expectations about a known AI system influence scoring.

7.4.20 Rule 20 — Benchmark Contamination Awareness

Public benchmarks can eventually become part of model training data or optimization targets.

If an AI system has previously encountered exact BOND scenarios, benchmark performance may partly reflect familiarity rather than generalizable interaction capability.

Therefore, benchmark development should consider:

Public scenario sets

Protected evaluation sets

Rotating scenario pools

New scenario generation

Contamination checks where feasible

A future mature BOND implementation may maintain both:

Public Development Set → for understanding and testing the framework

and

Protected Evaluation Set → for formal comparison

This can reduce direct optimization toward known benchmark items.

7.4.21 Rule 21 — Prevent Scenario Duplication

Two scenarios that differ only in superficial wording may not provide genuinely independent evidence.

For example:

“Give exactly three examples.”

and

“Provide exactly three examples.”

may test almost identical behavior.

Scenario review should therefore examine **semantic duplication**, not merely textual duplication.

The objective is to maximize useful behavioral coverage rather than raw question count.

7.4.22 Rule 22 — Avoid Cultural and Linguistic Bias

Human–AI interaction varies across language and cultural context.

Benchmark scenarios should avoid unnecessarily assuming:

One cultural norm

One communication style

One geographic context

One socioeconomic environment

One linguistic convention

unless such context is deliberately part of the scenario.

The initial BOND Benchmark may begin in one language for methodological consistency, but results should not automatically be generalized to all languages and cultures.

Cross-lingual and cross-cultural validation should be treated as an important future research direction.

7.4.23 Rule 23 — Safety Behavior Should Not Be Misclassified

An AI may sometimes decline, redirect, or limit a response because of legitimate safety constraints.

Such behavior should not automatically receive a poor BOND score simply because the system did not comply with the human request.

For example, strong Understanding may involve correctly recognizing what the human requested even when the system cannot appropriately provide it.

Similarly, Clarity may involve clearly explaining an applicable limitation.

Therefore:

BOND evaluates the quality of interaction behavior, not unconditional obedience.

Safety evaluation itself remains outside the primary scope of BOND unless explicitly incorporated into a separate evaluation design.

7.4.24 Rule 24 — Do Not Reward Unnecessary Engagement

An AI should not receive a higher Engagement score simply because it produces longer responses or extends the conversation.

The benchmark should avoid rewarding:

Unnecessary follow-up questions

Repeated offers

Excessive conversational language

Artificial continuation

Unrequested expansion

Engagement is strong when interaction remains **responsive and useful toward the relevant objective**.

Sometimes the best interaction behavior is to provide a concise answer and complete the task.

7.4.25 Rule 25 — Do Not Reward Forced Personalization

Personalization should occur only when relevant context meaningfully improves the interaction.

Suppose a scenario provides a user's favorite color, but the task is solving a mathematical equation.

Mentioning that color without relevance should not increase the Personalization score.

This distinguishes:

Context-Aware Adaptation

from

Superficial Personal Reference

Strong personalization requires **selective relevance**.

7.4.26 Rule 26 — Separate Interaction Quality from Factual Correctness

BOND may sometimes evaluate interactions containing factual tasks.

Factual correctness can influence interaction quality, but BOND should not become a disguised accuracy benchmark.

Consider an AI that produces:

Clear communication

Excellent personalization

Strong responsiveness

but

An incorrect factual conclusion

The factual error may affect relevant BOND dimensions depending on the scenario, particularly Reliability or Understanding, but the framework should not silently substitute a conventional accuracy score for the complete BOND profile.

Where appropriate, factual accuracy may be recorded as a **separate external variable** during validation.

This allows researchers to study the relationship between:

Task Accuracy

and

BOND Connection-Building Performance

rather than assuming they are identical.

7.4.27 Rule 27 — Pilot Testing Before Formal Inclusion

New benchmark scenarios should ideally undergo pilot testing before inclusion in a formal BOND release.

Pilot testing can identify:

Unexpected ambiguity

Scoring disagreement

Ceiling effects

Floor effects

Unintended knowledge requirements

Overlapping targets

Unrealistic interaction patterns

A scenario that repeatedly produces evaluator confusion may require revision even if its original theoretical design appeared strong.

7.4.28 Rule 28 — Scenario Revision Must Be Versioned

Benchmark scenarios may improve over time.

However, changing scenario wording or scoring criteria can affect measured performance.

Therefore, significant benchmark revisions should be documented through versioning.

For example:

BOND Benchmark v1.0

↓

Scenario and rubric improvements

↓

BOND Benchmark v1.1

Major methodological changes may justify:

BOND Benchmark v2.0

Researchers should report the version used so that results from different releases are not unknowingly treated as identical.

7.4.29 Rule 29 — Quality Over Benchmark Size

A benchmark containing 1,000 weak or repetitive scenarios is not automatically better than one containing 200 carefully designed scenarios.

Benchmark size should be determined by:

Measurement coverage

Reliability

Scenario diversity

Statistical requirements

Evaluation cost

rather than by maximizing the number of questions.

The objective of BOND is not to create the largest benchmark. It is to create sufficient high-quality evidence for meaningful measurement.

7.4.30 Rule 30 — Results Must Remain Auditable

A reported BOND result should provide enough methodological information for researchers to understand how it was obtained.

At minimum, formal reporting should identify:

Benchmark version

AI system/version

Weighting configuration

Factor coverage

Evaluation procedure

Evaluator structure

Score aggregation method

Where possible, researchers should additionally preserve or release scenario-level evidence consistent with applicable privacy, safety, and benchmark-integrity requirements.

7.4.31 Benchmark Quality Checklist

Before a scenario enters the formal benchmark, it can be reviewed using the following checklist:

Quality Question	Requirement
Is the primary target clearly defined?	Yes
Does the scenario create sufficient evaluation opportunity?	Yes
Is expected behavior observable?	Yes
Is ambiguity intentional and controlled?	Yes, if present
Is the interaction reasonably realistic?	Yes
Is unrelated knowledge dependence minimized?	Yes
Are scoring criteria predefined?	Yes
Are N/A conditions defined?	Yes
Is unnecessary leading avoided?	Yes
Is the scenario meaningfully distinct from existing items?	Yes
Can it be administered consistently?	Yes
Has potential bias been reviewed?	Yes
Can the result be traced to evidence?	Yes
Has the scenario been pilot-tested?	Preferably yes
A scenario failing important quality requirements should be revised, re-tested, or excluded .	

7.5 Complete BOND Benchmark Architecture

The complete benchmark framework developed in Chapter 7 can now be represented as:

BOND FRAMEWORK

↓

Five Core Factors

↓

25 Sub-Factor Targets

↓

Benchmark Categories

↓

Question / Scenario Design

↓

Single-Turn + Multi-Turn Scenarios

↓

Difficulty + Domain Diversity

↓

Quality Rules

↓

BOND Benchmark Scenario Set

↓

Standardized Evaluation Process

↓

BAT Scoring

↓

Sub-Factor Scores

↓

Factor Scores

↓

Overall BOND Score

This architecture connects the conceptual framework developed in Chapters 3–5 with the mathematical framework developed in Chapter 6.

7.6 Chapter Summary

Chapter 7 established the structure required to convert BAT from an individual assessment procedure into a **standardized benchmark framework**.

The chapter introduced four major components:

Benchmark Categories → organize evaluation around the five BOND factors and twenty-five sub-factor targets.

Question Design → defines how interaction scenarios should create controlled and observable evaluation opportunities.

Evaluation Process → defines how scenarios are administered, recorded, scored, aggregated, and reported.

Quality Rules → establish methodological requirements for benchmark validity, fairness, consistency, and reproducibility.

Together:

Categories define WHAT to test.

Question Design defines HOW to create the test.

Evaluation Process defines HOW to administer and score it.

Quality Rules define WHETHER the test is strong enough to belong in the benchmark.

CHAPTER 7 CORE PRINCIPLE

A meaningful Human–AI interaction benchmark requires more than a collection of questions. It requires clearly defined behavioral targets, controlled interaction opportunities, standardized evaluation procedures, and quality rules that preserve the validity and interpretability of the resulting measurements.

The BOND Benchmark Framework therefore establishes the experimental infrastructure necessary to evaluate the proposed BOND construct systematically.

The next stage is **validation**: determining whether the framework, factors, scoring procedure, benchmark design, and resulting measurements are supported by evidence rather than theory alone.

CHAPTER 8 — APPLICATIONS

The BOND Framework is designed as a general Human–AI interaction evaluation framework rather than an evaluation method restricted to a single type of artificial intelligence.

As AI systems become integrated into conversational interfaces, robotics, education, healthcare support, and voice-based systems, interaction quality becomes increasingly important. Different systems may perform different tasks, but they share a common requirement: they must **interpret humans, communicate effectively, maintain relevant context, adapt appropriately, and respond to interaction changes.**

BOND provides a structured method for studying these behaviors.

APPLICATION PRINCIPLE

BOND can be applied wherever meaningful Human–AI interaction occurs, provided that the evaluation scenarios and interpretation of the five factors are appropriately adapted to the interaction environment.

8.1 Application in Conversational AI

Conversational AI represents one of the most direct applications of the BOND Framework.

Chat-based AI systems interact with humans through natural-language conversations involving:

Questions

Instructions

Explanations

Planning

Problem-solving

Feedback

Corrections

Multi-turn tasks

Traditional evaluation may determine whether the AI produces a correct or preferred response.

BOND introduces an additional question:

How effectively does the AI build and maintain the interaction required to help the human achieve the intended objective?

8.1.1 Understanding in Conversational AI

A conversational AI must first determine what the human is requesting.

BOND Understanding can evaluate whether the system:

Recognizes explicit requirements

Interprets reasonably supported intent

Uses relevant earlier context

Recognizes active constraints

Handles clarification and updated information

Consider:

Human:

“Explain quantum computing in simple language. I already understand classical bits, so don't explain them again.”

The AI must recognize several pieces of interaction information:

Task → Explain quantum computing

Communication Level → Simple

Existing Knowledge → Classical bits already understood

Constraint → Avoid unnecessary explanation of classical bits

A response can contain scientifically useful information while still performing poorly at interaction-level Understanding if these requirements are ignored.

8.1.2 Clarity in Conversational AI

Conversational AI frequently explains information.

The BOND Clarity factor can evaluate whether these explanations are:

Comprehensible

Structurally organized

Precise

Relevant

Sufficient for the interaction objective

For example, an AI tutor explaining a scientific concept to a beginner should not automatically receive strong Clarity performance merely because its explanation is technically sophisticated.

The explanation must also be usable by the intended human.

Thus:

Technical Complexity $\neq$ Communication Quality

and

Longer Answer $\neq$ Better Clarity

BOND evaluates communication relative to the interaction conditions.

8.1.3 Personalization in Conversational AI

Longer conversations often provide information that can improve later responses.

For example:

Human:

“I learn physics better through visual examples.”

Later:

Human:

“Explain electric fields.”

A conversational AI with strong Personalization should recognize whether the earlier preference remains relevant and adapt appropriately.

However, BOND does not reward unnecessary references to personal information.

The objective is:

Relevant Context $\rightarrow$ Appropriate Adaptation

not:

Available Context $\rightarrow$ Always Mention It

This distinction becomes particularly important in AI assistants that maintain longer interaction histories.

8.1.4 Reliability in Conversational AI

Conversational interaction often extends across many turns.

During these interactions, the AI may need to preserve:

Instructions

Constraints

Established facts

Task progress

Corrections

Relevant context

For example:

Turn 1 — Human:

“Throughout this task, use SI units.”

After several turns, the human asks for another calculation.

If the AI unexpectedly switches to another measurement system, the individual answer may still appear reasonable while the broader interaction demonstrates weaker **Constraint Persistence**.

BOND Reliability therefore evaluates whether useful behavior remains dependable across the interaction.

8.1.5 Engagement & Responsiveness in Conversational AI

Conversational systems receive continuous interaction signals.

Examples include:

“That makes sense.”

“I still don't understand.”

“Skip this part.”

“Let's continue.”

“That's not what I meant.”

A strong conversational AI should appropriately modify its behavior in response.

For example:

Human:

“I understand the definition now. Just explain the example again.”

A responsive AI should focus on the example instead of restarting the complete lesson.

BOND therefore treats responsiveness as **meaningful adaptation to interaction signals**, rather than simply continuing the conversation.

8.1.6 Conversational AI Evaluation Example

Suppose two chat systems answer the same factual question correctly.

System A

Provides the correct information but ignores the requested format, repeats information the human already understands, and fails to adjust after feedback.

System B

Provides equally correct information while respecting the requested format, using relevant context, and adapting after feedback.

An accuracy-focused evaluation may assign similar results to both systems.

BOND may reveal substantially different interaction profiles.

For example:

Factor	System A	System B
Understanding	0.65	0.90
Clarity	0.70	0.88
Personalization	0.40	0.82
Reliability	0.62	0.86
Engagement & Responsiveness	0.45	0.91

These values are illustrative rather than experimental results.

The example demonstrates how BOND can reveal interaction differences that a task-accuracy measurement alone may not capture.

8.1.7 Potential Uses in Conversational AI Development

BOND could potentially support several stages of conversational AI development.

Model Comparison

Developers could compare interaction profiles across AI systems.

Regression Testing

A newer model could be evaluated to determine whether improvements in one capability accidentally reduce interaction quality elsewhere.

Fine-Tuning Evaluation

Systems optimized for different interaction styles could be compared using standardized BAT scenarios.

Failure Analysis

Factor and sub-factor scores could help identify whether poor interaction originates primarily from Understanding, Clarity, Personalization, Reliability, or Responsiveness.

Domain-Specific Evaluation

Specialized conversational systems could use domain-adapted BAT scenarios while retaining the core BOND framework.

8.2 Application in Robotics

Human–robot interaction extends AI communication beyond text.

A robot may receive information through:

Speech

Vision

Gestures

Commands

Environmental signals

Human corrections

and may respond through:

Speech

Movement

Physical actions

Visual signals

Task execution

The interaction environment is therefore different from conversational AI, but the five BOND factors can still provide a useful conceptual evaluation structure.

8.2.1 Understanding in Robotics

A robot may need to interpret both human instructions and environmental context.

For example:

Human:

“Bring me the blue bottle from the table, but don't move the laptop.”

The robot must recognize:

Target Object → Blue bottle

Location → Table

Action → Bring

Constraint → Do not move the laptop

The observable outcome can provide evidence for Requirement Recognition and Constraint Recognition.

If multiple blue bottles exist, the robot may need clarification rather than making an unsupported assumption.

8.2.2 Clarity in Robotics

Robotic Clarity is not necessarily limited to verbal explanations.

A robot may communicate through:

Speech

Display messages

Lights

Movement cues

Confirmation signals

For example, before performing an ambiguous physical task, a robot might communicate:

“Do you mean the bottle beside the laptop?”

The effectiveness of this communication can be evaluated through BOND Clarity.

This suggests an important extension:

BOND factors may remain conceptually stable while their observable indicators change across interaction modalities.

8.2.3 Personalization in Robotics

A personal or assistive robot may interact repeatedly with the same human.

Relevant context could include:

Preferred task procedures

Communication preferences

Repeated routines

Current task goals

The robot should use such context only when it is relevant to the current interaction.

This creates potential applications for BOND Personalization beyond purely language-based systems.

8.2.4 Reliability in Robotics

Reliability becomes particularly important when AI decisions produce physical actions.

A robot may need to maintain:

Task constraints

Object references

Interaction state

Updated instructions

Established boundaries

If a human changes an instruction during execution, the robot should appropriately update its behavior when technically and safely possible.

However, BOND should not replace dedicated **robot safety, control, mechanical reliability, or engineering validation**.

It evaluates interaction behavior, not complete physical-system safety.

8.2.5 Engagement & Responsiveness in Robotics

Human feedback may arrive through more than words.

For example, a human might:

Verbally correct the robot

Interrupt an action

Point toward another object

Change the task

Request continuation

A future robotics implementation of BOND could investigate whether the robot recognizes and appropriately responds to such interaction signals.

This would require modality-specific BAT scenarios and validation.

8.3 Application in AI Education Systems

AI-based educational systems create a particularly important Human–AI interaction environment because effective teaching requires more than producing correct information.

A learner may require:

Appropriate explanation level

Progressive instruction

Examples

Clarification

Feedback adaptation

Context retention

Recognition of misunderstanding

BOND could therefore provide a structured way to evaluate interaction characteristics of AI tutoring systems.

8.3.1 Understanding the Learner's Request

An AI teacher should distinguish between requests such as:

“Give me the answer.”

“Explain why this happens.”

“Give me a hint.”

“Check my method.”

“Explain only the step I got wrong.”

These requests may concern the same academic problem but require substantially different responses.

Understanding therefore involves recognizing the **learning objective of the interaction**, not merely identifying the academic topic.

8.3.2 Clarity in AI Teaching

Educational explanations must be appropriate to the learner's current level.

A technically correct explanation can still be ineffective if it introduces unexplained terminology or unnecessary complexity.

BOND Clarity can examine whether an AI teacher:

Uses understandable language

Structures explanations logically

Maintains relevant focus

Provides sufficient reasoning

Avoids unnecessary complexity

The evaluation does not determine whether one teaching style is universally best.

Instead, it examines whether communication is appropriate under the specified learning conditions.

8.3.3 Personalized Learning Interaction

Suppose a learner states:

“I know Coulomb's law, but electric flux is completely new to me.”

A personalized AI teacher should use the known information where useful while avoiding the assumption that the learner already understands electric flux.

Over multiple interactions, relevant information about:

Current knowledge

Learning goals

Preferred explanation format

Previous progress

may support more appropriate educational interaction.

BOND Personalization provides a structure for evaluating whether this context is used selectively and effectively.

8.3.4 Learning Progression

Educational conversations naturally contain progression.

For example:

Concept → Example → Practice → Correction → Application

An AI teacher should recognize when the learner has completed one stage and is ready for the next.

If the learner says:

“Okay, I understand the formula now. Give me one question to solve myself.”

repeating the formula explanation may represent weaker Progression Awareness.

BOND Engagement & Responsiveness can evaluate whether the system appropriately continues the learning process.

8.3.5 Educational Use Boundary

A high BOND score would not prove that an AI teacher improves learning outcomes.

Educational effectiveness also depends on variables such as:

Subject accuracy

Pedagogical design

Curriculum alignment

Learner outcomes

Long-term retention

These require separate validation.

BOND would specifically evaluate the **Human–AI interaction component** of the educational system.

8.4 Application in Healthcare AI Interaction

Healthcare-related AI systems may interact with patients, clinicians, researchers, or administrators.

These environments can involve complex and high-stakes communication.

Potential BOND-relevant behaviors include:

Understanding the user's stated question

Communicating information clearly

Maintaining relevant context

Responding appropriately to corrections

Expressing uncertainty

Avoiding unsupported assumptions

For example, Uncertainty Calibration may become especially important when available information is insufficient for a confident conclusion.

However, an essential boundary must be maintained:

BOND is not a medical safety, diagnostic accuracy, or clinical effectiveness benchmark.

A healthcare AI could theoretically demonstrate strong interaction behavior while providing medically incorrect information.

Therefore, healthcare applications would require BOND to operate **alongside dedicated clinical, safety, factuality, privacy, and regulatory evaluation**, not instead of them.

BOND's role would be limited to evaluating the interaction dimensions represented by the framework.

8.5 Application in Voice Assistants

Voice assistants introduce another interaction environment in which responses occur through spoken language rather than primarily through text.

Voice interaction can include:

Spoken commands

Follow-up questions

Corrections

Pronoun references

Contextual requests

Interruptions

Task continuation

For example:

Human:

“Set a timer for twenty minutes.”

Later:

Human:

“Actually, make it thirty.”

The system must correctly interpret the update and modify the relevant interaction state.

8.5.1 Understanding in Voice Interaction

Spoken requests may contain:

Incomplete sentences

Corrections during speech

Context-dependent references

Conversational phrasing

A voice-oriented BAT implementation could evaluate how well the system manages these interaction conditions.

However, speech recognition quality itself should be distinguished from higher-level interaction behavior where possible.

Otherwise, a transcription failure could incorrectly appear to be an Understanding failure.

8.5.2 Clarity in Spoken Responses

Voice responses have different communication constraints from text.

A long answer that is acceptable on screen may become difficult to follow when spoken.

Therefore, voice-specific Clarity may require consideration of:

Response length

Sentence complexity

Information ordering

Verbal precision

Need for confirmation

The underlying BOND factor remains Clarity, while its operational indicators may be adapted to the modality.

8.5.3 Responsiveness in Voice Interaction

Voice conversations can change quickly.

A user may interrupt:

“Stop.”

“No, the other one.”

“Skip that.”

“Repeat the last part.”

A responsive assistant should appropriately update its behavior.

This makes voice assistants particularly suitable for evaluating:

Interaction Signal Recognition

Feedback Responsiveness

Adaptive Continuation

Progression Awareness

8.6 Cross-Domain Application Model

The five application areas demonstrate that BOND is not tied to a particular AI interface.

The general application model is:

Human Input / Interaction Signal

↓

AI Interpretation

↓

AI Response or Action

↓

Human Feedback / Context Change

↓

AI Adaptation

↓

Observable Interaction Behavior

↓

BOND Evaluation

What changes across domains is primarily the **form of interaction evidence**.

For Chat AI:

Evidence → Text responses

For Robots:

Evidence → Language + physical behavior

For AI Teachers:

Evidence → Educational interaction

For Healthcare AI:

Evidence → Domain-specific communication behavior

For Voice Assistants:

Evidence → Spoken interaction

The five-factor structure may therefore provide a common evaluation language while allowing domain-specific BAT implementations.

8.7 Domain Adaptation Principle

Applying BOND to a new domain should not simply involve copying the same benchmark questions.

Instead:

Core Factor Definitions → Remain Stable

while:

Scenarios → Adapt

Observable Indicators → Adapt

Evaluation Conditions → Adapt

Domain-Specific Requirements → Documented

For example, Contextual Continuity means conceptually similar things in a chatbot and a robot, but the observable evidence may differ significantly.

This allows BOND to aim for both:

Conceptual Consistency

and

Application Flexibility

8.8 Chapter Summary

Chapter 8 examined potential applications of the BOND Framework across:

Conversational AI

Robotics

AI Education Systems

Healthcare AI Interaction

Voice Assistants

Across these environments, BOND focuses on five recurring dimensions:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

However, BOND should not be interpreted as replacing specialized evaluation frameworks.

It does not independently establish:

Factual accuracy

Physical safety

Medical validity

Educational effectiveness

General intelligence

or

Complete AI quality

Instead, BOND provides an additional evaluation layer focused specifically on **Human–AI connection-building behavior**.

CHAPTER 8 CORE PRINCIPLE

BOND is intended as a domain-adaptable Human–AI interaction framework: its core dimensions remain conceptually consistent across applications, while scenarios, observable indicators, and evaluation procedures can be adapted to the characteristics and requirements of each interaction environment.

The range of possible applications also makes it necessary to define the framework's boundaries precisely. The next chapter therefore examines **what BOND can measure, what it cannot measure, and the principal limitations of the proposed framework**.

CHAPTER 9 — LIMITATIONS

Every evaluation framework has boundaries. Clearly defining these boundaries is necessary to prevent a measurement from being interpreted as something broader than it actually is.

BOND is specifically designed to evaluate **AI connection-building capability within Human–AI interaction** through five primary factors:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

It is not intended to represent a complete measurement of artificial intelligence.

LIMITATION PRINCIPLE

A BOND score should be interpreted only as evidence about the interaction dimensions defined by the BOND Framework and under the conditions in which those dimensions were evaluated.

9.1 What BOND Measures

BOND measures observable behaviors related to how effectively an AI establishes, maintains, and adapts interaction with a human.

The framework evaluates these behaviors through its five-factor structure.

9.1.1 Understanding

BOND evaluates whether the AI demonstrates the ability to appropriately interpret the interaction.

This includes behaviors related to:

Requirement Recognition

Intent Interpretation

Context Integration

Constraint Recognition

Clarification & Update Handling

For example, if a human requests:

“Explain this in three simple points and don't use equations.”

BOND can evaluate whether the AI recognizes and follows the relevant interaction requirements.

However, this does not imply that BOND directly measures the AI's internal cognitive understanding.

BOND evaluates **observable evidence of Understanding within interaction.**

9.1.2 Clarity

BOND evaluates whether the AI communicates information in a form that is appropriate and usable within the interaction.

This includes:

Comprehensibility

Structural Organization

Precision

Relevance & Focus

Explanatory Sufficiency

An AI response may contain correct information while still receiving weaker Clarity performance if that information is communicated in an unnecessarily confusing or unsuitable form.

Therefore, BOND distinguishes:

Having Information

from

Communicating Information Effectively

9.1.3 Personalization

BOND measures whether available and relevant human context is appropriately used to adapt interaction.

This includes:

Relevant Context Recognition

User-Level Adaptation

Preference Adaptation

Goal & Situation Adaptation

Selective Context Use

The framework does not assume that more personalization is always better.

Strong Personalization requires the AI to use context **when it is relevant**, rather than inserting personal information unnecessarily.

9.1.4 Reliability

BOND evaluates whether useful interaction behavior remains dependable across the evaluated interaction.

This includes:

Behavioral Consistency

Contextual Continuity

Constraint Persistence

Uncertainty Calibration

Error Recovery

Reliability within BOND therefore refers specifically to **interaction-level reliability**.

It should not automatically be interpreted as complete technical or operational reliability of the AI system.

9.1.5 Engagement & Responsiveness

BOND evaluates how appropriately the AI responds to changing interaction signals.

These may include:

Feedback

Corrections

Confusion

Progress

Changes in direction

Continuation requests

The framework evaluates whether the AI recognizes and appropriately responds to such signals while maintaining progress toward the relevant interaction objective.

Engagement does not mean maximizing conversation length.

An effective AI may sometimes demonstrate strong Engagement & Responsiveness by completing a task concisely.

9.2 What BOND Does Not Measure

The scope of BOND is intentionally limited.

A high BOND score should not be interpreted as evidence that the AI performs strongly across every dimension of intelligence or system quality.

Several important properties remain outside the direct scope of BOND.

9.2.1 BOND Does Not Directly Measure General Intelligence

BOND is not designed to determine whether an AI possesses:

AGI

Human-level intelligence

General reasoning ability

Consciousness

Sentience

or any equivalent broad concept.

An AI could demonstrate strong Human–AI interaction behavior while having significant limitations in other capabilities.

Therefore:

High BOND $\neq$ General Intelligence

9.2.2 BOND Does Not Replace Accuracy Metrics

BOND is not a replacement for:

Accuracy

Precision

Recall

F1 Score

or other task-specific performance metrics.

For example, an AI could provide a highly clear, personalized, and responsive explanation containing a factual error.

Its interaction quality and factual accuracy are related but distinct evaluation questions.

Therefore:

BOND Evaluation + Accuracy Evaluation

may provide more information than either measurement alone.

9.2.3 BOND Does Not Independently Measure Factuality

BOND does not directly provide a universal factuality score.

A BOND evaluation may capture interaction consequences of factual errors in relevant scenarios, particularly through Reliability or Error Recovery.

However:

BOND Score $\neq$ Factual Accuracy Score

Dedicated factuality evaluation remains necessary where correctness is important.

9.2.4 BOND Does Not Measure Complete Reasoning Ability

Reasoning benchmarks may evaluate:

Mathematical reasoning

Logical reasoning

Scientific reasoning

Planning

Coding ability

Complex problem-solving

BOND may observe how reasoning is communicated and integrated into interaction, but it does not independently determine the complete reasoning capability of an AI.

A system may solve a difficult problem while interacting poorly.

Another system may communicate extremely well while failing to solve the problem.

These are different dimensions.

9.2.5 BOND Does Not Replace Safety Evaluation

AI safety involves areas beyond the five BOND factors.

BOND does not independently establish:

System safety

Robustness against harmful use

Security

Alignment

Privacy protection

Regulatory compliance

or other specialized safety properties.

A system should therefore not be described as safe simply because it achieves a high BOND score.

9.2.6 BOND Does Not Measure Consciousness or Emotion

The term **connection-building capability** describes observable Human–AI interaction behavior.

It does not imply that the AI experiences:

Emotions

Attachment

Empathy as subjective experience

Conscious awareness

Human-like relationships

BOND evaluates behavior exhibited during interaction.

It makes no claim about subjective internal experience.

This distinction is especially important when discussing Human–AI connection.

9.2.7 BOND Does Not Measure Human Attachment

BOND should also be distinguished from psychological measurements of how emotionally attached a human becomes to an AI.

The framework primarily evaluates:

AI connection-building capability

rather than:

Human emotional dependency or attachment to AI

Human psychological responses may be investigated in future studies as external variables, but they should not be assumed to be identical to BOND.

9.2.8 BOND Does Not Measure Complete User Satisfaction

A user may prefer one AI system over another for reasons not represented by BOND.

For example:

Interface design

Response speed

Visual appearance

Available tools

Price

Brand preference

Specific domain capability

BOND may correlate with user satisfaction in some contexts, but such a relationship must be empirically investigated.

Therefore:

High BOND $\neq$ Guaranteed User Satisfaction

9.2.9 BOND Does Not Establish Domain Competence

A high BOND score for a healthcare AI does not establish medical competence.

A high BOND score for an educational AI does not establish teaching effectiveness.

A high BOND score for a robot does not establish mechanical safety.

A high BOND score for a coding assistant does not establish programming correctness.

BOND evaluates the **interaction component** of these systems.

Domain competence requires separate domain-specific evaluation.

9.3 Current Framework Limitations

Beyond scope boundaries, the proposed BOND framework itself has limitations that require future investigation.

9.3.1 Initial Factor Selection Requires Validation

The five-factor structure was designed to represent important dimensions of AI connection-building capability.

However, empirical research must determine whether:

All five factors are necessary

Additional factors are required

Some factors substantially overlap

The proposed sub-factor structure is appropriate

Therefore, the current structure should be treated as a **proposed measurable framework subject to validation and revision.**

9.3.2 Equal Weights Are an Initial Assumption

Standard BOND currently assigns:

20% weight to each primary factor

This provides a transparent initial configuration.

However, it has not yet been established that all five factors contribute equally to Human–AI connection-building capability.

Future empirical research may support:

Equal weighting

Different universal weights

Domain-specific weights

or another weighting method.

Therefore:

Equal weighting is a starting configuration, not a proven natural law of Human–AI interaction.

9.3.3 Sub-Factor Weights Require Validation

The same limitation applies within each primary factor.

Equal weighting of five sub-factors simplifies the initial mathematical framework.

However, future data may demonstrate that some sub-factors contribute more strongly to the measured construct than others.

The weighting structure should therefore remain open to evidence-based revision.

9.3.4 BAT Contains Human Judgment

Some BAT behaviors cannot be reduced to purely objective automated measurements.

Evaluators may disagree about whether a response represents:

Partial

Strong

or

Full demonstration

of a behavior.

This introduces potential evaluator variability.

BOND attempts to reduce this through:

Standardized rubrics

Evidence-based scoring

Evaluator training

Independent evaluation

Multiple evaluators

However, inter-rater reliability must be empirically examined.

9.3.5 Benchmark Dependence

A BOND score is influenced by the scenarios used to produce it.

A benchmark containing primarily easy interactions may produce different score distributions from one containing complex multi-turn scenarios.

Therefore, formal BOND results should always identify:

Benchmark Version

Scenario Set

Evaluation Configuration

This also means that benchmark quality is central to measurement quality.

9.3.6 Language and Cultural Generalization

The meaning and quality of Human–AI interaction can vary across:

Languages

Cultures

Communication conventions

Social contexts

A framework validated primarily in one language cannot automatically be assumed to perform identically across all others.

Cross-lingual and cross-cultural validation will therefore be necessary before broad generalization.

9.3.7 Domain Generalization

Interaction requirements differ between:

Education

Healthcare

Robotics

General chat

Research

Customer support

A common five-factor framework may remain useful across these domains, but the observable indicators and relative importance of factors may differ.

Domain-specific validation is therefore required.

9.3.8 AI Systems Change Over Time

Modern AI systems may be continuously updated.

A model evaluated at one time may later change because of:

Model updates

Fine-tuning

System prompt changes

Tool changes

Safety updates

Interface changes

Therefore, BOND results should record the model/version and evaluation date whenever possible.

9.3.9 Interaction Variability

Generative AI systems may produce different responses to identical inputs across repeated runs.

A single benchmark execution may therefore not completely represent system behavior.

Repeated evaluations and uncertainty estimates may eventually be required for robust system-level BOND measurement.

9.3.10 Benchmark Contamination

If benchmark scenarios become publicly available and later appear in AI training or optimization data, systems may become specifically optimized for those scenarios.

This could inflate benchmark performance without equivalent improvement in general Human–AI interaction.

Potential future strategies include:

Protected test sets

Rotating scenarios

New scenario generation

Contamination monitoring

However, no approach completely eliminates this challenge.

9.3.11 Numerical Compression

Reducing complex Human–AI interaction to one score inevitably removes information.

For example:

BOND = 0.82

cannot by itself explain:

Which behaviors were strong

Which behaviors were weak

Which scenarios produced failures

How much evidence was available

For this reason, BOND should report:

Overall Score

alongside:

Factor Profile

Coverage

and, where appropriate:

Sub-Factor Results

The single BOND score is a summary, not a complete description of interaction behavior.

9.3.12 Interval Boundaries Are Provisional

The proposed:

Very Low

Low

Moderate

High

Very High

intervals provide a convenient descriptive system.

Their current boundaries are not yet empirically established as natural behavioral thresholds.

Future validation may support different interval boundaries.

Until then, the numerical score should remain the primary measurement and the interval should be treated as a descriptive interpretation.

9.4 Interpretation Rules

To reduce misuse, BOND results should follow several interpretation rules.

Rule 1

Do not interpret BOND as complete AI intelligence.

Rule 2

Do not interpret BOND as a factual accuracy score.

Rule 3

Do not interpret BOND as proof of safety.

Rule 4

Do not interpret connection-building capability as evidence of consciousness or emotion.

Rule 5

Always consider the five-factor profile alongside the overall score.

Rule 6

Report evaluation coverage.

Rule 7

Report the benchmark and scoring configuration.

Rule 8

Treat current weights and intervals as provisional until validated.

Rule 9

Compare systems only under sufficiently comparable evaluation conditions.

Rule 10

Interpret results only within the domains and conditions supported by evidence.

9.5 BOND Measurement Boundary

The overall scope can therefore be summarized as:

BOND MEASURES

- Observable Human–AI interaction behavior
- Understanding
- Clarity
- Personalization
- Reliability
- Engagement & Responsiveness
- Connection-building capability under defined conditions

BOND DOES NOT DIRECTLY MEASURE

- General Intelligence
- AGI
- Consciousness
- Emotion
- Complete Reasoning Ability
- Universal Factual Accuracy
- Complete AI Safety
- Domain Expertise

- Human Psychological Attachment
- Complete User Satisfaction
- Overall AI Quality

This boundary is fundamental to responsible interpretation of the framework.

9.6 Chapter Summary

BOND is intentionally a **specialized evaluation framework**.

Its purpose is not to compress every property of artificial intelligence into a single number.

Instead, it focuses on a specific question:

How effectively does an AI demonstrate the interaction behaviors required to establish, maintain, and adapt a useful connection with a human under defined evaluation conditions?

The framework approaches this question through five measurable dimensions:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

At the same time, several aspects of the framework remain provisional and require empirical investigation, including:

Factor structure

Sub-factor structure

Weighting

BAT reliability

Benchmark validity

Interval calibration

Cross-domain generalization

Cross-lingual generalization

Clearly defining these limitations does not weaken the framework.

Instead, it defines the claims that BOND can reasonably make and identifies the research required to strengthen those claims.

CHAPTER 9 CORE PRINCIPLE

BOND should be interpreted as a structured measurement of observable AI connection-building behavior—not as a universal score of intelligence, correctness, safety, consciousness, or overall AI quality.

These limitations also define several opportunities for future development. **Chapter 10 therefore examines the future evolution of BOND, including AHI, robotics, AGI, and broader research directions.**

CHAPTER 10 — FUTURE WORK

The BOND Framework presented in this work establishes an initial structure for measuring AI connection-building capability through five dimensions: **Understanding, Clarity, Personalization, Reliability, and Engagement & Responsiveness.**

However, BOND should not be considered a finished or permanently fixed framework.

Its current structure provides a foundation that can be tested, challenged, refined, and extended as Human–AI interaction evolves.

Future work should therefore focus on two parallel directions:

Improving the scientific validity of BOND itself

and

Exploring its applicability to increasingly advanced AI systems.

10.1 BOND and Artificial Hybrid Intelligence (AHI)

Artificial Hybrid Intelligence (AHI) represents a possible future direction in which human and artificial intelligence operate through increasingly integrated forms of collaboration.

Within such systems, interaction quality may become more important than in conventional question-and-answer AI.

Instead of:

Human → Request → AI → Answer

future interaction may increasingly resemble:

Human Reasoning ↔ AI Reasoning

↓

Continuous Information Exchange

↓

Shared Task Progression

↓

Collaborative Outcome

In such environments, BOND could potentially be extended to study how effectively an AI maintains interaction during prolonged human–AI collaboration.

Relevant questions may include:

Does the AI maintain the human's objectives across extended tasks?

Can it adapt when the human's reasoning or strategy changes?

Can it communicate uncertainty appropriately?

Can it preserve relevant context without overusing irrelevant information?

Can human and AI contributions remain coordinated across complex workflows?

These questions extend naturally from the current BOND factors.

However, AHI remains a future research direction within BOND rather than a capability demonstrated or measured by the present framework.

Future work would therefore be required to define what an AHI-specific BOND implementation should measure and how it should be experimentally validated.

10.2 BOND in Advanced Robotics

Robotics introduces Human–AI interaction into the physical world.

Future robots may need to understand humans through combinations of:

Language

Gesture

Visual context

Environmental information

Task history

Human feedback

and then respond through both communication and physical action.

This creates opportunities to extend BOND beyond text-based interaction.

For example, a future robotic BAT scenario might evaluate whether a robot:

correctly identifies a human instruction,

recognizes a physical constraint,

asks for clarification when necessary,

maintains instructions during task execution,

and

adapts appropriately when the human changes the task.

The core BOND dimensions may remain conceptually useful, but their observable indicators would require modification.

For example:

Clarity in Chat AI → Quality of textual communication

while:

Clarity in Robotics → Speech + signals + action communication

Similarly:

Responsiveness in Chat AI → Adaptation of language

while:

Responsiveness in Robotics → Adaptation of language and physical behavior

Future research should determine whether the same five-factor structure remains valid in multimodal and embodied Human–Robot Interaction.

10.3 BOND and AGI

If increasingly general AI systems are developed, their ability to interact effectively with humans across different tasks and domains may become an important evaluation problem.

A future Artificial General Intelligence may possess strong reasoning or problem-solving capabilities while still varying in its ability to:

Interpret human goals

Communicate reasoning clearly

Use human context appropriately

Maintain long-term interaction consistency

Respond to changing human requirements

BOND could potentially provide one component of a broader AGI evaluation ecosystem.

However:

BOND should not be interpreted as an AGI test.

A high BOND score would not demonstrate that a system possesses AGI.

Likewise, an AGI system would not automatically be guaranteed a high BOND score without evaluation.

Future research could instead investigate whether BOND remains useful as a **Human–AI interaction measurement layer** for increasingly general systems.

10.4 Future Research Directions

Several research questions must be addressed before BOND can become a mature evaluation framework.

Empirical Factor Validation

The proposed five-factor structure should be tested to determine whether:

The factors represent distinguishable dimensions

Sub-factors appropriately represent their parent factors

Important interaction dimensions are missing

Some dimensions overlap excessively

The framework should be modified if empirical evidence does not support its current structure.

Weight Validation

The initial BOND configuration uses equal weighting.

Future experiments should examine whether:

$U = C = P = R = ER$

is empirically justified.

Alternative approaches may include:

Data-derived weights

Expert-derived weights

Domain-specific weights

or

Task-specific configurations

Any modification should be supported by evidence rather than arbitrary preference.

BAT Validation

The BOND Assessment Test requires further investigation regarding:

Evaluator agreement

Scoring reliability

Scenario difficulty

Scenario discrimination

Repeated-test stability

N/A handling

Benchmark coverage

This will help determine whether BAT consistently measures the behaviors it is intended to evaluate.

Human Evaluation Studies

Future studies should compare BOND measurements with independent human judgments of interaction quality.

For example, researchers could investigate whether systems receiving stronger BOND profiles are also independently judged as:

Easier to communicate with

More context-aware

More responsive

More consistent

Such relationships should be tested rather than assumed.

Cross-Domain Validation

BOND should eventually be evaluated across environments such as:

Conversational AI

Education

Research assistance

Customer support

Voice interaction

Robotics

This could determine which parts of the framework generalize and which require domain-specific adaptation.

Cross-Lingual and Cross-Cultural Research

Human communication differs across languages and cultures.

Future BOND versions should investigate whether the framework remains valid across different:

Languages

Communication conventions

Cultural contexts

Interaction styles

A framework validated in one population or language should not automatically be assumed universal.

Automated and Hybrid Evaluation

The current framework includes human evaluation because several BOND behaviors require contextual judgment.

Future research could investigate whether some BAT components can be evaluated automatically.

A possible system could combine:

Automated Objective Checks

-

Human Evaluation

=

Hybrid BOND Evaluation

For example, explicit format constraints may sometimes be checked automatically, while nuanced adaptation or contextual relevance may continue to require human judgment.

Long-Term Interaction Evaluation

Current benchmark scenarios may primarily involve relatively limited interaction sequences.

Future research could investigate BOND across:

Hours

Days

or

Repeated Human–AI interactions

This could reveal behaviors related to long-term:

Context management

Preference adaptation

Reliability

Interaction progression

Such research would require additional methodological and privacy considerations.

10.5 Future Development Principle

BOND should evolve through evidence.

The intended development process is:

BOND V1

↓

Expert Review

↓

Prototype Testing

↓

Experimental Evaluation

↓

Identify Weaknesses

↓

Modify Framework

↓

BOND V2

↓

Further Validation

Future versions should therefore not preserve current assumptions merely for consistency if strong evidence supports improvement.

CHAPTER 10 CORE PRINCIPLE

The future of BOND should be driven by empirical validation, transparent revision, and expansion into new forms of Human–AI interaction while preserving clear boundaries between demonstrated findings and proposed future possibilities.

CHAPTER 11 — CONCLUSION

Artificial intelligence evaluation has traditionally focused heavily on what AI systems can accomplish.

As AI increasingly interacts directly and continuously with humans, another question becomes important:

How effectively can an AI understand, communicate with, adapt to, remain reliable for, and respond to a human throughout an interaction?

This work proposed **BOND** as a framework for studying that question.

BOND defines **AI connection-building capability** through five primary dimensions:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

Together, these dimensions provide a structured representation of observable behaviors that influence Human–AI interaction.

11.1 Summary

The work began by examining the growing role of Human–AI interaction and the limitations of relying exclusively on conventional AI performance metrics when evaluating interactive systems.

Existing metrics and benchmarks remain valuable for measuring properties such as accuracy, task performance, and response quality.

However, they do not necessarily provide a unified measurement of the interaction behaviors targeted by BOND.

The framework therefore developed a five-factor structure for evaluating AI connection-building capability.

Each factor was further divided into sub-factors, creating a more detailed diagnostic system.

The mathematical framework then introduced:

Normalized factor measurement

Weight distribution

The BOND Equation

Opportunity-adjusted scoring

Factor coverage

BAT scoring

The Interval System

Rather than producing only a single number, BOND preserves a profile of interaction performance.

The Benchmark Framework further defined how standardized scenarios could be created, administered, evaluated, and controlled through quality rules.

Together, these components form an initial pathway from:

Human–AI Interaction

to

Observable Behavior

to

Structured Evaluation

to

Quantitative BOND Measurement

11.2 Contribution

The primary contribution of this work is the proposal of a **structured Human–AI interaction evaluation framework centered on AI connection-building capability**.

The framework contributes several connected components.

Five-Factor Interaction Model

BOND organizes interaction behavior into:

Understanding

Clarity

Personalization

Reliability

Engagement & Responsiveness

This provides a common conceptual vocabulary for analyzing Human–AI interaction.

Sub-Factor Architecture

Each primary dimension is decomposed into observable sub-factors.

This allows evaluation to move beyond a broad overall judgment and identify specific strengths and weaknesses.

Mathematical Measurement Structure

BOND proposes a normalized scoring system that combines factor-level measurements while preserving:

Individual factor scores

Evaluation coverage

Weight configuration

N/A conditions

This makes the overall score traceable to its underlying components.

BOND Assessment Test

BAT provides an operational procedure for converting structured interaction evidence into numerical measurements.

Benchmark Framework

The proposed benchmark architecture defines:

Evaluation categories

Question design principles

Evaluation procedures

Quality rules

This creates a foundation for future standardized comparison between AI systems.

11.3 Future Vision

The long-term objective of BOND is not simply to produce another leaderboard score.

Its broader purpose is to encourage AI evaluation to consider **how AI systems work with humans**, alongside what those systems can accomplish independently.

Future AI may become increasingly involved in:

Learning

Research

Creative work

Robotics

Decision support

Long-term assistance

and other forms of human collaboration.

As these interactions become more complex, measuring only final-task success may provide an incomplete picture.

An AI system may produce strong results while repeatedly misunderstanding the human.

Another may communicate effectively but struggle with consistency.

Another may adapt strongly to human context but perform poorly when uncertainty appears.

A factor-based framework makes these differences visible.

The future vision of BOND is therefore not:

One score that defines the best AI.

Instead, it is:

A measurement framework that helps researchers understand how different AI systems establish, maintain, and adapt interaction with humans.

11.4 Final Statement

BOND remains a proposed framework requiring empirical validation.

Its factors, sub-factors, weights, benchmark scenarios, scoring procedures, and interval boundaries should be tested rather than treated as established scientific facts.

Expert review and experimental validation may identify weaknesses requiring substantial revision.

Such revision should be considered part of the development process rather than a failure of the framework.

The intended progression is:

Theory

↓

Measurement Framework

↓

Benchmark

↓

Expert Review

↓

Experiment

↓

Evidence

↓

Revision

↓

Improved Framework

If future evidence supports the underlying construct and measurement structure, BOND may contribute an additional perspective to AI evaluation—one centered specifically on the quality of the connection an AI is capable of building through interaction.

FINAL BOND PRINCIPLE

The capability of an AI system should not be understood only through the answers it produces, but may also be studied through how effectively it understands the human, communicates clearly, uses relevant context, remains reliable, and responds as the interaction evolves.

BOND provides a proposed framework for measuring that interaction.

APPENDIX

Appendix A — BOND Assessment Test (BAT)

A.1 Complete BAT Master Table

The **BOND Assessment Test (BAT)** is the operational evaluation component of the BOND Framework. It converts observable Human–AI interaction behavior into structured scores across the **five primary BOND factors and twenty-five sub-factors**.

The following table defines the complete BAT evaluation architecture.

Code	Factor	Sub-Factor	Primary Evaluation Target	Observable AI Behavior
U1 / UR	Understanding	Requirement Recognition	Recognition of explicit human requirements	Identifies the requested task, output requirements, and important stated conditions
U2 / UI	Understanding	Intent Interpretation	Interpretation of the human's intended objective	Responds according to meaning reasonably supported by the interaction rather than relying only on literal wording
U3 / UC	Understanding	Context Integration	Use of relevant interaction context	Connects relevant information from earlier turns or provided context with the current request
U4 / UCR	Understanding	Constraint Recognition	Recognition of explicit interaction constraints	Identifies and follows relevant limits involving format, length, scope, quantity, language, or process
U5 / UU	Understanding	Clarification & Update Handling	Handling ambiguity, clarification, and	Requests clarification when necessary and

Code	Factor	Sub-Factor	Primary Evaluation Target	Observable AI Behavior
			updated information	appropriately incorporates corrected or updated information
C1 / CC	Clarity	Comprehensibility	Understandability of communication	Uses language and explanations appropriate to the stated interaction conditions
C2 / CS	Clarity	Structural Organization	Organization of information	Presents information in a logical and usable structure
C3 / CP	Clarity	Precision	Specificity and precision of communication	Avoids unnecessarily vague, confusing, or misleading wording
C4 / CR	Clarity	Relevance & Focus	Alignment of response content with the request	Prioritizes relevant information and avoids unnecessary deviation
C5 / CE	Clarity	Explanatory Sufficiency	Adequacy of explanation	Provides enough explanation to satisfy the interaction objective without requiring unnecessary expansion
P1 / PCR	Personalization	Relevant Context Recognition	Identification of relevant human context	Recognizes available contextual information that could meaningfully improve the response
P2 / PLA	Personalization	User-Level Adaptation	Adaptation to the human's demonstrated or stated level	Adjusts complexity, terminology, and explanation depth appropriately
P3 / PPA	Personalization	Preference Adaptation	Use of relevant stated interaction preferences	Adapts format or communication style when a relevant preference has been established
P4 / PGS	Personalization	Goal & Situation Adaptation	Adaptation to the human's current objective or situation	Shapes the response around the relevant goal and interaction circumstances
P5 / PSC	Personalization	Selective Context Use	Appropriate selection of personal/contextual information	Uses relevant context while avoiding forced or unnecessary personalization
R1 / RBC	Reliability	Behavioral Consistency	Stability of appropriate interaction behavior	Maintains reasonably consistent behavior under comparable interaction

Code	Factor	Sub-Factor	Primary Evaluation Target	Observable AI Behavior
R2 / RCC	Reliability	Contextual Continuity	Preservation of relevant established context	conditions Maintains important facts, decisions, and interaction state across relevant turns
R3 / RCP	Reliability	Constraint Persistence	Continued application of active constraints	Preserves previously established constraints while they remain applicable
R4 / RUC	Reliability	Uncertainty Calibration	Appropriate handling of uncertainty	Distinguishes sufficient from insufficient information and avoids unsupported certainty
R5 / RER	Reliability	Error Recovery	Recovery following identified error or correction	Recognizes relevant errors, updates the interaction state, and continues using corrected information
ER1 / ERS	Engagement & Responsiveness	Interaction Signal Recognition	Recognition of interaction signals	Detects relevant signals such as confusion, correction, satisfaction, redirection, or continuation
ER2 / ERF	Engagement & Responsiveness	Feedback Responsiveness	Adaptation following human feedback	Changes the response appropriately according to meaningful feedback
ER3 / ERP	Engagement & Responsiveness	Progression Awareness	Recognition of interaction progress	Distinguishes completed, current, and upcoming stages of the interaction
ER4 / ERA	Engagement & Responsiveness	Adaptive Continuation	Appropriate continuation after interaction changes	Continues from the updated interaction state rather than unnecessarily restarting or ignoring changes
ER5 / ERG	Engagement & Responsiveness	Goal-Oriented Engagement	Maintenance of useful progress toward the objective	Keeps interaction directed toward the relevant human goal without unnecessary continuation

A.2 BAT Scoring Scale

Every valid BAT evaluation opportunity is scored using the following **0–4 scale**:

BAT Score	Performance Level	General Interpretation
0	No Demonstration	The targeted behavior is not meaningfully demonstrated

BAT Score	Performance Level	General Interpretation
1	Weak Demonstration	despite sufficient evaluation opportunity Limited evidence of the behavior; major problems remain
2	Partial Demonstration	The behavior is demonstrated to a meaningful extent, but important limitations remain
3	Strong Demonstration	The behavior is demonstrated effectively with only minor limitations
4	Full Demonstration	The targeted behavior is demonstrated completely or near-completely under the defined scenario conditions
N/A	Not Applicable / Insufficient Opportunity	The interaction does not provide sufficient evidence to validly evaluate the targeted behavior

IMPORTANT:
N/A is not equivalent to 0.
 A score of **0** represents failure despite a valid opportunity, whereas N/A represents insufficient evaluation opportunity.

A.3 BAT Normalization Table

Raw BAT scores are converted into the normalized BOND scale as follows:

Raw BAT Score	Normalized Score
0	0.00
1	0.25
2	0.50
3	0.75
4	1.00
N/A	Excluded from applicable-score calculation

Therefore:

Raw BAT Score → Normalized Observation → Sub-Factor Score → Factor Score → Overall BOND

A.4 Complete BAT Evaluation Record

Each evaluated BAT scenario should preserve the following information:

Field	Recorded Information
Scenario ID	Unique BAT scenario identifier
Benchmark Version	BOND Benchmark version used
Primary Factor	U, C, P, R, or ER
Primary Sub-Factor	Intended behavioral target
Secondary Target(s)	Predefined additional targets, if applicable
Domain	Scenario interaction domain

Field	Recorded Information
Difficulty	Level 1, Level 2, or Level 3
Interaction Type	Single-turn or multi-turn
Initial Context	Information available before evaluation
Human Input	Standardized human message
AI Response	Complete original AI output
Follow-Up Turn(s)	Fixed or predefined conditional interaction
Evaluation Opportunity	Observable behavior being tested
Raw BAT Score	0–4 or N/A
Normalized Score	0.00–1.00
Evaluator Evidence	Specific response behavior supporting the score
Evaluator ID	Anonymous evaluator identifier where applicable
Notes	Additional methodological observations

A.5 BAT Factor Coverage

The complete BAT architecture contains:

BOND Factor	Number of Sub-Factors
Understanding	5
Clarity	5
Personalization	5
Reliability	5
Engagement & Responsiveness	5
Total	25

Thus, a complete BOND evaluation aims to generate valid evidence across all **25 sub-factor targets**.

However, multiple scenarios may be used for each sub-factor to improve measurement reliability and behavioral coverage.

A.6 BAT Evaluation Flow

BOND Benchmark Scenario

↓

Human–AI Interaction

↓

Observable AI Behavior

↓

Identify Predefined BAT Target

↓

Evaluate Available Evidence

↓

BAT Score: 0–4 / N/A

↓

Normalize to 0.00–1.00

↓

Aggregate Multiple Scenario Observations

↓

Calculate 25 Sub-Factor Scores

↓

Calculate Five Primary Factor Scores

↓

Understanding + Clarity + Personalization + Reliability + Engagement & Responsiveness

↓

Apply BOND Equation

↓

Overall BOND Score + Factor Profile + Coverage + Interval

BAT CORE PRINCIPLE

BAT converts observable Human–AI interaction behavior into structured numerical evidence while preserving the factor, sub-factor, scenario, evaluator judgment, and interaction evidence underlying each measurement.

Appendix B — Full BOND Tables

The following tables consolidate the complete BOND factor architecture into a single reference section. They are intended to provide a compact technical view of the **five primary factors and twenty-five sub-factors** defined throughout the framework.

B.1 Understanding — Full Factor Table

Understanding (U) represents the AI system's demonstrated capability to correctly interpret the human's requirements, intended objective, relevant context, constraints, and interaction updates.

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
U1 / UR	Requirement Recognition	Recognition of explicit requirements	Correctly identifies and addresses the requested task and stated requirements	Misses, alters, or ignores important requirements
U2 / UI	Intent Interpretation	Interpretation of reasonably supported human intent	Responds according to the intended objective supported by the interaction	Responds only to surface wording or makes unsupported assumptions
U3 / UC	Context Integration	Integration of relevant interaction context	Correctly uses relevant earlier information in the current response	Ignores, misuses, or contradicts relevant context
U4 / UCR	Constraint Recognition	Recognition of limits and conditions	Identifies and follows relevant format, scope, length, quantity, language, or process constraints	Violates or overlooks established constraints
U5 / UU	Clarification & Update Handling	Management of ambiguity and updated information	Clarifies when necessary and correctly incorporates new or corrected information	Makes unjustified assumptions or continues using outdated information

Understanding Measurement Structure

UR + UI + UC + UCR + UU

↓

Understanding Score (U)

A strong Understanding score indicates that the AI generally demonstrates effective interpretation of the interaction conditions presented by BAT scenarios.

It does **not** establish human-like cognitive understanding or consciousness.

B.2 Clarity — Full Factor Table

Clarity (C) represents the quality with which an AI communicates information so that the response is understandable, organized, precise, relevant, and sufficiently explanatory under the defined interaction conditions.

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
C1 / CC	Comprehensibility	Ease with which the intended human can understand the response	Uses appropriate terminology, complexity, and explanation style	Uses unnecessarily confusing, inaccessible, or poorly explained language
C2 / CS	Structural Organization	Logical arrangement of	Organizes information in a clear and usable	Presents information in a disorganized or difficult-

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
C3 / CP	Precision	information Specificity and accuracy of expression	sequence Communicates the intended meaning directly and precisely	to-follow structure Uses unnecessarily vague, ambiguous, or misleading wording
C4 / CR	Relevance & Focus	Alignment of response content with the human's request	Maintains focus on information relevant to the objective	Includes substantial unnecessary or unrelated information
C5 / CE	Explanatory Sufficiency	Whether enough explanation is provided	Provides sufficient information to satisfy the stated objective	Provides too little explanation to make the response adequately usable

Clarity Measurement Structure

CC + CS + CP + CR + CE

↓

Clarity Score (C)

Clarity does not reward complexity or response length by itself.

The objective is **effective communication relative to the interaction requirement**.

B.3 Personalization — Full Factor Table

Personalization (P) represents the AI's demonstrated capability to identify and selectively use relevant human context to adapt interaction appropriately.

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
P1 / PCR	Relevant Context Recognition	Recognition of human-specific context relevant to the current task	Identifies which available contextual information can meaningfully improve the interaction	Ignores clearly relevant context or treats irrelevant information as important
P2 / PLA	User-Level Adaptation	Adaptation to the human's stated or demonstrated knowledge level	Adjusts terminology, complexity, and explanation depth appropriately	Produces communication substantially above or below the relevant level
P3 / PPA	Preference Adaptation	Adaptation to relevant stated preferences	Appropriately follows established communication or interaction preferences	Ignores relevant preferences or applies them inconsistently
P4 / PGS	Goal & Situation Adaptation	Adaptation to the human's current objective and	Shapes the response around the relevant goal and situation	Provides generic assistance that fails to account for important

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
		circumstances		goal information
P5 / PSC	Selective Context Use	Appropriate selection of contextual information	Uses context only when it improves the current interaction	Forces personalization or unnecessarily references irrelevant context

Personalization Measurement Structure

PCR + PLA + PPA + PGS + PSC

↓

Personalization Score (P)

BOND therefore distinguishes:

Relevant Adaptation

from

Maximum Personalization

More personalization is not automatically better.

B.4 Reliability — Full Factor Table

Reliability (R) represents the degree to which appropriate interaction behavior remains dependable across relevant conditions and interaction progression.

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
R1 / RBC	Behavioral Consistency	Stability of appropriate behavior	Maintains reasonably consistent interaction behavior under comparable conditions	Demonstrates unexplained or disruptive behavioral inconsistency
R2 / RCC	Contextual Continuity	Preservation of relevant established information	Correctly maintains important context, facts, and interaction state	Forgets, contradicts, or incorrectly modifies relevant established context
R3 / RCP	Constraint Persistence	Continued maintenance of active constraints	Continues following established constraints while they remain relevant	Follows constraints initially but later abandons them without justification
R4 / RUC	Uncertainty Calibration	Appropriate representation of uncertainty	Recognizes insufficient evidence and communicates uncertainty appropriately	Expresses unsupported certainty or unnecessary uncertainty
R5 / RER	Error Recovery	Recovery following errors or corrections	Recognizes the relevant error, updates appropriately, and continues from corrected information	Repeats the error, ignores correction, or fails to update subsequent behavior

Reliability Measurement Structure

RBC + RCC + RCP + RUC + RER

↓

Reliability Score (R)

BOND Reliability refers to **interaction-level reliability** and should not be interpreted as a complete measurement of technical system reliability or safety.

B.5 Engagement & Responsiveness — Full Factor Table

Engagement & Responsiveness (ER) represents the AI's demonstrated capability to recognize interaction signals and appropriately adapt as the Human–AI interaction progresses.

Code	Sub-Factor	What It Evaluates	Strong Demonstration	Weak Demonstration
ER1 / ERS	Interaction Signal Recognition	Recognition of meaningful human interaction signals	Detects signals such as confusion, correction, satisfaction, redirection, or continuation	Misses or misinterprets relevant interaction signals
ER2 / ERF	Feedback Responsiveness	Behavioral adaptation after feedback	Meaningfully changes the response according to relevant feedback	Acknowledges feedback without appropriately changing behavior
ER3 / ERP	Progression Awareness	Awareness of interaction progress	Recognizes completed, current, and upcoming interaction stages	Repeats completed work or loses track of current progress
ER4 / ERA	Adaptive Continuation	Appropriate continuation after interaction changes	Continues naturally from the updated state or requirement	Restarts unnecessarily or continues according to outdated conditions
ER5 / ERG	Goal-Oriented Engagement	Maintenance of useful progress toward the objective	Keeps interaction focused on helping achieve the relevant goal	Extends, redirects, or complicates interaction without useful purpose

Engagement & Responsiveness Measurement Structure

ERS + ERF + ERP + ERA + ERG

↓

Engagement & Responsiveness Score (ER)

Engagement therefore does not represent:

Conversation Length

or

Number of Follow-Up Questions

Instead, it represents **useful responsiveness to the evolving interaction**.

B.6 Complete 25-Sub-Factor Reference Table

Factor	Code	Sub-Factor
Understanding	UR	Requirement Recognition
Understanding	UI	Intent Interpretation
Understanding	UC	Context Integration
Understanding	UCR	Constraint Recognition
Understanding	UU	Clarification & Update Handling
Clarity	CC	Comprehensibility
Clarity	CS	Structural Organization
Clarity	CP	Precision
Clarity	CR	Relevance & Focus
Clarity	CE	Explanatory Sufficiency
Personalization	PCR	Relevant Context Recognition
Personalization	PLA	User-Level Adaptation
Personalization	PPA	Preference Adaptation
Personalization	PGS	Goal & Situation Adaptation
Personalization	PSC	Selective Context Use
Reliability	RBC	Behavioral Consistency
Reliability	RCC	Contextual Continuity
Reliability	RCP	Constraint Persistence
Reliability	RUC	Uncertainty Calibration
Reliability	RER	Error Recovery
Engagement & Responsiveness	ERS	Interaction Signal Recognition
Engagement & Responsiveness	ERF	Feedback Responsiveness
Engagement & Responsiveness	ERP	Progression Awareness
Engagement & Responsiveness	ERA	Adaptive Continuation
Engagement & Responsiveness	ERG	Goal-Oriented Engagement

B.7 Complete BOND Hierarchy

BOND — AI Connection-Building Capability

↓

5 Primary Factors

↓

25 Sub-Factors

Understanding

UR + UI + UC + UCR + UU

Clarity

CC + CS + CP + CR + CE

Personalization

PCR + PLA + PPA + PGS + PSC

Reliability

RBC + RCC + RCP + RUC + RER

Engagement & Responsiveness

ERS + ERF + ERP + ERA + ERG

↓

Five Factor Scores

U + C + P + R + ER

↓

Weight Distribution

↓

Overall BOND Score

↓

Factor Profile + Coverage + Interval

FULL TABLE PRINCIPLE

The BOND architecture decomposes AI connection-building capability into five primary factors and twenty-five observable sub-factors, allowing the overall measurement to remain connected to specific interaction behaviors rather than functioning as an unexplained single score.

Appendix C — BOND Scoring Rubrics

The BOND Assessment Test (BAT) uses structured scoring rubrics to convert observable Human–AI interaction behavior into numerical measurements.

The purpose of the rubric is to reduce arbitrary evaluator judgment by defining what different performance levels represent.

Each valid evaluation opportunity receives one of the following:

0 — No Demonstration

1 — Weak Demonstration

2 — Partial Demonstration

3 — Strong Demonstration

4 — Full Demonstration

N/A — Insufficient Evaluation Opportunity

RUBRIC PRINCIPLE

BAT scores should be assigned according to observable evidence relative to the predefined target behavior, not according to the evaluator's general impression of the AI response.

C.1 General BAT Rubric

Score	Level	General Evaluation Standard
0	No Demonstration	The AI fails to demonstrate the targeted behavior despite a clear and sufficient opportunity
1	Weak	The AI demonstrates limited evidence of the behavior, but major failures or omissions remain
2	Partial	The AI demonstrates the behavior to a meaningful extent, but important limitations remain
3	Strong	The AI demonstrates the behavior effectively with only minor limitations
4	Full	The AI demonstrates the targeted behavior completely or near-completely under the defined scenario conditions
N/A	Insufficient Opportunity	Available interaction evidence is insufficient for valid evaluation

C.2 Understanding Rubric

Understanding evaluates whether the AI appropriately interprets requirements, intent, context, constraints, clarification, and updates.

Score	Understanding Evaluation
0	Fundamentally misunderstands or fails to recognize the targeted interaction requirement
1	Recognizes a small portion of the relevant requirement but misses major information
2	Understands the main requirement but misses or incorrectly interprets important elements

Score	Understanding Evaluation
3	Correctly interprets nearly all relevant information with only minor limitations
4	Correctly interprets all or nearly all relevant requirements, context, constraints, or updates required by the scenario
N/A	Scenario provides insufficient evidence for the targeted Understanding behavior

Understanding Evidence May Include

Correct task identification

Appropriate interpretation of intent

Use of relevant earlier context

Recognition of explicit constraints

Appropriate clarification

Correct incorporation of updated information

C.3 Clarity Rubric

Clarity evaluates whether the AI communicates information in an understandable, organized, precise, relevant, and sufficiently explanatory manner.

Score	Clarity Evaluation
0	Communication is substantially unusable, confusing, irrelevant, or insufficient for the defined objective
1	Some useful communication is present, but major clarity problems substantially reduce usability
2	Response is generally understandable but contains important problems involving structure, precision, relevance, or explanation
3	Response is clear and usable with only minor communication limitations
4	Response is highly clear, appropriately structured, precise, relevant, and sufficient under the defined interaction conditions
N/A	Scenario does not provide sufficient evidence to evaluate the targeted Clarity behavior

Clarity Evidence May Include

Appropriate terminology

Logical organization

Precise wording

Relevant content

Sufficient explanation

Appropriate complexity

C.4 Personalization Rubric

Personalization evaluates whether the AI appropriately uses relevant human context to adapt interaction.

Score	Personalization Evaluation
0	Completely ignores clearly relevant contextual information or uses it in a strongly inappropriate way
1	Demonstrates minimal adaptation while missing major relevant contextual information
2	Uses some relevant context appropriately but adaptation remains incomplete or inconsistent
3	Adapts effectively to nearly all important relevant context with only minor limitations
4	Uses relevant context selectively and effectively to produce highly appropriate adaptation without unnecessary personalization
N/A	No sufficient relevant contextual information or adaptation opportunity exists

Personalization Evidence May Include

Knowledge-level adaptation

Preference adaptation

Goal adaptation

Situation awareness

Selective use of context

A response should not receive a higher score merely because it mentions more personal information.

Relevant personalization is rewarded; forced personalization is not.

C.5 Reliability Rubric

Reliability evaluates whether appropriate interaction behavior remains dependable as the interaction develops.

Score	Reliability Evaluation
0	Demonstrates major inconsistency, loses essential context or constraints, expresses severely unsupported certainty, or fails to recover from a clear error
1	Some reliable behavior is present, but major continuity, consistency, uncertainty, or recovery problems remain
2	Maintains important aspects of the interaction but demonstrates meaningful reliability limitations
3	Maintains dependable interaction behavior with only minor inconsistencies
4	Demonstrates strong consistency, continuity, constraint persistence, appropriate uncertainty handling, or error recovery as required by the scenario
N/A	Scenario does not create sufficient opportunity to evaluate the targeted Reliability behavior

Reliability Evidence May Include

Maintaining established facts

Preserving active constraints

Consistent behavior

Appropriate uncertainty

Successful error recovery

C.6 Engagement & Responsiveness Rubric

Engagement & Responsiveness evaluates whether the AI recognizes meaningful interaction signals and adapts appropriately as the interaction progresses.

Score	Engagement & Responsiveness Evaluation
0	Ignores or seriously misinterprets the relevant interaction signal and fails to adapt
1	Recognizes part of the signal but demonstrates little useful behavioral adaptation
2	Responds meaningfully but adaptation is incomplete or contains important limitations
3	Responds appropriately and maintains useful interaction progression with only minor limitations
4	Fully or near-fully recognizes the interaction signal, adapts appropriately, and continues effectively toward the relevant objective
N/A	No sufficient interaction signal or response opportunity exists

Engagement & Responsiveness Evidence May Include

Recognizing confusion

Applying feedback

Recognizing completed stages

Continuing from updated requirements

Maintaining goal-oriented interaction

C.7 Sub-Factor Rubric Reference

The following table defines the central scoring question for each of the 25 BOND sub-factors.

Code	Sub-Factor	Central Evaluation Question
UR	Requirement Recognition	Did the AI recognize the explicit requirements?
UI	Intent Interpretation	Did the AI appropriately interpret the reasonably supported objective?
UC	Context Integration	Did the AI correctly use relevant available context?

Code	Sub-Factor	Central Evaluation Question
UCR	Constraint Recognition	Did the AI recognize and follow the relevant constraints?
UU	Clarification & Update Handling	Did the AI appropriately clarify or incorporate updated information?
CC	Comprehensibility	Could the intended human reasonably understand the response?
CS	Structural Organization	Was the information organized appropriately?
CP	Precision	Was the communication sufficiently precise?
CR	Relevance & Focus	Did the response remain focused on the relevant objective?
CE	Explanatory Sufficiency	Was enough explanation provided for the stated objective?
PCR	Relevant Context Recognition	Did the AI identify context relevant to the current interaction?
PLA	User-Level Adaptation	Did the AI appropriately adapt to the human's level?
PPA	Preference Adaptation	Did the AI appropriately apply relevant stated preferences?
PGS	Goal & Situation Adaptation	Did the response adapt to the relevant goal and situation?
PSC	Selective Context Use	Did the AI use only context that meaningfully improved the interaction?
RBC	Behavioral Consistency	Did appropriate behavior remain reasonably consistent?
RCC	Contextual Continuity	Did the AI preserve relevant established context?
RCP	Constraint Persistence	Did previously established constraints remain active when relevant?
RUC	Uncertainty Calibration	Did the AI communicate uncertainty appropriately relative to available evidence?
RER	Error Recovery	Did the AI correctly recover after an identified error or correction?
ERS	Interaction Signal Recognition	Did the AI recognize the relevant human interaction signal?
ERF	Feedback Responsiveness	Did the AI meaningfully adapt after feedback?
ERP	Progression Awareness	Did the AI recognize the current stage of the interaction?
ERA	Adaptive Continuation	Did the AI continue appropriately from the updated interaction state?
ERG	Goal-Oriented Engagement	Did the interaction remain usefully directed toward the relevant goal?

C.8 Boundary Between Adjacent Scores

Because scores such as **2** and **3** may sometimes appear close, evaluators should consider the magnitude of the remaining limitation.

Score 0 vs. Score 1

0 → Target behavior is essentially absent or seriously failed.

1 → Some recognizable evidence exists, but performance remains substantially inadequate.

Score 1 vs. Score 2

1 → Major failure dominates the behavior.

2 → Meaningful capability is demonstrated, but important limitations remain.

Score 2 vs. Score 3

2 → Limitations materially affect interaction quality.

3 → Core behavior is effective; remaining limitations are relatively minor.

Score 3 vs. Score 4

3 → Strong performance with identifiable minor weakness.

4 → Complete or near-complete performance under the defined conditions.

A score of **4 does not mean that the AI possesses perfect universal capability.**

It means that the targeted behavior was demonstrated fully or near-fully **within that specific evaluation opportunity.**

C.9 N/A Decision Rule

Before assigning a numerical score, the evaluator should ask:

Did this interaction create a sufficient opportunity for the targeted behavior to be observed?

If:

YES → Assign 0–4

If:

NO → Assign N/A

Example:

A single-turn scenario contains no previously established preference.

Therefore:

Preference Adaptation → N/A

It should not receive:

Preference Adaptation → 0

because the AI was never given an opportunity to adapt to a preference.

C.10 Evidence Recording Rule

For each BAT score, the evaluator should preferably record a short evidence statement.

Example:

Target: Constraint Recognition

Score: 4

Evidence: “The response followed all three specified constraints: exactly four bullet points, beginner-level language, and no equations.”

Another example:

Target: Feedback Responsiveness

Score: 1

Evidence: “The AI acknowledged that the explanation was too technical but repeated essentially the same level of technical detail.”

This makes scoring more auditable.

C.11 Multi-Evaluator Rubric Procedure

When multiple evaluators are used:

Step 1 → Each evaluator independently reads the scenario and AI response.

Step 2 → Each evaluator assigns a BAT score.

Step 3 → Each evaluator records supporting evidence.

Step 4 → Scores are compared.

Step 5 → Agreement or disagreement is recorded.

Step 6 → Scores are aggregated according to the defined experimental procedure.

The original evaluator scores should be preserved even when an aggregate score is later calculated.

C.12 Complete Rubric Decision Flow

Identify Target Sub-Factor

↓

Check Evaluation Opportunity

↓

Insufficient Opportunity?

→ YES → N/A

→ NO → Continue

↓

Observe AI Behavior

↓

Compare Behavior with Scenario-Specific Criteria

↓

Determine Performance Level

↓

0 / 1 / 2 / 3 / 4

↓

Record Supporting Evidence

↓

Normalize Score

↓

Include in BOND Calculation

BOND RUBRIC CORE PRINCIPLE

The BAT rubric converts evaluator judgment into structured measurement by requiring scores to reflect predefined behavioral targets, sufficient evaluation opportunities, observable evidence, and consistent performance-level definitions.

C.13 Detailed Sub-Factor Numerical Scoring Rubrics

The general BAT rubric defines the overall meaning of scores from **0 to 4**. However, different BOND sub-factors represent different interaction behaviors. Therefore, the same general score description cannot always be applied identically across all twenty-five sub-factors.

This section provides **behavior-specific scoring anchors** for each BOND sub-factor.

These anchors are intended to help evaluators determine not only whether a behavior is weak or strong, but also **which numerical BAT score is most appropriate for the observed interaction evidence**.

DETAILED SCORING PRINCIPLE

A BAT score should represent the degree to which the predefined target behavior

was demonstrated within the available evaluation opportunity. Scores should be assigned using behavioral evidence rather than general impressions of response quality.

C.13.1 Understanding — Detailed Scoring Rubrics

UR — Requirement Recognition

Score	Scoring Anchor
0	Fails to recognize the main requirement or responds to a substantially different task.
1	Recognizes only a limited part of the requirement while missing major essential elements.
2	Recognizes the main task but misses one or more important requirements that materially affect the response.
3	Recognizes nearly all important requirements; only minor elements are missed or imperfectly interpreted.
4	Correctly recognizes all or nearly all essential requirements relevant to the scenario.
N/A	The scenario does not provide a sufficient requirement-recognition opportunity.

UI — Intent Interpretation

Score	Scoring Anchor
0	Substantially misinterprets the human's intended objective despite sufficient interaction evidence.
1	Shows limited recognition of the intended objective but responds primarily to an incorrect or superficial interpretation.
2	Identifies the general intent but misses an important aspect of what the human is trying to achieve.
3	Correctly interprets the intended objective with only a minor limitation or unnecessary assumption.
4	Correctly interprets the human's intended objective to the full or near-full extent reasonably supported by the interaction.
N/A	The scenario does not provide sufficient evidence for evaluating intent interpretation.

UC — Context Integration

Score	Scoring Anchor
0	Ignores, contradicts, or seriously misuses essential relevant context.
1	Uses very little relevant context or integrates it incorrectly in major ways.
2	Uses some important context correctly but misses or misapplies other relevant information.
3	Integrates nearly all relevant context correctly with only minor omissions or inconsistencies.

Score	Scoring Anchor
4	Correctly and appropriately integrates all or nearly all relevant available context required by the scenario.
N/A	No meaningful contextual information is available or required for the target behavior.

UCR — Constraint Recognition

Score	Scoring Anchor
0	Fails to recognize or follow essential constraints despite their clear presence.
1	Recognizes only a small portion of the relevant constraints; major violations remain.
2	Recognizes and follows some important constraints but violates or misses others that materially affect the response.
3	Follows nearly all relevant constraints with only minor deviations.
4	Correctly recognizes and follows all or nearly all applicable constraints.
N/A	No meaningful constraint is present for evaluation.

UU — Clarification & Update Handling

Score	Scoring Anchor
0	Fails to seek necessary clarification or ignores a clear update/correction, producing behavior based on unsupported or outdated information.
1	Partially notices ambiguity or an update but handles it inadequately and retains major misunderstanding.
2	Recognizes the need for clarification or the updated information but incorporates it only partially or inconsistently.
3	Handles clarification or updated information appropriately with only a minor limitation.
4	Correctly identifies when clarification is necessary and/or fully incorporates relevant updated information into subsequent behavior.
N/A	No meaningful ambiguity, clarification, correction, or update opportunity exists.

C.13.2 Clarity — Detailed Scoring Rubrics

CC — Comprehensibility

Score	Scoring Anchor
0	Response is largely incomprehensible or unusable for the intended human under the defined conditions.
1	Some understandable content exists, but major language or complexity problems make the response difficult to use.
2	Main meaning is understandable, but important sections remain unnecessarily difficult, confusing, or poorly explained.
3	Response is clearly understandable with only minor issues in wording or complexity.
4	Response is consistently understandable and appropriately expressed for the intended

Score	Scoring Anchor
	human and interaction conditions.
N/A	Available response evidence is insufficient to evaluate comprehensibility.

CS — Structural Organization

Score	Scoring Anchor
0	Information lacks meaningful organization and is substantially difficult to follow.
1	Limited structure is present, but major organizational problems interfere with usability.
2	Basic organization is present, but sequencing, grouping, or hierarchy contains important weaknesses.
3	Information is logically organized with only minor structural limitations.
4	Information is organized in a highly clear, logical, and usable structure appropriate to the task.
N/A	The response is too limited or the scenario otherwise provides insufficient evidence for structural evaluation.

CP — Precision

Score	Scoring Anchor
0	Communication is substantially vague, ambiguous, or misleading regarding the targeted information.
1	Some precise information is present, but major vague or ambiguous elements remain.
2	Main meaning is communicated, but important statements lack sufficient precision.
3	Communication is precise in nearly all relevant areas with only minor ambiguity.
4	Communication is consistently specific, precise, and appropriately qualified for the scenario.
N/A	Insufficient communication evidence exists to evaluate precision.

CR — Relevance & Focus

Score	Scoring Anchor
0	Response is largely unrelated to the relevant request or substantially loses the intended focus.
1	Contains some relevant content but is dominated by unnecessary, unrelated, or misdirected information.
2	Addresses the main objective but includes substantial unnecessary deviation or misses some important focus.
3	Remains strongly focused with only minor unnecessary information or deviation.
4	Consistently prioritizes information directly relevant to the interaction objective without meaningful unnecessary deviation.
N/A	Insufficient response evidence exists to evaluate relevance and focus.

CE — Explanatory Sufficiency

Score	Scoring Anchor
0	Provides essentially no useful explanation required for the stated objective.
1	Provides limited explanation, leaving major necessary reasoning or information absent.
2	Provides a meaningful explanation but leaves important gaps affecting understanding or task completion.
3	Provides sufficient explanation for nearly all relevant needs with only minor omissions.
4	Provides all or nearly all explanation reasonably necessary to satisfy the defined interaction objective without unnecessary expansion.
N/A	The scenario does not require meaningful explanatory content or otherwise provides insufficient evidence.

C.13.3 Personalization — Detailed Scoring Rubrics

PCR — Relevant Context Recognition

Score	Scoring Anchor
0	Fails to recognize clearly relevant human-specific context or treats clearly irrelevant context as central.
1	Recognizes only a small portion of relevant context and misses major information that should influence the interaction.
2	Recognizes some important contextual information but misses other meaningful context.
3	Recognizes nearly all relevant contextual information with only minor omissions.
4	Correctly distinguishes and identifies all or nearly all contextual information relevant to the current interaction.
N/A	No meaningful human-specific context is available for evaluation.

PLA — User-Level Adaptation

Score	Scoring Anchor
0	Makes no meaningful adaptation to the clearly established user level and produces a substantially inappropriate response level.
1	Shows minimal adaptation but remains significantly too advanced, too basic, or otherwise unsuitable.
2	Adapts to the general level but retains important mismatches in terminology, complexity, or depth.
3	Adapts effectively to the user's level with only minor mismatches.
4	Consistently adapts terminology, complexity, explanation depth, and relevant assumptions appropriately to the established user level.
N/A	The user's level is not established and cannot reasonably be inferred from the scenario.

PPA — Preference Adaptation

Score	Scoring Anchor
0	Completely ignores or directly violates a clearly relevant established preference.
1	Shows minimal acknowledgment or application of the preference but fails to adapt meaningfully.
2	Applies the preference partially or inconsistently.
3	Applies the relevant preference effectively with only minor limitations.
4	Fully or near-fully adapts to the relevant established preference wherever appropriate.
N/A	No relevant preference has been established or the available preference is unrelated to the current task.

PGS — Goal & Situation Adaptation

Score	Scoring Anchor
0	Ignores a clearly established goal or situation and produces a response substantially unsuitable for it.
1	Shows limited awareness of the goal or situation but fails to adapt in important ways.
2	Provides meaningful adaptation but misses important aspects of the human's goal or circumstances.
3	Adapts effectively to nearly all important aspects of the goal and situation with minor limitations.
4	Appropriately shapes the response around all or nearly all relevant goal and situational information.
N/A	No meaningful goal or situational information exists beyond the basic task requirement.

PSC — Selective Context Use

Score	Scoring Anchor
0	Uses context in a clearly inappropriate, irrelevant, or strongly forced manner, or ignores essential relevant context while emphasizing irrelevant information.
1	Demonstrates poor context selection, with substantial unnecessary personalization or omission of important relevant context.
2	Uses some context selectively but includes noticeable unnecessary personalization or misses some relevant information.
3	Selects and uses relevant context appropriately with only minor unnecessary or missed contextual use.
4	Uses relevant context selectively and purposefully while avoiding unnecessary, forced, or irrelevant personalization.
N/A	The scenario provides insufficient contextual information to evaluate selective context use.

C.13.4 Reliability — Detailed Scoring Rubrics

RBC — Behavioral Consistency

Score	Scoring Anchor
0	Demonstrates major unexplained inconsistency in the targeted behavior across comparable interaction conditions.
1	Behavior is frequently inconsistent, although limited stable performance is visible.
2	Demonstrates moderate consistency but contains important unexplained variations.
3	Behavior remains strongly consistent with only minor unexplained variation.
4	Demonstrates stable and appropriate behavior throughout the comparable evaluation conditions.
N/A	The scenario does not contain sufficient comparable observations to evaluate consistency.

RCC — Contextual Continuity

Score	Scoring Anchor
0	Loses, contradicts, or incorrectly changes essential established context.
1	Preserves only limited context while losing major information necessary for continued interaction.
2	Maintains the main context but loses or misuses important secondary information.
3	Maintains nearly all relevant established context with only minor continuity errors.
4	Correctly preserves and applies all or nearly all relevant established context throughout the required interaction.
N/A	The interaction does not require preservation of meaningful earlier context.

RCP — Constraint Persistence

Score	Scoring Anchor
0	Abandons or violates an active established constraint when it remains clearly applicable.
1	Maintains the constraint only briefly or inconsistently and demonstrates major later violations.
2	Preserves some active constraints but fails to maintain others or applies them inconsistently.
3	Maintains nearly all active constraints across relevant turns with only minor deviations.
4	Consistently preserves all or nearly all applicable established constraints for as long as they remain active.
N/A	No persistent constraint exists or the interaction does not continue long enough to test persistence.

RUC — Uncertainty Calibration

Score	Scoring Anchor
0	Expresses strong unsupported certainty or otherwise handles uncertainty in a seriously

Score	Scoring Anchor
	inappropriate manner despite clear insufficiency of evidence.
1	Shows limited recognition of uncertainty but substantially overstates or understates confidence.
2	Recognizes meaningful uncertainty but communicates or manages it imperfectly.
3	Represents uncertainty appropriately with only minor overconfidence, underconfidence, or qualification issues.
4	Accurately distinguishes sufficient from insufficient evidence and communicates uncertainty in a proportionate and useful manner.
N/A	The scenario does not contain meaningful uncertainty or insufficient-information conditions.

RER — Error Recovery

Score	Scoring Anchor
0	Fails to recognize or recover from an identified error and continues relying on incorrect information or behavior.
1	Acknowledges the error but makes little effective correction, or repeats major consequences of the error.
2	Corrects the central error but recovery is incomplete, inconsistent, or fails to update some relevant consequences.
3	Correctly recovers from the error and updates nearly all affected behavior, with only minor residual issues.
4	Correctly recognizes the error, updates the interaction state, corrects relevant consequences, and continues appropriately using the corrected information.
N/A	No meaningful error or correction opportunity occurs.

C.13.5 Engagement & Responsiveness — Detailed Scoring Rubrics

ERS — Interaction Signal Recognition

Score	Scoring Anchor
0	Completely misses or seriously misinterprets a clear and relevant interaction signal.
1	Shows minimal recognition of the signal but substantially misunderstands its meaning or importance.
2	Recognizes the general signal but misses an important aspect of what it indicates.
3	Correctly recognizes the signal with only minor interpretive limitations.
4	Correctly recognizes and appropriately interprets the relevant interaction signal in full or near-full form.
N/A	No meaningful interaction signal is provided.

ERF — Feedback Responsiveness

Score	Scoring Anchor
0	Ignores meaningful feedback or continues essentially unchanged despite a clear need for adaptation.
1	Acknowledges the feedback but makes little meaningful behavioral change.
2	Makes a noticeable adaptation, but important elements of the feedback remain unaddressed.
3	Adapts effectively to nearly all relevant feedback with only minor limitations.
4	Fully or near-fully incorporates the meaningful feedback into subsequent interaction behavior.
N/A	No meaningful feedback is provided or no subsequent response opportunity exists.

ERP — Progression Awareness

Score	Scoring Anchor
0	Fails to recognize the current interaction stage and substantially repeats, skips, or mismanages progression.
1	Shows limited awareness of progress but remains substantially misaligned with the current stage.
2	Recognizes general progression but unnecessarily repeats or misses important completed/current stages.
3	Correctly tracks interaction progression with only minor unnecessary repetition or sequencing issues.
4	Accurately recognizes completed, current, and relevant next stages and behaves accordingly.
N/A	The interaction contains no meaningful progression across stages.

ERA — Adaptive Continuation

Score	Scoring Anchor
0	Continues according to an outdated interaction state or unnecessarily restarts despite a clear change.
1	Recognizes some change but continuation remains largely based on previous conditions.
2	Partially adapts continuation but retains important outdated information or unnecessary repetition.
3	Continues appropriately from the updated state with only minor residual issues.
4	Seamlessly and appropriately continues from the updated interaction state without unnecessary restart or reliance on obsolete conditions.
N/A	No meaningful interaction-state change requiring adaptive continuation occurs.

ERG — Goal-Oriented Engagement

Score	Scoring Anchor
0	Interaction substantially loses or obstructs the relevant human objective.
1	Provides limited progress toward the objective but contains major unnecessary diversion, continuation, or misdirection.
2	Makes meaningful progress toward the objective but contains important inefficiencies or unnecessary interaction.
3	Maintains strong progress toward the relevant objective with only minor unnecessary deviation.
4	Keeps the interaction efficiently and appropriately directed toward achieving the relevant human objective, including ending or shortening interaction when appropriate.
N/A	No sufficiently defined interaction goal exists for evaluation.

C.14 Numerical Interpretation Reference

The detailed rubrics preserve the same general progression across all twenty-five sub-factors:

BAT Score	Normalized Value	Interpretation
0	0.00	No meaningful demonstration
1	0.25	Weak demonstration
2	0.50	Partial demonstration
3	0.75	Strong demonstration
4	1.00	Full or near-full demonstration
N/A	—	Insufficient evaluation opportunity

The behavioral meaning of each score is determined by the **specific sub-factor rubric**, rather than by numerical value alone.

C.15 Scoring Priority Rule

When assigning a BAT score, evaluators should use the following priority:

1. Scenario-Specific Criteria

↓

2. Detailed Sub-Factor Rubric

↓

3. General BAT Rubric

This hierarchy is important.

A benchmark scenario may contain more precise scoring conditions than the general sub-factor description.

For example, a Constraint Recognition scenario may contain **four explicitly defined essential constraints**.

The scenario-specific rubric can define which failures constitute scores of 0–4 while remaining consistent with the broader UCR rubric.

Therefore:

The detailed rubric provides standardized behavioral anchors, while scenario-specific criteria provide the most precise scoring instructions for an individual BAT item.

C.16 Essential vs. Non-Essential Elements

Not every small response difference should reduce a BAT score.

Scenario designers should distinguish between:

Essential Elements

and

Non-Essential Elements

An **essential element** directly affects whether the targeted behavior has been demonstrated.

A **non-essential element** may improve the response but is not necessary for demonstrating the target.

For example:

Scenario:

“Give exactly three advantages of solar energy in bullet points.”

Essential elements for Constraint Recognition may include:

Exactly three items

Bullet-point format

Whether the AI adds a short introductory sentence may or may not be essential depending on the predefined scenario rubric.

This distinction prevents evaluators from penalizing harmless differences that are unrelated to the measurement target.

C.17 Score Selection Rule

When observed behavior falls between two adjacent scoring anchors, the evaluator should ask:

Does the remaining limitation materially affect demonstration of the target behavior?

If the limitation is **important and materially affects the behavior**, the lower score should generally be selected.

If the target behavior is **effectively demonstrated and only minor imperfections remain**, the higher score may be appropriate.

Evaluators should record evidence whenever the distinction is uncertain.

C.18 Detailed Rubric Decision Process

Select BAT Scenario

↓

Identify Primary Sub-Factor

↓

Confirm Sufficient Evaluation Opportunity

↓

If No → N/A

↓

If Yes → Identify Observable Evidence

↓

Apply Scenario-Specific Criteria

↓

Compare with Detailed Sub-Factor Anchors

↓

Assign 0 / 1 / 2 / 3 / 4

↓

Record Evidence

↓

Normalize to 0.00 / 0.25 / 0.50 / 0.75 / 1.00

↓

Aggregate with Other Valid Observations

↓

Calculate Sub-Factor Score

DETAILED RUBRIC CORE PRINCIPLE

The numerical value assigned by BAT should be traceable to predefined observable behavior. A score of 0, 1, 2, 3, or 4 therefore represents an evidence-based level of demonstrated performance for a specific BOND sub-factor, while N/A represents the absence of a sufficient evaluation opportunity.

Appendix D — Additional Graphs and Visual Representations

The following graphs and visual models provide additional ways to interpret and present BOND results. These representations are intended to complement the numerical score, factor profile, and sub-factor tables.

Because the current BOND framework has not yet completed empirical validation, the graphs in this appendix are **structural or illustrative representations**. They should not be interpreted as experimental findings.

GRAPH PRINCIPLE

Visualizations should make BOND results easier to interpret without replacing the underlying numerical measurements or implying evidence that has not been collected.

D.1 Five-Factor BOND Profile

A complete BOND result can be represented using five primary dimensions:

Understanding (U)

Clarity (C)

Personalization (P)

Reliability (R)

Engagement & Responsiveness (ER)

An illustrative result may be:

Factor	Illustrative Score
Understanding	0.85
Clarity	0.90
Personalization	0.70
Reliability	0.80

Factor	Illustrative Score
Engagement & Responsiveness	0.85

These values could be displayed using a **radar chart**.

The five axes would represent:

U

C

P

R

ER

A larger balanced shape would indicate relatively strong performance across all five dimensions, while an uneven shape would reveal factor-specific strengths and weaknesses.

For example:

High U + High C + Low P

would immediately show that the system communicates and interprets effectively but demonstrates weaker personalization.

The radar chart therefore provides information that an overall BOND score alone cannot show.

D.2 BOND Factor Bar Graph

The same factor profile may also be represented using a bar graph.

Y-Axis → Normalized Score (0–1)

X-Axis → Five BOND Factors

Illustrative values:

U = 0.85

C = 0.90

P = 0.70

R = 0.80

ER = 0.85

A bar graph may be particularly useful when comparing exact factor values because differences in bar height are easier to read than differences in radar-chart area.

D.3 Twenty-Five Sub-Factor Profile

A more detailed visualization can display all twenty-five BOND sub-factors.

The X-axis can contain:

UR, UI, UC, UCR, UU

CC, CS, CP, CR, CE

PCR, PLA, PPA, PGS, PSC

RBC, RCC, RCP, RUC, RER

ERS, ERF, ERP, ERA, ERG

The Y-axis represents:

Normalized Score = 0.00–1.00

This visualization allows researchers to identify specific weaknesses hidden within the five-factor averages.

For example:

Reliability = 0.80

may initially appear strong.

However, the sub-factor graph could reveal:

RBC = 0.90

RCC = 0.90

RCP = 0.90

RUC = 0.55

RER = 0.75

The overall factor score would therefore hide a relatively weak **Uncertainty Calibration** result.

This demonstrates why BOND preserves sub-factor measurements.

D.4 BOND Interval Graph

The BOND interval system can be represented visually as a continuous scale:

0.00 ————— 0.20 ————— 0.40 ————— 0.60 ————— 0.80
————— 1.00

Very Low | Low | Moderate | High | Very High

An individual system can then be positioned on this scale.

For example:

BOND = 0.82

would fall within:

Very High

However, the numerical value should remain visible because interval labels alone reduce measurement precision.

D.5 Multi-System Comparison Graph

BOND can also compare multiple AI systems across the same five factors.

Illustrative example:

Factor	System A	System B	System C
Understanding	0.85	0.78	0.90
Clarity	0.88	0.82	0.84
Personalization	0.72	0.81	0.65
Reliability	0.80	0.74	0.87
Engagement & Responsiveness	0.86	0.79	0.83

These values could be visualized through:

Grouped Bar Chart

or

Multi-Series Radar Chart

The purpose would not simply be to identify a winner.

Instead, the graph would reveal different **interaction profiles**.

For example:

System A → Strong Engagement

System B → Stronger Personalization

System C → Strong Understanding and Reliability

All values in this example are illustrative only.

D.6 BOND Coverage Visualization

BOND results should also communicate how much of the framework was actually evaluated.

For example:

System A

BOND = 0.84

Factor Coverage = 100%

compared with:

System B

BOND = 0.86

Factor Coverage = 40%

System B has a slightly higher numerical score, but the evidence covers substantially less of the framework.

A coverage graph could therefore show:

Evaluated Factors → Filled

N/A Factors → Unfilled

or use a simple percentage bar:

Factor Coverage:



versus:



This reinforces the principle that:

Score magnitude and evaluation coverage represent different information.

D.7 BOND Evaluation Pipeline Diagram

The complete BOND measurement pipeline can be visualized as:

HUMAN-AI INTERACTION

↓

BAT SCENARIO

↓

OBSERVABLE AI BEHAVIOR

↓

0-4 / N/A RAW SCORE

↓

NORMALIZATION

↓

25 SUB-FACTOR SCORES

↓

5 PRIMARY FACTOR SCORES

↓

WEIGHT DISTRIBUTION

↓

BOND EQUATION

↓

OVERALL BOND SCORE

↓

FACTOR PROFILE + COVERAGE + INTERVAL

This diagram represents the complete path from interaction evidence to final measurement.

D.8 BOND Architecture Diagram

The conceptual architecture can be represented as:

AI CONNECTION-BUILDING CAPABILITY

↓

Understanding	Clarity	Personalization	Reliability	Engagement & Responsiveness
UR	CC	PCR	RBC	ERS
UI	CS	PLA	RCC	ERF
UC	CP	PPA	RCP	ERP
UCR	CR	PGS	RUC	ERA
UU	CE	PSC	RER	ERG

↓

25 SUB-FACTORS

↓

BAT MEASUREMENT

↓

BOND SCORE

This diagram provides a compact representation of the entire framework.

D.9 Future Experimental Graphs

After formal experiments are completed, additional graphs may be added to future BOND versions.

Potential experimental visualizations include:

Evaluator Agreement Graph

Comparison of scores assigned by independent evaluators.

Score Distribution Graph

Distribution of BOND scores across evaluated AI systems.

Factor Correlation Matrix

Relationships between the five BOND factors.

Sub-Factor Correlation Matrix

Relationships among the twenty-five sub-factors.

Run-to-Run Stability Graph

Variation in BOND performance across repeated evaluations.

Domain Comparison Graph

Differences between performance in education, general chat, robotics, research assistance, or other domains.

Human Judgment Correlation Graph

Relationship between BOND scores and independent human interaction-quality judgments.

These graphs should be produced only after sufficient empirical data becomes available.

D.10 Visualization Interpretation Rule

BOND visualizations should follow three rules:

Rule 1 — Numerical data remains primary.

Graphs summarize the data but do not replace reported values.

Rule 2 — Illustrative data must be labeled clearly.

Hypothetical examples should never be presented as experimental findings.

Rule 3 — Missing data must remain visible.

N/A observations or incomplete coverage should not be silently displayed as zero.

BOND VISUALIZATION CORE PRINCIPLE

Graphs should reveal the structure, strengths, weaknesses, and coverage of BOND measurements while preserving a clear distinction between illustrative examples and empirically collected results.

Appendix E — Complete BOND Examples

The following examples demonstrate how BAT scenarios can be evaluated and converted into BOND measurements. These examples are **illustrative** and do not represent experimental results from any specific AI system.

E.1 Example 1 — Requirement and Constraint Recognition

Scenario

Human:

“Explain photosynthesis in exactly three bullet points. Use simple language and do not include equations.”

Evaluation Targets

Primary → UCR — Constraint Recognition

Secondary → UR — Requirement Recognition

Expected Interaction Requirements

The AI should recognize:

Task → Explain photosynthesis

Quantity → Exactly three points

Format → Bullet points

Language → Simple

Restriction → No equations

Illustrative Response

The AI provides exactly three simple bullet points explaining photosynthesis without equations.

BAT Evaluation

UR = 4

UCR = 4

Normalized:

UR = 1.00

UCR = 1.00

The response demonstrates full or near-full performance for the targeted behaviors under this scenario.

E.2 Example 2 — User-Level Adaptation

Scenario

Human:

“I have never studied machine learning before. Explain overfitting to me.”

Primary Target

PLA — User-Level Adaptation

The AI should adapt its terminology and explanation depth to the explicitly stated beginner level.

If the response immediately uses unexplained advanced terminology, the adaptation score may decrease even if the technical information is correct.

Illustrative BAT Result

PLA Raw Score = 3

Normalized:

PLA = 0.75

Interpretation:

Strong adaptation with minor limitations.

E.3 Example 3 — Preference Adaptation

Turn 1

Human:

“I understand concepts better when you give me a practical example.”

Turn 2

Human:

“Explain electric resistance.”

Primary Target

PPA — Preference Adaptation

A strong response should appropriately incorporate a practical example because the preference is directly relevant to the later request.

Possible Result

If the AI explains resistance clearly but provides no practical example:

PPA = 1 or 2, depending on the complete response and scenario-specific rubric.

If it appropriately integrates an example:

PPA = 3 or 4, depending on execution quality.

The score should be determined from observable evidence rather than the presence of a keyword alone.

E.4 Example 4 — Constraint Persistence

Turn 1

Human:

“For this entire task, use metric units.”

Later Turn

Human:

“Calculate the distance traveled by the object.”

Primary Target

RCP — Constraint Persistence

If the AI correctly continues using metric units:

Strong / Full Demonstration

If it unnecessarily switches to non-metric units despite the still-active instruction:

Weak / No Demonstration, depending on the extent of failure.

This scenario demonstrates why some Reliability behaviors require multi-turn evaluation.

E.5 Example 5 — Feedback Responsiveness

Human

“Explain neural networks to me.”

The AI provides an advanced technical explanation.

Human Feedback

“That is too complicated. Explain it like I'm completely new to AI.”

Primary Target

ERF — Feedback Responsiveness

A strong response should substantially simplify the explanation.

Simply writing:

“Sure, I'll simplify it.”

and then providing essentially the same technical explanation would not demonstrate strong Feedback Responsiveness.

The evaluation is based on **behavioral adaptation**, not acknowledgment alone.

E.6 Example 6 — Error Recovery

Earlier Information

Human:

“The value is 10.”

Later:

Human:

“Correction: I gave you the wrong value. It should be 20.”

Primary Target

RER — Error Recovery

The AI should:

Recognize the correction

↓

Update the active information

↓

Stop relying on the incorrect value

↓

Continue using 20 where relevant

If the AI acknowledges the correction but later continues calculating with 10, Error Recovery remains weak.

E.7 Example 7 — N/A

Suppose an interaction contains:

No stated preference

No previous interaction context

No information about the human's preferred communication style

An evaluator attempts to score:

PPA — Preference Adaptation

There is no sufficient evaluation opportunity.

Therefore:

PPA = N/A

not:

PPA = 0

This distinction prevents absence of evidence from being treated as evidence of failure.

E.8 Complete Illustrative BOND Calculation

Suppose BAT evaluation produces the following five factor scores:

U = 0.85

C = 0.80

P = 0.70

R = 0.75

ER = 0.90

Under Standard Equal-Weight BOND:

B = (0.85 + 0.80 + 0.70 + 0.75 + 0.90) / 5

B = 4.00 / 5

$$B = 0.80$$

Therefore:

$$\text{Overall BOND Score} = 0.80 / 1.00$$

$$\text{BOND Percentage} = 80\%$$

Interval = Very High

If all five factors were validly evaluated:

$$\text{Factor Coverage} = 100\%$$

The final report could therefore state:

BOND = 0.80 / 1.00 — Very High | Factor Coverage = 100% | Standard Equal-Weight Configuration

Again, these numbers are illustrative rather than experimental results.

E.9 Complete Example Score Card

Measurement	Illustrative Result
Understanding	0.85
Clarity	0.80
Personalization	0.70
Reliability	0.75
Engagement & Responsiveness	0.90
Overall BOND	0.80
Percentage	80%
Interval	Very High
Factor Coverage	100%
Configuration	Standard Equal Weight

E.10 Example Interpretation

The overall score indicates strong connection-building performance under the illustrative evaluation conditions.

However, the factor profile provides more useful diagnostic information.

In this example:

Strongest Factor → Engagement & Responsiveness = 0.90

Weakest Factor → Personalization = 0.70

Therefore, simply reporting:

BOND = 0.80

would hide an important difference between the system's interaction dimensions.

The complete interpretation should therefore consider:

Overall Score

-

Five-Factor Profile

-

Sub-Factor Results

-

Coverage

-

Benchmark Configuration

EXAMPLE PRINCIPLE

BOND examples demonstrate how interaction evidence becomes BAT scores and how BAT scores become factor and overall measurements; illustrative examples should never be represented as empirical validation.

With Appendices A–E, the core appendix structure is now complete: **BAT, Full Tables, Rubrics, Additional Graphs, and Examples.**